C A R S E M

ENGINEERING CHANGE NOTICE

E.C.N. NO : 2508037C

TITLE : CORRECTIVE AND PREVENTIVE ACTION PROCEDURE

SPECIFICATION NO: CRQA100M01

REVISION: CM

✓ PERMANENT ECN :

- [ ] NEW SPECIFICATION
- ✓ SPECIFICATION REVISION
- [ ] REVIEW FOR COMPLIANCE
- [ ] OBSOLETE SPECIFICATION

[ ] TEMPORARY ECN

- [ ] *PERMANENT CHANGE
- [ ] TEMPORARY CHANGE

*PERMANENT CHANGE TO BE ENCLOSED WITH ECN TO SPEC DEPARTMENT THROUGH E-MAIL.

TRAINING NEEDS:

(TO BE FILLED BY ORIGINATOR)

- ✓ ORIGINATOR REQUIRE UPDATE TRAINING MANUAL INFORMATION AND CONDUCT TRAINING
- [ ] TRAINING NOT REQUIRED.
- [ ] TRAINING TO BE DONE BY TRAINING DEPT AND REQUIRE TO UPDATE THE TRAINING MANUAL
- [ ] JUSTIFICATION IF NO REQUIRE TO UPDATE TRAINING MANUAL:

TRAINING DEPT'S ACKNOWLEDGEMENT : **MAHANI/CHEN XH**

JUSTIFICATION FOR CHANGE:

NOTE : - THOSE MARKED * MUST REQUIRE ECN TO BE RAISED OTHERWISE TECN WILL BE REJECTED.
EXCEPTION IS CONSIDERED PROVIDED EVIDENCE ATTACHED FOR THOSE ITEM MARKS '*' ONLY.

- [ ] * QUALIFICATION BUILD
- [ ] AUDIT RESPONSE
- [ ] NEW MACHINE / PACKAGE / MATERIAL / PROCESS
- [ ] NEW SPECIFICATION / PROCEDURE
- ✓ PROCESS / QUALITY IMPROVEMENT
- [ ] 8D ACTIONS, CARSEM REF #:
- [ ] *EVALUATION / RISK BUILD
- [ ] SPECIFICATION COMPLIANCE
- [ ] 3D ACTIONS, CARSEM REF #:_____
- [ ] OTHERS, PLEASE

SPECIFY: _____

CHANGE TYPE :

- [ ] CUSTOMER REQUEST
- ✓ CARSEM INITIATED

THE SPECIFICATION IS APPLICABLE TO :

- CARSEM (M) : ✓ M-SITE ✓ S – SITE
- CARSEM **CHINA :** ✓ **Suzhou Site** ✓ **Suxiang Site**

TIME FRAME (FOR TECN ONLY)- PLEASE SPECIFY :

_____ WEEK / S or _____ MONTHS / S

TECN FOR MORE THAN 1 MONTH CAN BE ALLOWED FOR EVALUATION PURPOSE ONLY

EXPIRY DATE : [ ][ ][ ][ ] - [ ][ ][ ] - [ ][ ]

- [ ] NEW
- [ ] EXTENSION, _____TIME

THE CHANGE(S) IS APPLICABLE TO (FOR TECN ONLY):

- CARSEM (M) : [ ] M – SITE [ ] S – SITE
- CARSEM **CHINA:** [ ] **Suzhou Site** [ ] **Suxiang Site**

** COUNTER-PART APPROVAL [ ] REQUIRED [ ] NOT REQUIRED

** CHANGES REQUIRED PCN [ ] YES [ ] NO

IF YES , PCN NUMBER : _____

*NOTE: ** DETERMINED BY ORIGINATOR*

DESCRIPTION OF CHANGE :

FROM : AS PER HISTORY OF CHANGES

TO : AS PER HISTORY OF CHANGES (PAGE 151 OF 151)

RELATED/CROSS REFERENCE SPEC TO BE REVISED:

- ✓ NO
- [ ] YES. SPEC NO._____

ORIGINATOR	JAMES YAN
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HEAD OF QUALITY	R. MANIVANAN

(TO BE FILLED BY ORIGINATOR)

CUSTOMER APPROVAL: [ ] REQUIRED ✓ NOT REQUIRED

COMPANY NAME: _____ NAME / SIGNATURE _____

APPROVAL FOR SPECIFICATION RELEASE: JAMES YAN EFFECTIVE DATE: 2025-SEP-17

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1.0 Title :

Corrective and Preventive Action Procedure.

2.0 Purpose :

2.1 To define the procedures how the corrective and preventive action system operates for non-conforming products and non-conformity.

3.0 Scope :

3.1 Applies to all fault feedback External (VOC) and Internal (VOP).

3.2 Applies to Non-conformance due to Non-compliance to procedure/ admin issues/line audit failures, Lot Rejection at QA Gate and Quality Feedback from Internal Customer.

3.3 Applies to Corrective Action Request to Supplier (SCAR)

4.0 Reference :

Applicable forms

4.1 8D Corrective action report.

4.2 Supplier corrective action request.

4.3 Recurring deficiency report.

4.4 Pertinent QA Acceptance/Test Specifications

4.5 CRQA100N02 – Customer Complaint And Product Return

4.6 CRQA100H07 – Material Review Board (MRB).

4.7 CRQA100G02 – Document/Record Retention Procedure.

4.8 CRQA100H03 – Incoming Material Quality Control.

4.9 QCG000046 – Continuous Improvement Process

4.10 ISO 9001:2015 – Quality Management Systems Requirements

4.11 EMSWK0004 – Environmental, Safety & Health Accidents / Incidents Investigation Procedure

4.12 QRA0044 – e8D System User Manual

4.13 IATF 16949:2016 – Automotive Quality Management System Standard

4.14 CRQA100 - Quality Assurance Manual

4.15 QCG0000133 – Best Known Method (BKM)/Lesson Learnt (LL) Proliferation Procedure

4.16 EMS000024 – Hazardous Substances Management System Manual.

4.17 ESH000014 - Occupational Safety, Health & Environmental Incident Investigation Procedure

4.18 VDA6.3 Standard

5.0 Definition :

5.1 8D–8D discipline Corrective action report.

5.2 Recipient – Person who receives the customer fault feedback is directed to.

5.3 VOC – Voice of Customer

5.4 VOP – Voice of Process

5.5 NCR – Non-Conformance Report

5.6 MRB – Material Review Board

5.7 24 Hours – 1 Working day

5.8 NTF – No Trouble Found designation applied to a part replaced during a service event that, when analysed by the vehicle or parts manufacturer, meets all the requirements of a ”good part”(also referred to as”No Fault Found” or ”Trouble Not Found”).

5.9 0 km complaints - Complaints about mass-produced products being transported to customers, incoming material inspection, and quality problems being found without assembly or during the assembly process.

6.0 Responsibilities :

6.1 C.A.R Co-Ordinator

6.1.1 To track on the external & internal customer fault feedback and request for Corrective action.

6.1.2 To track the 8D C.A.R closure (Attachment 8).

6.1.3 To circulate weekly the relevant C.A.R summary report to the respective General Manager, Operation Manager, Dept Manager, QA Manager & QA Engineer who has been identified. Reporting to include 3D and 8D cycle time, Open and close status for all Customers.

6.2 8D Team Leader

6.2.1 To form a cross functional team (a team of individuals with diverse skills and knowledge related to the problem) to identify root cause and provide corrective actions when non-conformance is found to prevent recurrence (This may include an interim corrective action and a final corrective action, and subsequent audits to ensure that the required corrective action measures are in place and effective in preventing a recurrence of the non-conformance).

6.2.2 Conduct regular team meetings to review progress, discuss findings, and make decisions.

6.2.3 Monitor and track the progress of the 8D process to ensure that it is on schedule. Identify and address any obstacles or challenges that may hinder progress.

6.2.4 To ensure all corrective and preventive action are being implemented accordingly.

6.3 Corrective Action Champion (Person responsible for committed actions)

6.3.1 Contribute ideas in process of preparing the 8D C.A.R

6.3.2 To ensure all corrective and preventive action are being implemented accordingly and timely.

6.3.3 Emphasis shall be placed on identifying the root cause and the preventive of reoccurrence on the non-conformance. This may include an interim corrective action and a final corrective action, and subsequent audits to ensure that the required corrective action measures are in place and effective in preventing a recurrence of the non-conformance.

6.3.4 To ensure all actions closed timely and validate effectiveness accordingly.

6.4 QA Engineer/Executive-Herein after referred to as QA Engineer

6.4.1 To request for corrective actions for discrepancies/non-conformance.

6.4.2 To follow up on the corrective action to prevent similar recurrence

6.4.3 Reserve the authority to reject any corrective action response that is deemed to be not adequate to prevent recurrence.

6.4.4 To ensure all corrective and preventive action are being implemented accordingly.

6.4.5 To validate effectiveness of the containment and corrective actions.

6.4.6 Emphasis shall be placed on identifying the root cause and the preventive of reoccurrence on the non-conformance. This may include an interim corrective action and a final corrective action, and subsequent audits to ensure that the required corrective action measures are in place and effective in preventing a recurrence of the non-conformance

6.4.7 To issue the NCR for all non- conforming material, except for Quality Feedback by Internal Customer which is issued by affected process engineers or production personnel.

6.4.8 To ensure that the procedures are adhered by all parties and initiate an MRB Meeting when necessary for material disposition.

6.4.9 Authorized to 'stop' operation when non-compliance is observed with the approval from the QA Manager or his /her deputy during his / her absence.

6.4.10 Emphasis shall be placed on identifying the root cause and the preventive of reoccurrence on the non-conformance. This may include an interim corrective action and a final corrective action, and subsequent audits to ensure that the required corrective action measures are in place and effective in preventing a recurrence of the non-conformance

6.5 Quality Manager / Engineering & Production Manager

6.5.1 Oversee, guide, and ensure the effectiveness of the 8D report writing process. Responsibilities encompass various aspects of quality management compliance, communication, and continuous improvement to drive successful outcomes in addressing identified issues.

6.5.2 Review and approve the 8D report, ensuring that it is comprehensive, well-documented, and meets the required standards. Verify that the report includes all relevant details, such as the problem description, containment actions, root cause analysis, and corrective actions.

6.5.3 Promote a culture of continuous improvement within the organization.

6.5.4 To review the 8D C.A.R provided and approve / reject where appropriate.

6.5.5 To identify systemic learning from the 8D CAR for proliferation.

7.0 Procedures :

7.1 Customer fault feedback External and Internal (refer to Section I for detail procedures).
Section 1A : Voice of Customer (VOC)
Section 1B : Voice of Process (VOP)

7.2 Non Conformance Report (refer to Section II for detail procedures).
A : Non-conformity found due to non-compliance to procedure / admin issues / line audit failures.
B : Lot rejection at QA Gates
C : Quality feedback from Internal customer- eg. Final Test to Assembly, Metal finishing to Assembly, Assembly to Metal finishing, Assembly to Die store/ Back grind, etc..

7.3 Supplier corrective action request (refer to section III for detail procedures).

7.4 Preventive Action
The procedure for preventive action shall include:

7.4.1 Actions to eliminate causes of potential non-conformities in order to prevent their occurrence, shall be derived from appropriate sources of information such as processes and work operations which effect product quality, concessions, audit result, quality records, service reports and customer complaints. Preventive actions shall be appropriate to the effects of the potential problem.

7.4.2 Initiation at preventive action and application of controls to ensure that it is effective.

7.4.3 Preventive actions shall include the followings, but not limited to:

a) FMEA

b) Poka Yoke (Error-proofing)

c) Proliferation/ Lesson Learned /Standardization

7.4.4 Actions carried out shall be made known to management.

7.5 Risk Analysis

7.5.1 a) Carsem conducts the risk assessment per 8D procedure to ensure potentially affected lots are identified and traced back as part of the 8D process requirements.

b) To ensure risk scope is complete, QA engineer should work with 8D leader, case owner, production, engineering team and IT team to ensure the lot or raw material or others traceability data is complete.

c) 8D leader & QA engineer should highlight to QA Manager / QA Director if there is any HIGH risk lots that potentially result in material scrap or charge from customer.

7.5.2 Lot recapturing procedures are incorporated in MRB affecting non-conforming lot having critical defect.

7.5.3 Furthermore, "Lesson Learned" will be initiated to gain learnings from;

Product recalls related to field returns from customers triggered by customer complaint VOC-SCAR.

b) Product audits failures trigger line down issues during internal product audit

c) Whole lot scrap due to RMA triggered by customer complaint VOC- SCAR

d) Customer audit major findings (inter-sites).

7.5.4 Documented information of 8D, MRB and lessons learnt will be retained as evidence of risk analysis (risk assessment).

7.6 Customer Special Requirement (CSR)

7.6.1 Customer special requirements as per in Appendix to be referred to ensure compliance to the requirement.

8.0 Document / Records :

8.1 All NCR , MRB , SCAR , VOC & VOP shall be retained for the period as defined in CRQA000G02

8.2 Records to be retained :

i) 8D report (all relevant Attachments to the 8D) till closure.

ii) M.R.B (if any).

iii) Supplier corrective action request closure.

9.0 Material / Equipments :

N/A

10.0 Safety Precautions :

N/A

11.0 Maintenance / Schedule :

N/A

No	Attachment	Title
	Section I	
1	Attachment 1A	8D Format (Sample)
2	Attachment 1B	8D Guideline
3	Attachment 1C	Containment Lot Management Flow
4	Attachment 1D	Containment Actions Results Recording Format
5	Attachment 2	Customer Fault Feedback Management Flow
6	Attachment 3	Internal Fault Feedback Management Flow
7	Attachment 4	8D Administration Flow
8	Attachment 5	C.A.R. Verification Closure Flow
9	Attachment 6A	Customer Special Requirement on C.A.R. Cycletime
10	Attachment 6C	Customer Special Requirement on C.A.R. Cycletime
11	Attachment 6D	Customer Special Requirement on C.A.R. Cycletime-Renesas
12	Attachment 6E	Customer Special Requirement on C.A.R. Cycletime
13	Attachment 6F	Customer Special Requirement on C.A.R. Cycletime
14	Attachment 6G	Customer Special Requirement for Infineon
15	Attachment 6H	Customer Special Requirement on C.A.R. Cycletime-Melexis
16	Attachment 6I	Customer Special Requirement on C.A.R. Cycletime-Vitesco
17	Attachment 6J	Customer Special Requirement on C.A.R. Cycletime-ICS/IDT
18	Attachment 6K	Customer Special Requirement on C.A.R. Cycletime-Littlefuse
19	Attachment 6L	Customer Special Requirement on C.A.R. Cycletime-Macom
20	Attachment 6M	Customer Special Requirement on C.A.R. Cycletime-LEM
21	**Attachment 6N**	**Customer Special Requirement on C.A.R. Cycletime-JOU**
22	Attachment 7	List of Customer with Their Own 8D Format & 8D Cycle Time Matrix
23	Attachment 8	8D Closure Tracking
24	Attachment 11	Guideline for process mapping and commonality analysis
25	Attachment 12	Guideline on VOC/VOP categorization for Internal Reporting

No	Attachment	Title
	Section II	
1	Attachment 1	Non-Conformance Report (NCR)
2	Attachment 2	Flow Chart
3	Attachment 3	Red Flag at Final Test
4	Attachment 4	Red Flag Definition
5	Attachment 5	NCR Tracking for QA Inspection
6	Attachment 6	NCR Tracking for Test IQA
7	Attachment 7	NCR Tracking for Audit
8	Attachment 8	Customer Special Requirement for Avago (All Devices)
9	Attachment 9	Test NCR flow for ADI customer

No	Attachment	Title
	Section III	
1	Attachment 1	Material Non-Conformance, Corrective Action Request
2	Attachment 2	8D Corrective Action Report
3	Attachment 3	8-Discipline Evaluation Checklist

No	Appendix	Title
1	Appendix 1A	Customer Special requirement for ST
2	Appendix 1B	Customer Special requirement for ST
3	Appendix 1C	Customer Special requirement for ST
4	Appendix 1D	Customer Special requirement for ST
5	Appendix 1E	Customer Special requirement for ST
6	Appendix 1F	Customer Special requirement for ST
7	Appendix 1G	Customer Special requirement for ST
8	Appendix 2	Customer Special Requirement for IFX
9	Appendix 3A	Customer Special Requirement for Robert Bosch
10	Appendix 3B	Customer Special Requirement for Robert Bosch
11	Appendix 3B-1	Customer Special Requirement for Robert Bosch
12	Appendix 3C	Customer Special Requirement for Robert Bosch
13	Appendix 4	Customer Special Requirement for ALLEGRO
14	Appendix 5	Customer Special Requirement for TI
15	Appendix 6	Customer Special Requirement for TI

SECTION 1

SECTION 1A:

1.0 Title :
Customer fault feedback – Voice of Customer.

2.0 Purpose
The purpose of this documented procedure is to detail how the customer fault feedback is addressed.

3.0 Scope
Applicable to all customers fault feedback.

4.0 Applicable form :
- 4.1 8D Corrective Action Report (Attachment 1A).
- 4.2 Containment Action Results Recording Format (Attachment 1D)
- 4.3 8D Closure Tracking Format (Attachment 8)

5.0 Procedure :
8D Corrective Action Report (Refer to Attachment 1A for Quality incidence)
The 8D Corrective action report shall be used for responding to all customer fault feedback, unless customer specified.

5.1 Refer to C.A.R. Flow (Attachment 4)

5.2 The 8D Corrective action report format shall be used for responding to customer on quality incidences or customer fault feedback, unless customer specified format is required. Refer to Attachment 7 for the list of customers with type of 8D format to use and Attachment 1B for 8D guideline.

5.3 Customer specified 8D format
- 5.3.1 For customer's specific requirement on 8D format, refer to QA public folder to extract customer specific 8D format.

Note: The 8D corrective action form (Customer format & Carsem format) shall be living document and will be maintained by the C.A.R Coordinator .The template of the 8D form can be obtained from the QA Public Folder.

5.4 Containment action response requirements
- 5.4.1 A containment action to customer fault feedback (assembly/test related) is required to be sent within 24 hours (1 working day) to customers, electrical verification within 24 hours (1 working day). A 3D (till step 3 of the 8D) will be sent to the customer as containment action response.
- 5.4.2 Containment actions shall be lifted only upon implementation and proof of corrective actions effectiveness validation.
- 5.4.3 Containment action records to be captured in 8D report and retained until 8D closure. Please refer to attached Attachment 1D, (Containment action results recording- as reference guideline) and to be used as and when needed. Containment action management flow as per Attachment 1C.
- 5.4.4 Containment action on increasing QA sample size through "Sampling Instruction" shall be updated in MES system (if required) as per below guideline:
  - 5.4.4.1 Include the "Start Time" and "End Time" of the containment actions.

5.4.4.2 The "End Time" should be the target date of the implementation of corrective actions effectiveness validation.
Example: To increase sample from 125 to 200 effective from 31-Mar-2023 until corrective actions effectiveness is validated.

5.5 Final 8D report

5.5.1 The corrective action shall be determined by cross-functional team, comprising the QA Engineer, Process Engineering, Production personnel or any other group related to the complaint nature.

5.5.2 QA Engineer required following up with customer on sample status. Upon sample receipt, the 8D will be responded within 7 working days. Customer specified response time supersedes this requirement. (See special requirement Attachment 6 series). For those case confirmed by customer that sample will not be provided, the 8D response time will start base on customer confirmation date. For those case sample not required, the response time will start from date of complaint receipt.

5.5.3 After the actions had been agreed upon to be adequate and appropriate, it shall be reviewed and approved by the Dept and QA Manager.

5.5.4 The 8D C.A.R shall be valid only when it carries the QA Manager's signature or his/her deputy in his/her absence.

5.5.5 The 8D Team Leader / QA Engineer shall send the 8D response to the customers copying all the parties involved in the corrective action implementation.

5.5.6 All response to customer fault feedback shall be responded within 7 working days from date of receipt of the complaint or after verification of defect samples, (if any). Customer specified response time supersedes this requirement.
(See special requirement Attachment 6 series). In the event, the recipient requires additional time to complete the 8D response, customer is to be notified for approval.

5.5.7 8D response cycle time calculation will be stopped upon 8D submission to customer. Customer specified 8D response cycle time calculation method will supersedes this requirement.

5.5.8 When the customer has accepted the 8D C.A.R, the corrective action champion is responsible to follow-up on the corrective action implementations and effectiveness. QA Engineer is responsible to update customer if the 8D actions requires any recommit of the completion date.

5.5.9 If the 8D response is not accepted by the customer, the 8D C.A.R will be re-opened again and the 8D team leader will have to revisit the items highlighted by the customer. If the 8D is revised with changes on root cause or to committed actions or additions of new actions, QA Manager approval is mandatory.

5.5.10 Customer has to be notified if any changes made upon submission of the 8D report to customers on timely basis by QA Engineer Every 8D report revision to be captured in the 8D report.QA Engineer will verify and update the status of the 8D. Hence determine the 8D Closure (See Attachment 8 for the 8D-closure tracking).

5.5.11 The QA Engineer shall lift the increase sampling size setting in system (if require), before close the 8D. Upon verification for closure as of target date; if all actions are found to be in place and effectively carried out the 8D C.A.R shall then be defined as 'closed'. Any C.A.R found not effective or actions not carried out shall be defined as 'Open' or 'Pending'.

5.5.12 Any C.A.R that had been closed earlier on by corrective action champion but found not effective or not implemented by the QA Engineer upon verification, shall be 'reopened' for enforcement of actions. Any changes on the root cause or committed action or additions of new actions, customer to be notified

5.5.13 The action closure verification will be done after expiry of the corrective action

target date or in any circumstances to be within 30 working days from action target date.

5.5.14 Customers will be updated on the closure status by the QA Engineer on a timely manner.

5.5.15 The 8D closure tracking shall be updated from time to time to reflect the verification status by the CAR Coordinator.

5.5.16 C.A.R Coordinator shall provide weekly report to management on customer dissatisfaction trend.

5.6 Escalation of 8D C.A.R

5.6.1 Any 8D 's found not responded, not effective or 8D unable to 'close' timely, QA Engineer is responsible to escalate to Department Manager and HOD of the action leader with copies to Quality Manager and Director prior to the expiry of the action target date. The action leader is responsible to acknowledge to the escalation notice within 1 working day and react accordingly from date of receipt.

5.6.2 Weekly reporting also works as escalation platform.

Note :

1. 8D Team Leader / QA Engineer must ensure to copy C.A.R Coordinator for all customer complaint communication.
2. 8D team leader / QA Engineer must ensure if there is requirement for proliferation, actions to be proliferated to be closed and validated prior to proliferation request (Validation results to be attached in 8D reports as confirmation of effectiveness of the provided action).
3. 8D team leader / QA Engineer must clearly identify engineer in-charge and site/department/process/product line to be proliferated to.
4. Process engineer must update TECN (temporary engineering change) within 2 working days after corrective action submitted if corrective action involved engineering change. 8D actions will be considered close upon approval of ECN/PFMEA. Evidence of the approved document to be attached as action closure evidence by action leaders.

SECTION 1B

1.0 Title
Internal Fault Feedback (VOP – Voice of process)

2.0 Purpose
The purpose of this documented procedure is to detail how issues detected internally are addressed.

3.0 Scope
Applicable to all internal fault feedback

4.0 Applicable Form :

4.1 8D Corrective Action Report (Attachment 1A)
4.2 Containment Action Results Recording Format (Attachment 1D)
4.3 8D Closure Tracking Format (Attachment 8)

5.0 Procedure

5.1 Sample of 8D Corrective Action Report (Refer to Attachment 1A) for Quality incidence
5.2 The 8D Corrective action report shall be used for responding to all internal fault feedback, unless customer specified. Refer to Attachment 7 for the list of customers with their own 8D format.
5.3 For customer's specific requirement on 8D format, refer to QA public folder to extract customer specific 8D format.
5.4 Containment action response requirements
5.4.1 A containment action required to be sent out within 24 hours (1 working day) if there is request from customers. A 3D (till step 3 of the 8D) will be sent to the customer as containment action response.
5.4.2 Containment actions shall be lifted only upon implementation and proof of corrective actions effectiveness validation.
5.4.3 Containment action records to be captured in 8D report and retained until 8D closure. Please refer to attached Attachment 1D, (Containment action results recording- as reference guideline) and to be used as and when needed. Containment action management flow as per Attachment 1C.
5.4.4 Containment action on increasing QA sample size through "Sampling Instruction" shall be updated in MES system (if required) as per below guideline:
5.4.4.1 Include the "Start Time" and "End Time" of the containment actions.
5.4.4.2 The "End Time" should be the target date of the implementation of corrective actions effectiveness validation.
Example: To increase sample from 125 to 200 effective from 31-Mar-2023 until corrective actions effectiveness is validated

5.5 Final 8D Report
5.5.1 After the actions had been agreed upon to be adequate and appropriate, it shall be reviewed and approved by the Dept and QA Manager.

5.5.2 QA Engineer shall send the 8D response to the customers copying all the parties involved in the corrective action implementation.
All response on internal fault feedback shall be responded within 7 working days from date the issue detected internally. Customer specified response time supersedes these requirements. (See special requirement Attachment 6 series). In the event, the recipient requires additional time to complete the 8D response, customer is to be notified for approval.
When the customer has accepted the 8D C.A.R, the corrective action champion will champion the follow-up on the corrective action implementations and effectiveness.
If corrective action is taken on one product/process, the 8D-implementation review must be done on other similar processes/products.
Appointed quality assurance personnel will verify and update the status of the 8D. Hence determine the 8D Closure (See Attachment 8 for the 8D closure tracking).

5.5.3 The corrective action shall be determined by cross-functional team, comprising the QA Engineer, Process Engineering, Production personnel or any other group related to the complaint nature.

5.5.4 After the actions had been agreed upon to be adequate and appropriate, it shall be reviewed and approved by the Dept and QA Manager.

5.5.5 The 8D C.A.R shall be valid only when it carries the QA Manager's signature or his/her deputy in his/her absence.

5.5.6 8D Team Leader / QA Engineer shall send the 8D response to the customers copying all the parties involved in the corrective action implementation.

5.5.7 When the customer has accepted the 8D C.A.R, the corrective action champion is responsible to follow-up on the corrective action implementations and effectiveness.

5.5.8 If the 8D response is not accepted by the customer, the 8D C.A.R will be re-opened again and the 8D team leader will have to revisit the items highlighted by the customer. If the 8D is revised with changes on root cause or to committed actions or additions of new actions, QA Manager's approval is mandatory.

5.5.9 When the customer has accepted the 8D C.A.R, the corrective action champion will champion the follow-up on the corrective action implementations and effectiveness.

5.5.10 Customer has to be notified if any changes made upon submission of the 8D report to customers on timely basis. Every 8D report revision to be captured in the 8D report.

5.5.11 QA Engineer or the will verify and update the status of the 8D. Hence determine the 8D Closure (See Attachment 8 for the 8D closure tracking).

5.5.11.1 The QA Engineer shall lift the increase sampling size setting in system (if require), before close the 8D. Upon verification for closure as of target date; if all actions are found to be in place and effectively carried out, the 8D C.A.R shall then be defined as 'closed'. Any C.A.R found not effective or actions not carried out shall be defined as 'Open' or 'Pending'.

5.5.11.2 Any C.A.R that had been closed earlier on by corrective action champion but found not effective or not implemented by the QA Engineer upon verification, shall be 'reopened' for enforcement of actions. Any changes on the root cause or committed action or additions of new actions, customer to be notified.

5.5.11.3 The action closure verification will be done after expiry of the corrective action target date or in any circumstances to be within 30 working days from action target date.

5.5.11.4 Customers will be updated on the closure status by the QA Engineer on a timely manner (only upon customer's request).

5.5.12 The 8D closure tracking shall be updated from time to time to reflect the verification status by the C.A.R Coordinator.

5.5.12.1 The 8D actions shall be track in QA CAT system. (Applicable to Carsem Suzhou only)
Procedure:
QA auditor put 8D actions into QA CAT system;
QA auditor verified the 8D action status;
QA auditor insert the closure evidence in QA CAT system before target date;
The QA CAT system will auto email reminder the owner before 3 days of action target date;
The QA CAT system will auto email escalation to department manager when delay 1 day.

5.5.12.2 The 8D actions tracking system procedure for ST customer requirement refer to Appendix 1C. (Applicable to Carsem Malaysia only).

5.5.13 C.A.R Coordinator shall provide weekly report to management on internal dissatisfaction trend.

5.6 Escalation of 8D C.A.R

5.6.1 Any 8D found not responded, not effective or 8D unable to 'close' upon verification; after the expiry of the action target date, shall be escalated to the corrective action champion and the Dept Manager with copies to the Operation Manager for appropriate actions. The corrective action champion is responsible to acknowledge to the escalation notice within 1 working day and react accordingly from date of receipt.

Note :

1. 8D team leader / QA Engineer must ensure to copy C.A.R Coordinator for all Internal complaint communication.
2. 8D team leader / QA Engineer must ensure if there is requirement for proliferation, actions to be proliferated to be closed and validated prior to proliferation request (Validation results to be attached in 8D report as confirmation of effectiveness of the provided action).
3. 8D team leader / QA Engineer must clearly identify engineer in-charge and site/department/process/product line to be proliferated to.
4. Process engineer must update TECN (temporary engineering change) within 2 working days after corrective action submitted if corrective action involved engineering change. 8D actions will be considered close upon approval of ECN/PFMEA. Evidence of the approved document to be attached as action closure evidence by action leaders.
5. Disclaimer footnote on the 8D format (Form # CRQA100M01-1) need to be formatted using font type Verdana and the font size must be using size 4.

SECTION 1 : Attachment 1A

8D FORMAT (SAMPLE)

CARSEM
8D Corrective Action Report

Carsem Confidential

Type of complaint	Carsem Reference #
Customer Feedback Date	Customer / Supplier Reference #
3D Report Submission Date	Department
Sample Receipt Date	Package type / Part description :
8D Report Submission Date :	Part Date Code / :
Point of Detection :	Device / Part # :
End Customer :	Reject Qty :
Re-Submission (if applicable) :	Failure Rate :

Problem Statement : Customer – Package – Defect

D1) Team Formation	Name	Job Title	Area of Responsibility
A) Team Leader:			
B) Team Member:			

D2) Problem / parts defect description.

A)Background :

B) Customer FA analysis :

FA / Sample Photo :

Photo 1 : (Description)	Photo 2 : (Description)	Photo 3 : (Description)

CARSEM
8D Corrective Action Report
Carsem Confidential

C) Carsem FA analysis :

Photo 4 : (Description)	Photo 5 : (Description)	Photo 6 : (Description)

D) Problem Description Summary:

D3) Interim containment / short term corrective action.

A) Pipeline Analysis

1) Check the existing (potential affected) material in the pipeline for similar defect. Notify customer when there is a need to return potential affected materials.

Date	Location	Sorted qty	Rej qty	% of Rej	Disposition
	Assembly:				
	Final Test: *Note:The checking must include partial or residue of affected lot.*				
	FG Store:				
	In Transit				
	Shipped to Customer				

B) Other short term corrective action *(only to be lifted upon implementation of permanent corrective action effectiveness validation – for tighten monitoring and gating items only).*

Actions	Responsible	Date

1) Alert QA, production and engineering on the feedback from customer.
Status:

2) Display OPL (one point lesson) and reject photo in production floor to increase people awareness for similar defect.
Status:

3) Check the existing process / machine to ensure no similar defect is generated.
Status:

4) Increase SPC/Visual monitoring frequency from _____ to _____.
Status:

5) Increase QA gate inspection sampling size from _____ to _____.
Status:

6) Retrieve the assembly history of the affected lot.
Status:

7) Inform customer on the potential affected material.
Status:

8) Inspection effectiveness validation
(min. 1 sublot or lot size min 3000ea (whichever tighter) for re-100% inspection to determine effectiveness of containment action method installed)

Results:
Production (100%):
QA (100%):

Note:
1. Records for the containment actions must be retained until 8D closure *(Any documents used for collection of containment action data).*

Containment Action Record (Y/N)	*Retention (Y/N)*

2. Specify and validate the inspection method against the reported failure mode

CARSEM
8D Corrective Action Report

Carsem Confidential

D4) Define and verify root causes *(Refer to defect mode of FMEA for FTA & fish bone analysis)*:

A) Fault Tree Analysis : Brain Storm Date :

Problem Defect: XXXXXXXXX ← state the defect

	Primary Contributing Factors			Secondary Contributing Factors		Analysis
1)						
2)						
3)						

B) Why-Why Analysis on the possible root cause :

Why 1:
Because

Why 2 :
Because

Why 3:
Because

Why 4:
Because

Why 5:
Because

C) Escape Point:
(Remark : Use Why Why Tools to analyse the escape point, if applicable)

D) *Summary of root cause analysis:*

Note:
FMEA is mandatory to be reviewed

D5) Chosen Corrective Actions/Preventive Actions (Systemic/POKA YOKE Solutions):

Root Cause	Action Taken	Remark
	Stated the action Resp : Date : Status :	
	Stated the action Resp : Date : Status :	

CARSEM
8D Corrective Action

D6) n of ss:

(Criteria: 100% inspection of 3 sublots or 15k whichever tighter)

To monitor ppm for coming weeks upon implementation of permanent corrective action.

Resp:

Status:

D7) Preventive Actions:

A) Documentation:

FMEA : (Mandatory to be reviewed)

Spec # :

Spec title :

Item to be review/update:

Resp:

Date:

Control Plan : (Based on the need)

Spec # :

Spec title :

Item to be review/update:

Resp:

Date:

Others Relevant Specification:

Spec # :

Spec title :

Item to be review/update :

Resp:

Date:

B) Proliferation of Lessson Learnt:

To proliferate the selected corrective actions from D5 & D7 with other Carsem product line. ☐ Yes ☐ No

Item to be proliferated :

Resp: Site/Department/Process/Product Line:

Date:

D8) Reviewed and approved by :

A) Quality incident		B) Environmental, safety & health incident	
Dept. Manager:		Dept. Manager:	
Or			
Date :		Date :	
Eng. Manager:		Safety officer:	
And		And	
Quality Manager:		EMR / safety chairman:	
Date :		Date :	

Note: 1) Mandatory signatories:

2024) Quality incident

Signature of the Quality Assurance Manager or his representative during his absence is mandatory.

B) Environmental, safety & health incident

Signature of EMR (Environmental Management Representative) / Safety Chairman or his representative during his absence is mandatory.

2) Documents to support the closure for an 8D must be attached and verified during verification of closure.

3) All dates shall be written in the format (yyyy-mmm-dd)

"The contents of this Report are strictly confidential and is addressed to you for your sole benefit and may not without our prior written consent (a) be relied upon by any other person; (b) be disseminated, copied or disclosed (except to persons within your company's employ, who in the ordinary course of business, have access to your papers and records); and if it is so disseminated, copied or disclosed in such manner, it is without any liability on our part and the recipients will make no further disclosure; (c) be filed with any person or quoted or referenced to in any private or public document; or (d) be used for any purpose other than those expressly stated in this Report. While reasonable care has been taken in the preparation of this Report, this Report and all information, assumptions, and recommendations herein, if any, are published, given, made, or expressed without any responsibility whatsoever on our part, whether arising by way of negligence, breach of contract, breach of statutory duty or otherwise. No warranty or representation of accuracy or reliability in respect of this Report is given by us or our directors, employees, servants, agents, consultants, successors in title and assigns. This disclaimer shall apply to liability to any person whatsoever, irrespective of how such liability arises, whether by use of this Report by that person or you or any other person or otherwise."

SECTION 1 : Attachment 1B
8D GUIDELINE

Note:
This 8D guideline may be used by the corrective action leader, as a guide to check the effectiveness of the 8D.

The following must be followed:

1. Type of complaint : External
   Complaint category is fixed to 2 which are External or Internal
   a) External – Refers to feedback from Carsem's customers
   b) Internal - Refers to Feedback from internal source eg: test, EOL, etc.

2. Customer Feedback Date : YYYY MMM DD
   State what was the date the customer informed Carsem regarding the problem.
   Comply to yyyy-mmm-dd.

3. 3D Report Submission Date : YYYY MMM DD
   a) Indicate the date, 3D was sent out.
   b) This must be 24hrs from the date of the official request for 8D.
   c) DO NOT SEND 3D WITHOUT CAR #.

4. Sample Receipt Date : YYYY MMM DD
   State the date of sample receipt. CAR coordinator must be informed of this date.

5. 8D Report Submission Date : YYYY MMM DD (Rev A)
   Indicate date the 8D was formally sent out to the customer. Indicate the Rev as "A".
   Subsequent revisions to be capture as Rev "B", "C", etc until final acceptation by customer.

6. Point of Detection: e.g. : Customer Final Test
   a) State the area where the defect was first detected.
   b) Indicate one of the following :-
      i) Customer Incoming
      ii) Customer Final Test/Subcon
      iii) End Customer
      v) Field Failure
      vi) Carsem Final Test/Assembly

7. End Customer: e.g.: Samsung
   State who is the end customer of the direct customer of Carsem

8. Re-Submission (if applicable): YYYY DDD MM (Rev B)
   If the 8D is revised due to changes on root cause or to committed action or additions of new actions, need to indicate the new re-submission date with the Rev as "B".

9. Carsem Reference # :
   Obtain the CAR number from CAR coordinator.

10. Customer / Supplier Reference # : e.g. CAR1234
    a) The CAR reference number will be provided by the customer.
    b) In the event there is no number provided, indicate "NIL" in the column.

11. Department : To indicate business BU (e.g. SOIC)
    State department that is responsible for the complaint

12. Package Type / Part Description : eg. 24 Ld MQFP (10 X 10)
    a) State the lead count and package effected.
    b) Additional information such as body size shall also be stated in (xxx).
    c) State the raw material description

13. Part Date Code : e.g. 0634
    a) Indicate date code of the effected part.
    b) Find out from the customer if not provided.
    c) If the date code cannot be determined, state the closest date code to the device that may have been built or process date

14. Device / Part # : e.g. ATT7461ARMC-A12Z
    a) Indicate the full device type, include the extension.
    b) For raw material state the part # or commercial ID #

15. Reject Qty : e.g. 2
    State how many units are affected

16. Failure Rate : e.g. 0.5%
    Calculate the failure rate based on the reject qty vs lot size from the customer information. If no lot size is given, then this will be "N/A".

17. Problem Statement : e.g. MLX – 8 SOIC – Chip Package
    State customer abbreviation, package type and defect / problem seen.

D1

8D team Formation

- A Cross Functional Team (CFT) is made up of members from many disciplines – with product and process knowledge.
- Indicate the job designation of each member & work function.
-

D2

Describe the problem description needs to be objectively defined in a way that captures all of the key information. 4 key parts to be covered but not limited:

i) Background
- Indicate the information provided by the customer and if needed provide more information

ii) Customer FA analysis
- State the customer FA findings, if none received based on comments indicate "no customer FA report provided".

iii) Carsem FA analysis
- State Carserm FA finding if there is sample return by customer.

iv) Problem summary
- Using the background information, customer FA analysis and Carsem FA analysis, describe the problem in summary use the data and information to quantify and clarify the problem into a statement.

D2(b)
Supporting Concept /Methodology
Helpful concept/methodology in defining problem description (for use, as and when needed):
- 5W/2H – During this analysis, it can be helpful to ask what, who, where, when, how, and how much in order to develop a clear description.
- Is / Is Not Analysis.
- Other analysis tools (check sheets, Pareto diagram, Control Charts).
- Process mapping – to map out the process as much as possible

Refer to below for process mapping guideline
a) For lot info, retrieve at E-list
b) For SPC data, retrieve at E-list, eportfolio (S site),SPC Data Query Site. (M site)
c) For CEMS downtime, retrieve at Info Hub.
d) Other info, eg: magazine Id.
e) Refer to flow in Section1:Attachment 11 for process mapping and commonality analysis.

Detection and occurrences – review detection and occurrences of the reported failure/customer feedback in existing relevant PFMEA.

D3(a)
Containment /short term corrective action (line purge data)
- line purge data's must be indicated in this section
- date when the line purge was done
- location (Including Assembly, Final Test, FG Store, In Transit & Shipped to Customer)
- sorted out quantity
- rejected quantity
- Disposition

D3 (b)
Other containment /short term corrective action
- what other action are taken beside section 3a, to protect customer(s) from receiving defective parts.
- temporary corrective actions to be validated to monitor effectiveness.
- temporary corrective action(s) to be in-place and is typically removed after the permanent corrective action(s) implemented and validated in term of the effectiveness.
- records for the containment actions must be retained until 8D closure.
- responsible person name who is responsible to carry out the task – date implementation.

D4
Define and verify root cause
- the brain storming date must be indicated
- state the problem defect
- need to recognize occurrence, escapee and system root cause.
- identify potential root cause (using why-why,etc where appropriate)
- The Cause Effect diagram is used to brainstorm potential analysis)
- utilize problem solving tools during brainstorming (FTA , Fish Bone Diagram factor that may cause the failure.
- FTA is used to validate factor listed identified at cause effect diagram. This is to identify possible factor that may cause the failure. Failure mechanism shall be repeat and demonstrated.
- The 5 Why is used to identify the final root cause for possible/potential factor identified during FTA.
-

D5
Permanent corrective action.
- root cause to be identified
- permanent corrective action taken to be indicated (utilize mistake proofing methodology).
- responsible person
- date of implementation of activity

*Note: The maximum allowable time frame for implementation of actions in a 8D report are:-
a) For procedure change – within 30 working days from date 8D was generated, otherwise per agreement with customer.
b) For system change – within 90 working days from date of 8D was generated, otherwise per agreement with customer.

D6

Verification of permanent corrective action's effectiveness

- data's of monitoring after implementation of actions to be indicated
- to provide internal / external data for effectiveness of corrective action
- validation results to be recorded in 8D report.

D7

Preventive actions

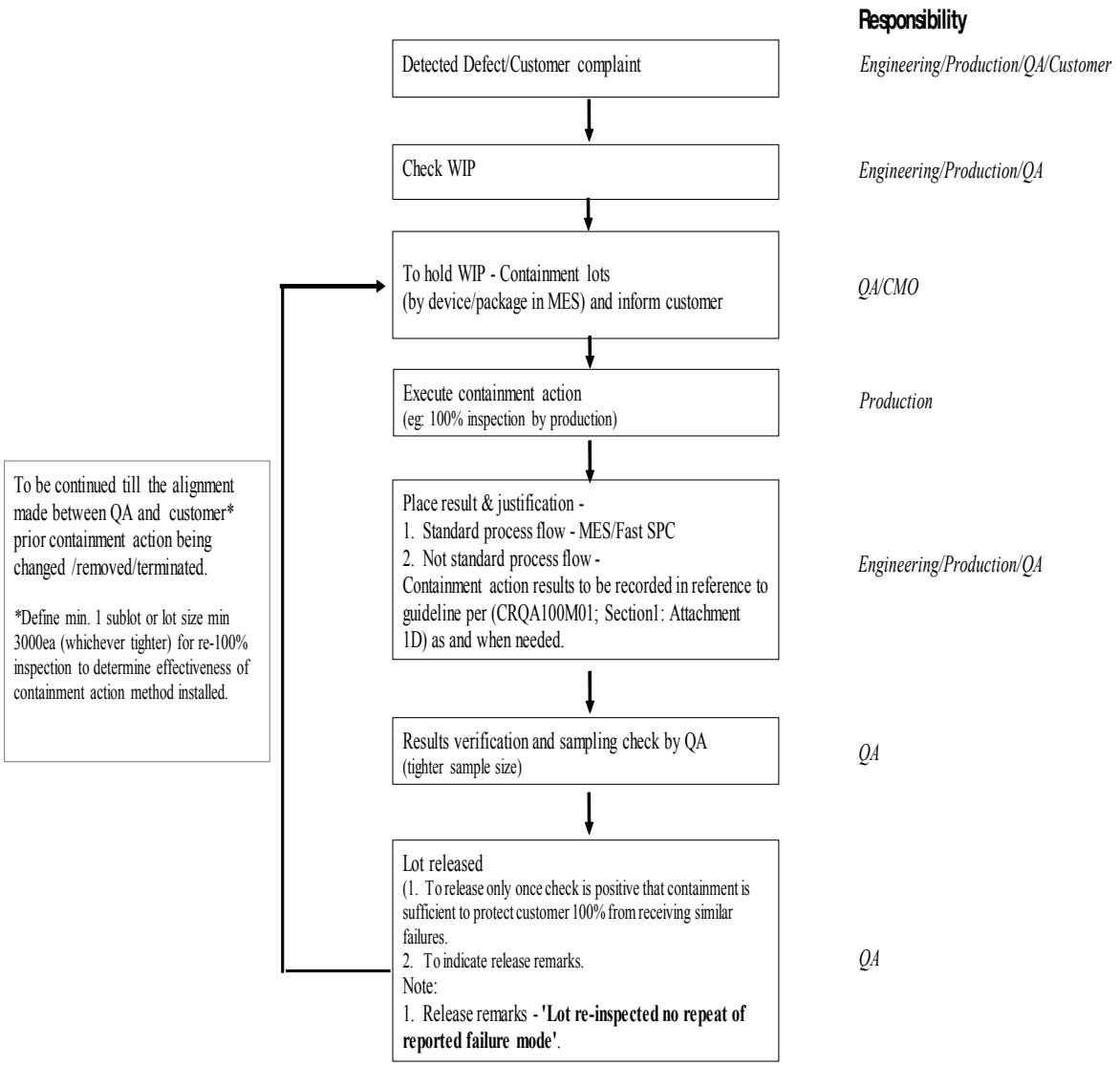
- Long-term risk weighted and systemic actions taken to prevent a problem from occurring, based on an understanding of the product or process. Actions to be considered but not limited to:-
- Specification / control plan review
- FMEA review and updates
  - to list which FMEA and specification was reviewed.
  - FMEA is mandatory to be filled
- Poka Yoke (Error-proofing) solutions
- Proliferation/ Lesson Learned /Standardization
  - To indicate the need for proliferation such as responsible personnel, which item to proliferate, processes, area, department involved, inter-site and other related information.

D8

Reviewed and approved

- The 8D must be signed off by the Dept Manager followed by the QA Manager approval. Approval of the Quality Assurance manager or his representative during his absence is mandatory. (If the 8d is revised with changes on root cause or to committed actions or additions of new actions, Dept Manager & QA Manager approval is Mandatory.)

SECTION 1 : Attachment 1C
Containment Lot Management Flow



SECTION 1: Attachment 1D

Containment Action Results Recording Format

Containment Action Results Recording Format (to be used as guideline and as and when needed)

Containment Action Results Recording Format

Customer: _____

PIC (QA/Production/Engineering): _____

8D No.: _____

Defect Mode: _____

Start Date : _____

Device/Package: _____

Desciption: _____

Containment Action :

Machine to use: _____

Magnification: _____

No.	Date	Mother lot #	Sublot #	Quantity	Machine ID	Operator ID	Lot Hold Status	Result	Remark

Note:

1. This checklist has to be retained until the 8D closure.
2. It's responsibility of supervisor (Operation) to retain the checklist until 8D closure.

Form # : CRQA100M01-20 Rev : A Retention Required

SECTION 1 : Attachment 2

CUSTOMER FAULT FEEDBACK MANAGEMENT FLOW

1. Customer complaint – Voice of Customer
   1.1 Every SCAR # from customer shall be registered as VOC regardless if it is with the same root cause.
   1.2 Consolidation of VOCs only allowed with customer approval/waiver/request. Evidence of the customer requirement to be shared with CAR coordinator.

2. Recipient (any person to whom the C.A.R. was directed to by the customer.)

   2.1 Customer service personnel /person the C.A.R is attention to, must copy the C.A.R Co-Ordinator.
   2.2 Verify Carsem (M) Sdn. Bhd. M-Site or Carsem (M) Sdn. Bhd. S-Site issue.
   2.3 Acknowledge customer within 24 hr. (QA Engineer).
   2.4 QA Engineer shall request for defect samples status once a week if it is required.
   ACKNOWLEDGMENT of receipt of sample is to be done within 24 hours
   (1 working day) upon sample received, with the expected target date of the final 8D response.
   2.5 QA Engineer shall request for complaint log in and provide needed information to CAR coordinators upon receipt of the complaint.

3. Log into C.A.R tracking system (C.A.R Coordinator).

   3.1 System auto assign V.O.C number.
   3.2 System auto distribute Customer Feedback notification to Team leader, QA Engineer & Dept Manager.

4.0 Identify 8D Team Leaders (Co-ordination by QA Engineer).

   4.1 QA Engineer to identify corrective 8D team leader to champion the issue or relevant Dept Manager within 24 hr. upon receipt of complaint not requiring samples or when samples are received. In any case, whereby, the 8D team leader can only be identified after the results of the FA report, then, it shall be identified with 24 hours upon receipt of the FA results.

5.0 Containment action

   5.1 QA Engineer to response to customer with containment action (3D response) within 24 hrs (1 working day) or as specified by customer is mandatory. It is mandatory the QA Engineer to copy C.A.R Co-Ordinator on 3D response.
   5.1.1 Lots shall be released only upon completion of item 5.2 and 5.3.
   5.2 QA Engineer to validate effectiveness of containment actions.
   5.3 Evidence include the following:
   i)do's and don't list (if required)
   ii) line purge data's (if required)
   iii) line briefing (if required)
   iv) Lot History or mapping (if required)
   v) Records of containment action inspection machine and method used.
   vi) Records of containment action data (to be captured in 8D report and retained till the 8D closure).
   vii) Containment action effectiveness validation results (Define min. 1 sublot or lot size min 3000ea (whichever tighter) for re-100% inspection to determine effectiveness of containment action method installed).
   Note : Verify the accuracy of compiled information vs original database. Eg E-list.

5.4 If more samples are required, leader or QA Engineer to request from customers copying the C.A.R Co-Ordinator. Besides that, the leader or the QA Engineer required to request customer more information on the feedback for problem solving if necessary.

6.0 8D Team leader (final 8D)

6.1 To form team

6.2 To determine root causes/verification of root causes

6.3 Response to customer with 8D by 8D Team Leader / QA Engineer. 8D required to be responded to customer within 7 working Days upon sample receipt. Customer specified response time supersedes this requirement. Refer to clause 5.5.2 for more details.

6.4 Copy the 8D to relevant personnel involved and to the C.A.R Co-Ordinator (Responsible: QA Engineer / 8D team leader)

6.5 Upon D4 outcome- decision/ agreement shall be obtained either to drop / proceed to complete 8D. To be captured in summary of root cause (Applicable for 8D report in Carsem format only).
In the event if the failure is not Carsem related, customer waiver/approval to be obtained for invalidation of the reported failure.
QE to update CAR Coordinator to record and update changes in weekly VOC/VOP tracker. Changes to be captured as 'Invalidation'.

7.0 Update C.A.R Tracking program (responsible: C.A.R Co-Ordinator)

7.1 C.A.R Co-Ordinator to update C.A.R tracking program upon 8D response.

7.2 To key in last target date for verification recall.

8.0 Verification of 8D (Corrective action champion)

8.1 Verification of 8D action per committed date.

8.2 To ensure effectiveness of implementation of 8D corrective actions. Sample size of min. 3 sublots / wafers or lot size of 15K (whichever tighter) with no repeat of same defect mode during 100% inspection to be used. (Responsible: Corrective action champion)

9.0 Verification of closure (By QA Engineer)

9.1 Action leaders must have all evidence pertaining to the particular C.A.R. i.e.: briefing evidence, QA buy-off data, copy of approved TECN / ECN if a spec is updated, etc. before he / she can close the said C.A.R.

9.2 QA Engineer will do the same before closing the 8D.

9.3 For ST 8D action validation, refer to guideline stated in attachment 8C. The validation to be performed by personnel from other group serves as 3rd pair of eyes.

10.0 8D Closure

10.1 Note :

1) QA Engineer must provide final updated 8D report with closure evidences to the C.A.R Co-Ordinator so that the C.A.R tracking program and 8D database can be updated.
2) Action leaders must ensure that all evidence are kept and should be able to produce when and as required by customer or the QA Engineer for verification for closure purpose.
3) 8D response which needs in depth investigation before an 8D can be established, which do not meet the specific in house or customers response cycle time requirements, must be communicated to the customer for approval and all this communication must be copied to the C.A.R Co-Ordinator for filing purpose.
4) 8D consider closed upon closure of all committed actions and enclosed with evidence of implementation and verification. All documents need to be approved prior to 8D closure.

SECTION 1 : Attachment 3

INTERNAL FAULT FEEDBACK MANAGEMENT FLOW

1.0 Internal Complaint / Feedback – Voice of Process

1.1 Internal feedback to customer whereby :

- There is material being scrap due to error make
- Internal feedback where customer required for 8D

2.0 Recipient (QA Engineer or the 8D Team Leader who feedback the issue to Customer)

2.1 The respective QA Engineer or the 8D Team Leader have to feedback the 8D request to CAR-Coordinator

3.0 CAR Coordinator will log into CAR Tracking System

3.1 System auto assign V.O.P number.

3.2 System auto distribute Internal Feedback notification to Team leader, QA Engineer & Dept Manager.

4.0 Identify 8D Team Leaders

4.1 8D team leader will be identify by the QA engineer or respective Dept Manager.

5.0 Containment Actions

5.1 Assigned QA Engineer to response to customer with containment action (3D Response) within 24 hrs (1 working day) or as specified by customer if there is a request from the customer. It is mandatory for the QA Engineer to copy C.A.R Coordinator on 3D response.

5.1.1 Lots shall be released only upon completion of item 5.2 and 5.3.

5.2 QA Engineer to validate effectiveness of containment actions.

Evidence includes the following:

i)do's and don'ts list (if required)
ii) line purge data's (if required)
iii) line briefing (if required)
iv) Lot History or mapping (if required)
v) Records of containment action inspection machine and method used.
vi) Records of containment action data (to be captured in 8D report and retained till the 8D closure)
vii) Containment action effectiveness validation results (Define min. 1 sublot or lot size min 3000ea (whichever tighter) for re-100% inspection to determine effectiveness of containment action method installed).

Note: Verify the accuracy compiled information vs original database. Eg E-list.

6.0 8D leader (final 8D)

6.1 The Corrective action should be responded within 7 working days from date of receipt of the internal feedback request. Customer specified response time supersedes this requirement. (See special requirement Attachment 6 series). In the event, the recipient requires additional time to complete the 8D response, customer is to be notified for approval.

7.0 Update C.A.R Tracking System (responsible : C.A.R Coordinator Ordinator)

7.1 C.A.R Coordinator to update C.A.R tracking upon 8D response.

8.0 Verification of 8D (Corrective action champion)

8.1 Verification of 8D action per committed date.

8.2 To ensure effectiveness of implementation of 8D corrective actions. Sample size of min. 3 sublots / wafers or lot size of 15K (whichever tighter) with no repeat of same defect mode during 100% inspection to be used. (Responsible: Corrective action champion)

9.0 Verification of closure (By QA Engineer)

9.1 Corrective action champion must have all evidence pertaining to the particular C.A.R. i.e.: briefing evidence, QA buy-off data, copy of TECN / ECN if a spec is updated, etc. before he / she can close the said 8D

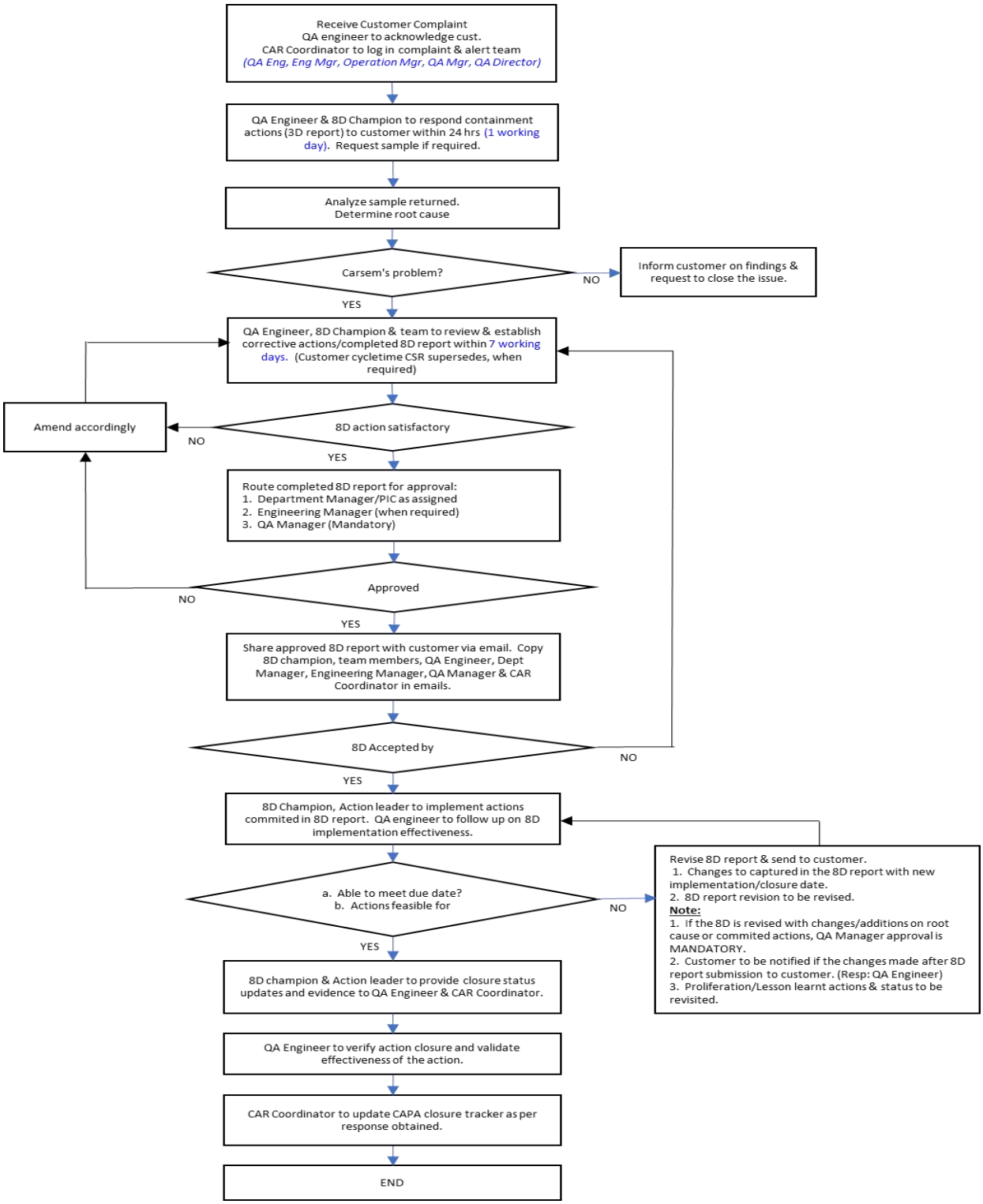
9.2 QA Engineer will do the same before closing the 8D.

Note :

1) QA Engineer must provide final updated 8D report with closure evidences to the C.A.R Coordinator so that the C.A.R tracking program and 8D database can be updated.
2) Action leaders must ensure that all evidences are kept and should be able to produce when and as required by customer or the QA Engineer for verification for closure purpose.
3) 8D response which needs in depth investigation before an 8D can be established, which do not meet the specific in house or customers response cycle time requirements, must be communicated to the customer for approval and all this communication must be copied to the C.A.R Coordinator for filing purposes.
4) 8D consider closed upon closure of all committed actions and enclosed with evidence of implementation and verification. All documents need to be approved prior to 8D closure.
5) Refer to Appendix 1 series for ST requirements.

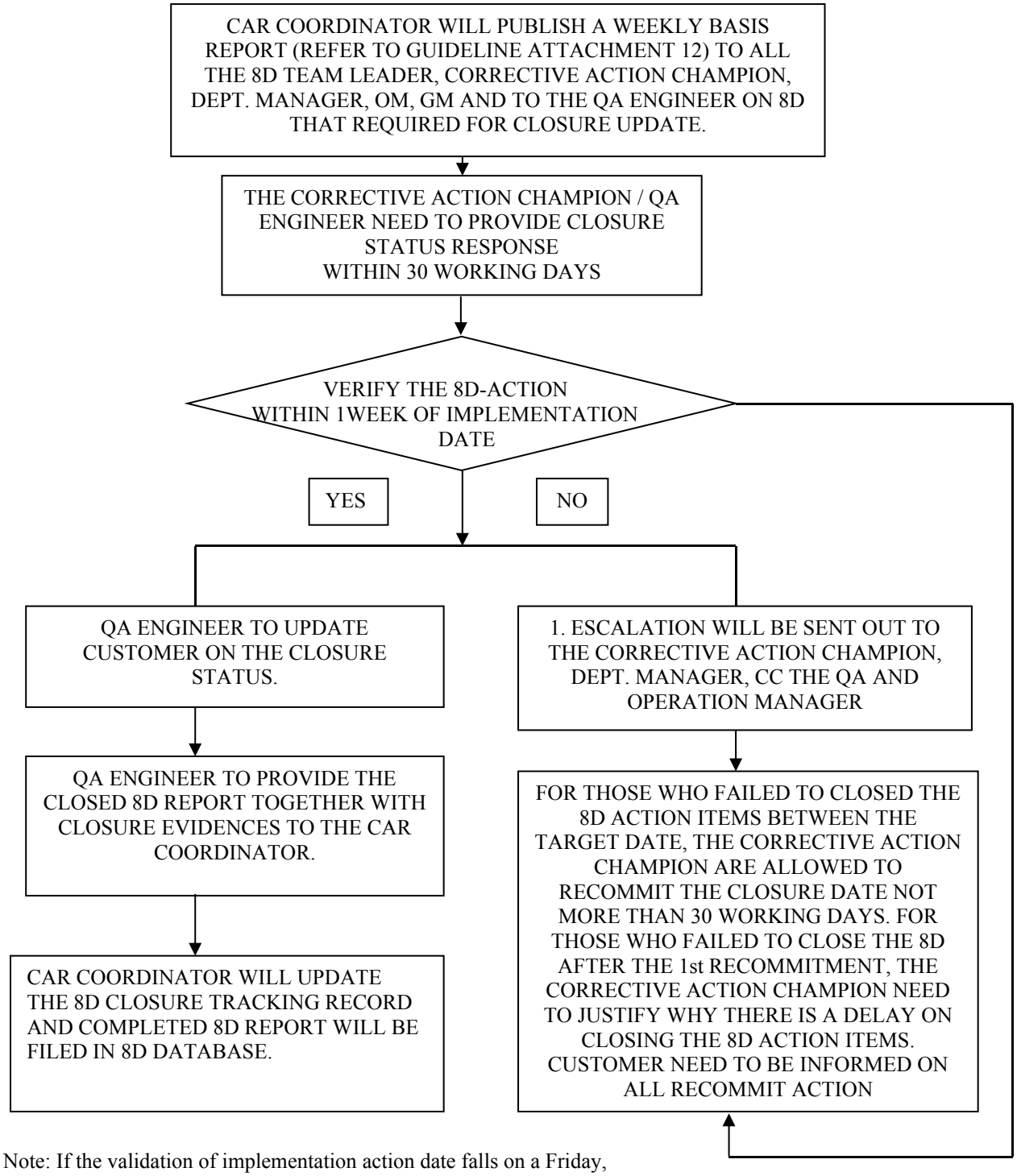
SECTION 1 : Attachment 4

8D ADMINISTRATION FLOW (RECEIPT OF CUSTOMER COMPLAINT OR INTERNAL FEEDBACK)



SECTION 1 :Attachment 5

C.A.R. VERIFICATION CLOSURE FLOW



Note: If the validation of implementation action date falls on a Friday, then the activity will be followed through the following week.

SECTION 1 : Attachment 6A

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME

4&5 WORKING / CALENDAR DAYS

CUSTOMERS	SOURCES OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.	2ND 8D REQ. IF INITIAL 8D IS REJECTED
AOS	Email from customer on 2011-Jan-17	C.A.R	24 HRS	4 WORKING DAYS	----
ST MICROELECTRONIC (ALL SITES)	7085545	C.A.R	24 HRS.	5 WORKING DAYS OR 7 CALENDAR DAYS WHICEVER COME FIRST	----

NOTE :

1. FOR CUSTOMER ST IN D3: LOT BLOCKING, RESCREEN RESULT OF WIP LOT OR ON SITE OF ST / ST CUSTOMER, EFFECTIVE AND TIMELY FIRE WALL.
2. IMPLEMENTATION OF ACTIONS IN D3 IS TO BE WITHIN 48 HOURS OF 8D REQUEST.

SECTION 1 : Attachment 6C

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME

5,7 & 10 WORKING / CALENDAR DAYS

CUSTOMERS	SOURCES OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.	2ND 8D REQ. IF INITIAL 8D IS REJECTED
ON SEMI	N/A	C.A.T.S.	24 HR	7 CALENDAR DAYS	_____
ANALOG (ALL SITES)	ADI0339	SCAR	24 HRS	10 CALENDAR DAYS	REFER NOTE 3
SEMTECH	SFBN-4JPT6M	C.A.R	24 HR.	10 WORKING DAYS FOR NON-AUTOMOTIVE CUSTOMERS 5 WORKING DAYS FOR AUTOMOTIVE	----
TEXAS INSTRUMENT	QSS 020-005	S.C.A.R.	24 HR.	10 CALENDAR DAYS FOR NON AUTOMOTIVE CUSTOMERS 5 CALENDAR DAYS FOR AUTOMOTIVE & HUGE IMPORTANT CUSTOMERS	_____
MAXIM	10-0082	S.C.A.R	24 HR	10 CALENDAR DAYS	_____
MICREL	N\A	C.A.R	24 HR.	7 WORKING DAY	_____
WOLFSON	N\A	C.A.R	24 HRS	7 CALENDAR DAYS	----
DIALOG	CMO1023	C.A.R	24HRS	7 WORKING DAYS	
UPI	Audit requirement	C.A.R	24HRS	7 WORKING DAYS	----

Note

1. WHEN RESPONDING TO MAXIM & ON SEMI CUSTOMER COMPLAINT, CARSEM SHALL USE THE RESPECTIVE 8D FORMS WHICH CAN BE OBTAINED FROM THE QA PUBLIC FOLDER.
2. 8D TEAM LEADER \ QA ENGINEER REQUIRED TO ATTACHED CARSEM (M) 8D APPROVAL FORMAT (TOGETHER WITH ADI , MAXIM AND ONSEMI 8D RESPONSE WHEN SEEK FOR APPROVAL FROM DEPARTMENT & QA MANAGER.
3. ALL MAXIM 8D REPORTS MUST BE SUBMITTED WITH SUBCONASSEMBLY 8-DISCIPLINE CHECKLIST ONLY FOR SCAR.
4. WHEN RESPONDING TO MAXIM CUSTOMER COMPLAINT, 3X5 ANALYSIS WILL BE USED IN ANALYSIS PROCESS OF EVERY MAXIM SCAR
5. WHEN RESPONDING TO TI CUSTOMER COMPLAINT, CARSEM MUST INCLUDE 3X5Y IN THE 8D REPORT TO ADDRESS THE OCCURRENCE, ESCAPEE & SYSTEMETIC ROOT CAUSE.
6. CARSEM REQUIRED TO ATTACHED TI SCORING FORMAT TOGETHER WITH 8D REPORT WHEN RESPONDING TO TI END CUSTOMER COMPLAINT.
7. TI WILL TRIGGER CARSEM WHEN THERE IS A END CUSTOMER COMPLAINT FROM AUTO & HUGE
8. CARSEM REQUIRED TO ATTACHED ONSEMI EVALUATION SHEET TOGETHER WITH 8D REPORT WHICH SCORE MORE THAN 80 WHE N RESPONDING TO ONSEMI CUSTOMER COMPLAINT
9. FOR ADI, CORRECTIVE ACTION REQUEST (CAR) CYCLE TIME (CALENDAR DAYS): RECEIPT OF SAMPLES BY SUB-CONTRACTOR UNTIL RECEIPT OF ACCEPTABLE 8D REPORT. IF FIRST SUBMISSION IS UNACCEPTABLE, TIMER WILL RESTART ON NOTIFICATION AND STOP ON RECEIPT UNTIL AN ACCEPTABLE REPORT IS SUBMITTED.
10. WHEN RESPONDING TO ADI CUSTOMER COMPLAINT, CARSEM SHALL USE ADI 8D FORMAT WHICH CAN BE OBTAINED FROM THE QA PUBLIC FOLDER.

CUSTOMER'S SPECIAL REQUIREMENTS

ELMOS

10 WORKING DAYS

CUSTOMER	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.
ELMOS	20VT0001E.01	C.A.R	24HRS	7 WORKING DAYS (PRIORITY) 10 WORKING DAYS (STANDARD)

*Note :

Failure analysis milestones (8D approach)	Standard	Priority
AoR : Acknowledgement of receipt	1 day*	1 day
D3 : Implement containment actions	3 days	1 day
D4 : Define root cause (s)	3 days	2 days
D5 : Define permanent corrective actions	7 days	5 days
D6 : Implementation of corrective actions	7 days	5 days
D7 : Define actions to prevent recurrence	10 days	7 days
D8 : Implementation of preventive actions	Per plan	Per plan
*"Days" indicate cumulative working days after problem submission by customer.		

SECTION 1 : Attachment 6D

CUSTOMER'S SPECIAL REQUIREMENTS ON C.A.R. CYCLETIME

RENESAS

D5 – 7 CALENDAR DAYS – AUTOMOTIVE
8D – 10 WORKING DAYS – COMMERCIAL

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.	2ND 8D REQ. IF INITIAL 8D IS REJECTED
RENESAS	SUB CORRECTIVE ACTION – CORRECTIVE ACTION SYSTEMS	C.A.R	24 HRS	FOR PRIORITY RETURNS (WHEN COMMUNICATED TO THE FACTORY) AN 8D COMPLETED THROUGH D5 WILL BE SUBMITTED NO LATER THAN 7 CALENDAR DAYS AND THE COMPLETED 8D WITHIN 14 CALENDAR DAYS (UNLESS COMMUNICATION WITH THE CUSTOMER HAS MODIFIED THE CYCLE TIME)	----
					----

1. WHEN RESPONDING TO RENESAS CUSTOMER COMPLAINT, CARSEM SHALL USE THE RESPECTIVE 8D FORMAT WHICH CAN BE OBTAINED FROM THE PUBLIC FOLDER.
2. 8D TEAM LEADER / QA ENGINEER REQUIRED TO ATTACHED CARSEM (M) 8D APPROVAL FORM TOGETHER WITH RENESAS 8D REPORT WHEN SEEKING FOR APPROVAL FROM DEPARTMENT & QA MANAGER.
3. RENESAS AUTOMOTIVE RATED CUSTOMER CORRECTIVE ACTION REQUIREMENTS
   - 3.1 CUSTOMERS ARE NOTIFIED WITHIN 24 HOURS AFTER PARTS ARE RECEIVED FOR ANALYSIS THAT PARTS ARE RECEIVED
   - 3.2 VERIFICATION AND CONTAINMENT IS TARGETED FOR 48 HOURS AFTER RECEIPT.
   - 3.3 THE 8D MUST BE SUBMITTED TO THE CUSTOMER NO LATER THAN 10 WORKING DAYS AFTER RECEIVING THE PRODUCT. IF A COMPLETED 8D REPORT CANNOT BE DELIVERED WITHIN THIS TIME FRAME, THE CUSTOMER WILL BE SUPPLIED A DETAILED INTERIM 8D REPORT.THE INTERIM 8D REPORT WILL IDENTIFY A DEADLINE FOR SUBMISSION OF THE COMPLETED 8D REPORT (OR FOR THE NEXT INTERIM 8D REPORT). THE TIME BETWEEN TWO INTERIM 8D REPORTS MAY NOT EXCEED 10 WORKING DAYS.
   - 3.4 FOR PRIORITY RETURNS (WHEN COMMUNICATED TO THE FACTORY) AN 8D COMPLETED THROUGH D5 WILL BE SUBMITTED NO LATER THAN 7 CALENDAR DAYS AND THE COMPLETED 8D WITHIN 14 CALENDAR DAYS (UNLESS COMMUNICATION WITH THE CUSTOMER HAS MODIFIED THE CYCLE TIME)
   - 3.5 THE 8D REPORT WILL BE CLOSED WHEN ALL CORRECTIVE ACTION ITEMS HAVE BEEN COMPLETED AND VERIFIED.

THE FOLLOWING INFORMATION IS MANDATORY BEFORE CARSEM LAUNCH AN INVESTIGATION :

1. CUSTOMER LOT #
2. QUANTITY AFFECTED
3. CONDITION OF FAILURE

ALL SECTIONS OF THE 8D REPORT WILL NEED TO BE FULFILLED. FOR CASES IN WHICH A LONGER TIME IS REQUIRED TO FIRM UP THE ROOT CAUSE AND CORRECTIVE ACTIONS, THE PRELIMINARY 8D REPORT THAT IS TO BE SUBMITTED TO **RENESAS** WITHIN 48 HOURS WILL NEED TO INCLUDE:-

A) D4 – POSSIBLE ROOT CAUSE(S) AND OUTLINE OF SIMULATIONS/EVALUATIONS TO BE DONE TO FIRM UP THE R/CAUSE
B) D5 TO D7 – TENTATIVE CORRECTIVE ACTIONS TO ADDRESS THE PLAUSIBLE ROOT CAUSE WHICH CARSEM CAN INSERT A NOTATION TO INDICATE THAT IT WILL BE FIRMED UP UPON CONFIRMATION OF ROOT CAUSE

6. ALL ACTIONS IN THE 8D REPORT NEED TO BE COMMITTED AND CLOSED WITHIN 35 CALENDAR DAYS FROM THE END OF THE 8D C/TIME

SECTION 1 : Attachment 6E

CUSTOMER'S SPECIAL REQUIREMENTS ON C.A.R. CYCLETIME

10 WORKING / CALENDAR DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.	2ND 8D REQ. IF INITIAL 8D IS REJECTED
SIPEX / EXAR	280-7043	F.A. (C.A.R)	24 HRS	10 WORKING DAYS (RESPOND < THAN OR WITHIN 5 DAYS – 2 PTS, RESPOND WITHIN 5- 10 DAYS – 1 PT, MORE THAN 10 DAYS – 0 PT)	-----
ALLEGRO	PRC-0001658	C.A.R.	24 HRS	10 CALENDAR DAYS FOR FIELD RETURN 20 CALENDAR DAYS FOR INTERNAL ALLEGRO FAILURE	----
SKYWORKS	SQ02-0020	C.A.R.	24 HRS	10 WORKING DAYS	----
PII	QA-055	C.A.R	24 HRS	10 WORKING DAYS	---
MIP	N\A	C.A.R	24 HRS	10 CALENDAR DAYS	
ATMEL	CPM-2013	C.A.R	24HRS	10 WORKING DAYS	
NSC	MPS-3-100	C.A.R	24HRS	10 WORKING DAYS	
AUS	ASSY – 266	C.A.R	24HRS	4D 2 WORKING DAYS 8D – 10 CALENDAR DAYS	
CYPRESS	22-00005	C.A.R	24 HRS	10 CALENDAR DAYS FOR NON AUTOMOTIVE CUSTOMERS 7 CALENDAR DAYS FOR AUTOMOTIVE CUSTOMERS	

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME

14 WORKING/ CALENDAR DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.	2ND 8D REQ. IF INITIAL 8D IS REJECTED
FREESCALE	CONTI GOA REQUIREMENTS	C.A.R	24 HRS	STANDARD REQ – ROOT CAUSE ANALYSIS – 3 CALENDAR DAYS FOR IMPLEMENTATION OF CA AND PA AS PER AN AGREED UPON PLAN – 14 CALENDAR DAYS HIGH PRIORITY ISSUES (CUSTOMER LINE DOWN , RELIABILITY AND SAFETY) 4D – 3 CALENDAR DAYS 6D – 7 CALENDAR DAYS 8D – AS PER AGREED UPON PLAN	-----
SMSC	PS – 331	SCAR	2 WORKING DAYS	14 WORKING DAYS	_____

CUSTOMER'S SPECIAL REQUIREMENTS INFINEON

D4+D5 10 WORKING DAYS

D6+D7+D8 13 WORKING DAYS

CUSTOMER	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.
IFN	SCOP 800-025	C.A.R.	max of 24 hours (D0-D3)	D4+D5 within 10 working days D6+D7+D8 within 13 working days otherwise specified per agreement

NOTE

1. WHEN RESPONDING TO IFX CUSTOMER COMPLAINT, CARSEM SHALL USE THE RELEVANT IFX 8D FORM WHICH CAN BE OBTAINED FROM THE PUBLIC FOLDER. ALL RESPONSES ON CUSTOMER COMPLAINT SHALL BE DOCUMENTED THROUGH FORMAL 8D REPORT BASED ON THE REQUIREMENT SPECIFIED AT SCOP 800-035 "8D REPORT HANDLING".

2. ALL IFX 8D FILE NAME MUST BE STANDARDIZE AS BELOW: CAR# _8D RESPONSE CATEGORY_(SUBCON-IFX CUSTOMER NAME_IFX PART NO_PACKAGE LEAD TYPE_ LOT CODE_DATE CODE_FAILURE MODE)_DATE OF REPORT.

3.8D TEAM LEADER \ QA ENGINEER \ EXECUTIVE REQUIRED TO ATTACHED CARSEM (M) 8D APPROVAL FORMAT TOGETHER WITH IFX 8D RESPONSE WHEN SEEKING FOR APPROVAL FROM DEPARTMENT & QA MANAGER.

4. WHEN THERE IS A NEED TO EXTEND THE TIMELINE FOR THE D5 REPORT FOR DEEPER ANALYSIS ON THE COMPLAINT, IFX NEEDS TO BE INFORMED AND PROVISIONAL D5 REPORT NEED TO BE SUBMITTED. SUBCONTRATOR/SUPPLIER SHALL PROVIDE D0 – D3 RESPONSE WITHIN 24 HOURS UPON RECEIVING OF FAILURE ANALYSIS REPORT.

5. D4 TO D5 RESPONSE SHALL BE PROVIDED WITHIN 10 WORKING DAYS AND D6 TO D8 WITHIN 13 WORKING DAYS OR OTHERWISE AS SPECIFIED PER AGREEMENT FROM THE TIME THE COMPLAINT NOTIFICATION WAS RECEIVED BY THE SUBCONTRACTOR/SUPPLIER. A PROVISIONAL D5 ACTION PLAN IS ACCEPTABLE UPON APPROVAL OF OSAT QM.

6. PROVISIONAL D5 RESPONSE SHALL BE PROVIDED IF THE FACTORY NEED ADDITIONAL TIME NECESSITATED BY DEEPER ANALYSIS REQUIRED.

7. TIME FRAME SHALL START WITH THE INITIAL NOTIFICATION OF A COMPLAINT OR PROBLEM BY IFX.

8. COMPLETE 8D REPORT SHALL BE SUBMITTED WITHIN 8 WORKING DAYS FROM THE COMPLAINT/SAMPLE RECEIPT DATE. IN CASE OF 8D REPORT REQUIRING MORE TIME FOR CLOSURE SHALL HAVE APPROVAL FROM OSAT QM.

9. ALL 8D REPORT GENERATED SHALL CONTAIN PIPELINE ANALYSIS, IS-IS NOT ANALYSIS, FAULT-TREE ANALYSIS INFORMATION. 3X5 WHY ANALYSIS IS REQUIRED FOR ALL COMPLAINTS ADDRESSING ROOT CAUSE, SYSTEMS FAILURE AND ESCAPE POINT.

10. CLOSURE OF 8D SHALL HAVE EFFECTIVENESS VALIDATION PLAN FOR CORRECTIVE ACTIONS FORM THE CORRECTED PRODUCTION LOT AT PLANNED INTERVAL OF 1-MONTH, 3-MONTHS, 6-MONTHS & 9-MONTHS AS REQUIRED BY SCOP 800-34 "8D PCA FOLLOW-UP AUDIT".

11. D6 – D8 FINAL ACTION RESPONSE MUST BE REVIEWED AND APPROVED FOR SUBMISSION BY THE SUB-CONTRACTOR QA DIRECTOR / MANAGER. IT IS RECOMMENDED TO ASSESS THE REPORT USING BOSCH-TOYOTA 8D EVALUATION TEMPLATE. THIS TEMPLATE IS REQUIRED FOR ALL BOSCH CASES.

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME
MELEXIS

10 CALENDAR DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.	2ND 8D REQ. IF INITIAL 8D IS REJECTED
MELEXIS	2014-06_Q	QUALITY ASSURANCE AGREEMENT REV.008	24HRS	10 CALENDAR DAYS-8D response time – time between receiving of failed devices from Melexis and submission of final report.	----

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME
VITESCO

7/14 CALENDAR DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.
VITESCO	A2C00052917AAA_VT, REV C	QUALITY PROCESS REQUIREMENTS (QPR) FOR COMPLAINT MANAGEMENT FOR PURCHASED COMPONENTS	STANDARD & PRIORITY/HIGH D2+ D3 – WITHIN 24HRS	STANDARD D4 – WITHIN 3 CALENDAR DAYS D5 – WITHIN 14 CALENDAR DAYS D6+D7+D8 – PER AGREED PLAN PRIORITY/HIGH D4 – WITHIN 24 HOURS D5 – WITHIN 7 CALENDAR DAYS D6+D7+D8 – PER AGREED PLAN

NOTE

1. WHEN RESPONDING TO VITESCO CUSTOMER COMPLAINT, CARSEM SHALL USE THE RELEVANT VITESCO 8D FORM WHICH CAN BE OBTAINED FROM THE PUBLIC FOLDER. ALL RESPONSES ON CUSTOMER COMPLAINT SHALL BE DOCUMENTED THROUGH FORMAL 8D REPORT BASED ON THE REQUIREMENT SPECIFIED AT A2C00052917AAA_VT, REV C

2.THE FOLLOWING 8D EVALUATION SHEET SHALL BE USED BY THE SUPPLIER FOR A PROPER 8D REPORT. THE VITESCO REQUIREMENT FOR A 8D EVALUATION IS TO REACH A MINIMUM SCORE OF 90%. CHANGES CAN BE DONE AND RELEASED BY THE PROCESS OWNER/SYSTEM OWNER AT VITESCO. IN ADDITION VITESCO CAN PROVIDE THE SUPPLIER THE VITESCO 8D CHECKLIST. THE 8D CHECKLIST HELPS THE SUPPLIER TO PERFORM THE 8D EVALUATION.

3.TIME FRAME SHALL START WITH THE INITIAL NOTIFICATION OF A COMPLAINT OR PROBLEM BY VITESCO.

4. 1ST STEP OF ROOT CAUSE ANALYSIS SHALL BE BASED ON 1ST LEVEL INFORMATION SHARE BY VITESCO SUCH AS PHOTO OR CUSTOMER FA OF THE DEFECT.

5. TO NOTIFY VITESCO IF UNABLE TO MEET GIVEN TIMELINE.

CUSTOMER'S SPECIAL REQUIREMENTS

ON C.A.R. CYCLETIME

ICS/IDT

+D4 ½ CALENDAR DAYS

D8 10/14 CALENDAR DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.	2ND 8D REQ. IF INITIAL 8D IS REJECTED
ICS / IDT	QCP-1000	QUALITY MANAGEMENT AGREEMENT REV.12	4D WITHIN 48HRS-2 CALENDAR DAYS	NON AUTOMOTIVE – 14 Calendar Days AUTOMOTIVE 10 Calendar Days	----

SECTION 1 : Attachment 6K

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME
LITTLEFUSE

7/12 CALENDAR DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.
LITTLEFUSE	SBU-OUT25-QA-007	GENERAL QUALITY AGREEMENT FOR SBU OUTSOURCED SUPPLIERS	STANDARD D2+ D3+D4 – WITHIN 4 CALENDAR DAYS & PRIORITY/HIGH WITHIN 24HRS	STANDARD D5+D6+D7 – WITHIN 12 CALENDAR DAYS PRIORITY/HIGH D5+D6+D7 – WITHIN 7 CALENDAR DAYS

NOTE

1.WHEN RESPONDING TO LITTLEFUSE CUSTOMER COMPLAINT, CARSEM SHALL USE LITTLEFUSE 8D FORM WHICH CAN BE OBTAINED FROM THE PUBLIC FOLDER. ALL RESPONSES ON CUSTOMER COMPLAINT SHALL BE DOCUMENTED THROUGH FORMAL 8D REPORT BASED ON THE REQUIREMENT SPECIFIED AT SBU-OUT25-QA-007.

2.ANY REJECT SAMPLES MUST BE KEPT AT LEAST 12 MONTHS AFTER CLOSED THE CASE.

3. TIME FRAME SHALL START WITH THE INITIAL NOTIFICATION OF A COMPLAINT OR PROBLEM BY LITTLEFUSE.
4. TO NOTIFY VITESCO IF UNABLE TO MEET GIVEN TIMELINE.

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME
MACOM

1/5/15/23 WORKING DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.
MACOM	ECP-QM-1000	SUPPLIER QUALITY MANUAL	IMMEDIATE CONTAINMENT ACTION WITHIN 1 WORKING DAY INTERIM CONTAINMENT ACTION STANDARD D2+D3–WITHIN 5 WORKING DAYS & PRIORITY/HIGH D2+ D3–WITHIN 3 WORKING DAYS Note: Immediate containment action - immediate action to stop a problem from spreading further. Interim containment action - temporary measure implemented to manage the problem's impact until a permanent solution put in place.	STANDARD D4 WITHIN 15 WORKING DAYS D8 WITHIN 23 WORKING DAYS PRIORITY/HIGH D4 WITHIN 5 WORKING DAYS D8 WITHIN 9 WORKING DAYS

NOTE

1. WHEN RESPONDING TO MACOM COMPLAINT, CARSEM SHALL USE CARSEM 8D FORM WHICH CAN BE OBTAINED FROM THE PUBLIC FOLDER. ALL RESPONSES ON CUSTOMER COMPLAINT SHALL BE DOCUMENTED THROUGH FORMAL 8D REPORT BASED ON THE REQUIREMENT SPECIFIED AT ECP-QM-1000.

2. ANY REJECT SAMPLES MUST BE KEPT AT LEAST 12 MONTHS AFTER CLOSED THE CASE.

3. TIME FRAME SHALL START WITH THE INITIAL NOTIFICATION OF A COMPLAINT OR PROBLEM BY MACOM
4. CARSEM TO ACKNOWLEDGE RECIPT OF ISSUE NOTIFICATION AND IMMEDIATE CONTAINMENT ACTIONS WITHIN 1 WORKING DAY AND INTERIM CONTAINMENT PLAN WITHIN 5 WORKING DAYS (3 WORKING DAYS FOR URGENT ISSUES).
5. CORRECTIVE AND PREVENTIVE ACTION PLAN IMPLEMENTED AND VERIFIED WITHIN MAXIMUM OF 90 WORKING DAYS.
6. TO NOTIFY MACOM IF UNABLE TO MEET GIVEN TIMELINE.

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME
LEM

10/20 WORKING DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.
LEM	CO.11.10.097.0	LEM SUPPLIERS GENERAL REQUIREMENTS MANUAL	D1-D3 WITHIN 48 HOURS	D5 WITHIN 10 WORKING DAYS D8 WITHIN 20 WORKING DAYS

NOTE

1. WHEN RESPONDING TO LEM COMPLAINT, CARSEM SHALL USE LEM 8D FORM WHICH CAN BE OBTAINED FROM THE PUBLIC FOLDER. ALL RESPONSES ON CUSTOMER COMPLAINT SHALL BE DOCUMENTED THROUGH FORMAL 8D REPORT BASED ON THE REQUIREMENT SPECIFIED AT CO.11.10.097.0.
2. ANY REJECT SAMPLES MUST BE KEPT TILL TO HAVE LEM'S APPROVAL FOR SCRAPPING.
3. TIME FRAME SHALL START WITH THE INITIAL NOTIFICATION OF A COMPLAINT OR PROBLEM BY LEM.
4. CARSEM TO ACKNOWLEDGE RECIPT OF ISSUE NOTIFICATION AND INITIATE CONTAINMENT ACTIONS WITHIN 1 WORKING DAY AND REPORT D3 WITHIN 48HOURS.
5. UPON SUBMISSION OF COMPLETE 8D REPORT (WHITHIN 20 WORKING DAYS), THE EFFICIENCY OF ACTIONS DEFINED IN D5 MUST BE CHECKED. LESSON LEARNED TO HAVE FORMALIZED IN THE PFMEA. ANY OTHER RELEVANT DOCUMENTS LINKED TO THE PROCESS/PRODUCT SHALL BE UPDATED. ACTIONS TO BE STANDDARDIZED TO SIMILAR PROCESSESS AND PRODUCTS.
6. TO NOTIFY LEM IF UNABLE TO MEET GIVEN TIMELINE AND CONDITIONS.

CUSTOMER'S SPECIAL REQUIREMENTS
ON C.A.R. CYCLETIME
JOU
5 CALENDAR DAYS

CUSTOMERS	SOURCE OF REQUIREMENT	TYPE OF REPORT ISSUED	CONTAINMENT ACTION RESPONSE REQ.	FINAL 8D RESPONSE REQ.
JOU	JW-Q-3-8.4-62	JW ASSEMBLY BASELINE FOR JWQ PRODUCT AT CARSEM 杰华特汽车级产品对嘉盛的封装要求	1.Follow 2485 rule to dispose customer complaint 对于客户抱怨响应遵守 2485 原则 2.First response: within 2 hours 首次回应在 2 小时以内 3.Second response: contamination action within 24 hours 紧急措施及二次回应在 24 小时以内 4.Third response: root cause and corrective action within 48 hours 原因分析分析、制定对策及三次回应在 48 小时以内 5.Fourth response: implement action and verify effectiveness with 5 days 对策实施、验证对策的有效性及四次回应在 5 天内	5 Calendar days.

SECTION 1: Attachment 7

CUSTOMER 8D CYCLE TIME & 8D REPORT FORMAT SUMMARY MATRIX

CUST	CAPA SPEC #	8D CYCLETIME	8D FORMAT
ADI/LTC	AD10339	10 Calendar Days	CUSTOMER
ALL	PRC-0001658	10 Calendar Days	CARSEM
ELM	20VT0001E	7 Working Days (Priority) 10 Working Days (Standard)	CARSEM
EXAR	280-7043	10 Working Days	CARSEM
ICS / IDT	QCP-1005	14 Calendar Days (Non Auto) 10 Calendar Days (Auto)	CUSTOMER
IFN	SCOP 800-025	13 Working Days	CUSTOMER
RENESAS	SUB-CORRECTIVE ACTION	10 Working Days	CUSTOMER
MAXIM	10-0082	10 Calendar Days	CUSTOMER (REF : 18-0100)
MCT	SPI-45528/QCI-40013	10 Working Days	CARSEM
MELEXIS	2014-06_Q	10 Calendar Days	CUSTOMER
ONS	N/A	7 Working Days	CUSTOMER
PII	QA-055	10 Working Days	CARSEM
SEMTECH	SFBN-4JPT6M	5 working Days (Auto) 10 working days (Non-Auto)	CARSEM
SITIME	QPR-18	7 working Days	CARSEM
ST	7085545 (Per Customer alignment email) email.pdf	5 Working Days or 7 Calender Days which ever comes first	CUSTOMER
RBG, BST	QAA_EN	2-14-60 Calendar Days	CARSEM format with special requirements
TIM	QSS 020-005	5 Calendar Days (Auto) 10 calendar days (Non-Auto)	CUSTOMER
UBQ	N/A	7 Working Days	CUSTOMER
VSP	N/A	7 Working Days	CUSTOMER
AUS	ASSY-266	10 Calendar Days	CARSEM
NXP	NX3-00014	7 working Days	CARSEM
PANASONIC	N/A	7 Working Days	CUSTOMER
LTC	N/A	7 Working Days	CARSEM
SKYWORKS	SQ02-0020	10 Working Days	CARSEM
CREE	N/A	7 Working Days	CARSEM
CSE	N/A	7 Working Days	CARSEM
VITESCO	A2C00052917AAA_VT, REV C	3-14 Calendar Days 24hrs-7 Calendar Days	CUSTOMER
EPC	N/A	14 Calendar Days	CARSEM
LITTLEFUSE	SBU-OUT25-QA-007	7 Calendar Days(Priority) 12 Calendar Days (Standard)	CUSTOMER
MACOM	ECP-QM-1000	9 Working Days (Priority) 23 Working Days (Standard)	CARSEM
LEM	CO.11.10.097.0	10 working days	CUSTOMER
JOU	**JW-Q-3-8.4-62**	**5 Calendar days (JWQ)**	**CARSEM**

NOTE:
3D Cycle time = 24 Hrs from date receipt of complaint. Applicable for all customers. 8D cycle time = based on above CSR, starts upon date receipt of complaint or date receipt of sample/rejects or upon Customer's request.
NOTE: customer dedicated 8d format will be available in *csmfile\m:\8d reports (Pic QA system) \8D format* (master list)

SECTION 1: Attachment 8
8D CLOSURE TRACKING

CARSEM - CAPA ACTION CLOSURE TRACKER

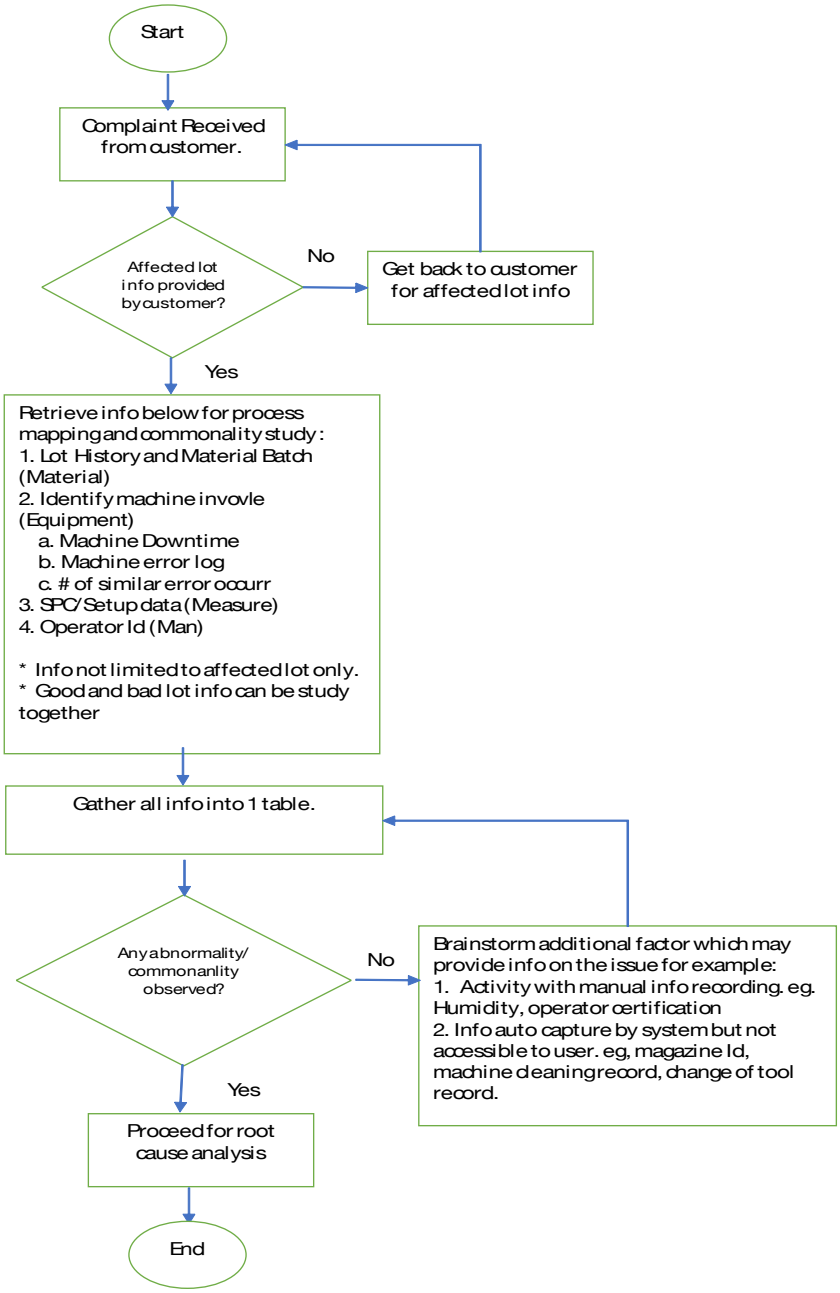
8D DETAILS										PROBLEM ACTION ITEMS DETAILS					8D Rev Alignment			CLOSURE STATUS				RECOMMITMENT		
MONTH	CAR CATEGORY	CUST	CAR#	DEPT / BU	8D RESPONDED DATE	8D CHAMPION	DATE CODE	PKG	QA ENGINEER / CQE	LOT #	DEVICE	FAILURE MODE	PROBLEM DESCRIPTION	ROOT CAUSE	CUST ACCEPTANCE DATE	8D REV (CAR COORVERSION)	8D REV (QE VERSION)	ACTION CLOSURE STATUS (DONE / IN PROGRESS)	IMPLEMENTATION DATE	8D STATUS (OPEN / CLOSED / PENDING)	EVIDENCES	1ST	2ND	REASON FOR 2ND RECOMMITMENT

FORM # CPQA100M01-15 REV : C RETENTION REQUIRED

Note:

1. Tracker format to have minimum requirement as in above table.

Attachment 11
Guideline for process mapping and commonality analysis



Attachment 12
Guidelines on VOC and VOP Categorizing for Internal Reporting

Definition of Major and Repeated

1. Major VOC – Any complaint received from customer due to –
   a. Field failure resulting to a product recall by end customer.
   b. Causing line disruption at customer, end user.
   Note: Only upon declaration from customer, complaints will be considered as major.

2. Major VOP– Internal fault feedback–
   a. Causing 100% of lot affected and lot scrapped.
   b. If affected all the recapture lots and/or require recall from customer.

3. Repeated VOP/VOC
   a. Customer complaint or internal detection will be considered repeated if the same failure mode and root cause repeated for lots with later date code after implementation of Corrective Actions (CA) and Preventive Actions (PA).

Responsibility for handling major/repeated VOC/VOP

2. The QA Engineers are responsible
   a. To ensure the guideline on handling the major VOC/VOP is being complied and facilitated.
   b. To determine the VOC/VOP category and liaise with respective team on major VOC/VOP's reaction plan and risk assessment.

3. Engineering Managers are responsible
   a. The respective area engineering manager to lead the overall assessment.

Guideline for handling major/repeated VOC/VOP

1. Major VOC/VOP
   a. Upon receiving a Major VOC/VOP QA Engineer to notify respective team including top management (QA Manager, Assembly OM or Test OM) within 24 hours upon confirmation of the feedback.
   b. 8D team leader shall be the Engineering Manager or as identified by the Operations Mgr.
   c. 8D report shall be reviewed by Management before sending to customer.
   d. Management must provide special attention for any support needed.

2. Repeated VOC/VOP
   a. VOC/VOP that shows a Root Cause of failure mode, repeating. Eg. Failure mode: Marking defect.
      i. Root Cause for the 1st VOC/VOP is – wrong marking diagram used.
      ii. Root Cause for the 2nd VOC/VOP is – marking rubber Pad worn off. The above is not a Repeated Root Cause. (The above example is an illustration of NON-REPEATED ROOT CAUSE FOR A FAILURE MODE)
   b. If the root cause for the marking failure mode was wrong mark diagram used (2nd time), then it is categorized as a REPEATED VOC/VOP.
   c. VOC/VOP with later date code after implementation of CAPA.

Reporting criteria

1. VOC
    a. Customer feedback confirmed it is Carsem process related issue.
    b. Customer feedback confirmed it is Carsem's supplier related issue.
    c. If there is any disruption/recall at customer site, the VOP will be classified as VOC.
    d. Customer request to upgrade from VOP with Carsem QA manager approval.

2. VOP
    a. Internal feedback to customer and there is material being scrap due to error made.
    b. Internal feedback where customer required 8D.
    c. Internal feedback where Carsem management decided to be in 8D.
    d. Customer request to downgrade from VOC.

3. CIP
    a. Not Carsem related issue and for continuous improvement.
    b. Carsem initiative on the continuous improvement.
    c. Customer request for continuous improvement action report.

Requirement for VOC and VOP Invalidation:

1. For VOC, in the event if the failure is not Carsem related, customer waiver/approval to be obtained for invalidation of the reported failure.

2. For VOP, invalidation only with respective area QA Manager's acknowledgement/approval.

3. QE to update CAR Coordinator to record and update changes in weekly VOC/VOP tracker. Changes to be captured as 'Invalidation'.

SECTION II

1.0 Title :

Non Conformance Report (in-house corrective action request).

2.0 Purpose :

To detail how the internal Non-Conformance Report (NCR) is used.

3.0 Scope :

Applies for the following areas:

A : Non-conformity found due to non-compliance to procedure / admin issues / line audit failures.
Non- conforming admin failures which lead to customer complain CAR / quality incidences will be define as RED FLAG at Final Test. (Refer to attachment 3 & 4)

B : Lot rejection at QA Gates

C : Quality feedback from Internal Customer- eg. Final Test to Assembly

4.0 Applicable forms:

4.1 Non Conformance report (NCR) (Attachment 1)
4.2 NCR tracking for QA Inspection (Attachment 5)
4.3 NCR tracking for IQA Internal Quality Alert (Attachment 6)
4.4 NCR tracking for audit failures . (Attachment 7)

5.0 Procedure:

5.1 Non Conformance Report (NCR)

The Non Conformance Report shall be issued to the appropriate personnel of the area by any employee when a discrepancy, non-conformance or any issue that may affect the company's image or Product Quality. . The format is divided into 8 elements (see attachment 1). Based on the NCR issuance, the appropriate contents must be completely filled. The use of hard copy or system NCR or equivalent is permitted.

NCR shall be used under this 3 categories & column to be fill up as below:

A) Non-conformity found due to non-compliance to procedure / admin issues / line audit failures. (Fill up ALL elements, except 6.)
B) Lot rejection at QA Gate – To reject lots that is non-conformance or not meeting to the QA Acceptance Specification, Customer Specifications / Drawings.
C) Quality feedback from Internal Customer – To feedback / alert any quality issues from internal customers, eg. Final test to assembly.(Fill up ALL elements, except 6.)
D) Specific customer requirement on Non Conformance Report (NCR) please refer to Section II attachment 8 and attachment 9 (ADI Test NCR Flow)

5.1.1 Non-Conformity found to non- compliance to procedure.

5.1.1.1 The NCR Closure Tracking. (N.C.R Co-ordinator / action leader).

5.1.1.2 When discrepancies are found in a process, procedure or in a material/product in term of workmanship or an out of control situation is observed, the operation shall be 'stopped' for corrective action. The operation shall only resume after necessary actions are in place and approved by the Quality Engineer after re-qualifying the operation and found to be acceptable. (Para 5.1.1.2 is applicable for QA line audit activity)

5.1.1.3 Corrective and preventive actions will be documented on the NCR (all section's of the NCR must be filled up accordingly) the Quality Personnel shall verify the actions taken for effectiveness and implementation before closure.

5.1.1.4 NCR issuance and closure shall be tracked by the respective personnel as below :-

5.1.1.4.1 Attachment 5 for QA inspection by the Line QA personal.

5.1.1.4.2 Attachment 6 for IQA (Internal quality alert) by centralized CAR Coordinator . Note : The closure will be supported by Assembly QA .

5.1.1.4.3 Attachment 7 for audit failures by respective QA

The action closure verification will be done after expiry of the corrective action target date or in any circumstances to be within 30 working days from action due date.

5.1.1.5 The recipient shall respond and return the NCR to the initiator within 2 working days (where practical) from date of issue; with all sections of the NCR filled accordingly. Except Resubmission result column. During holidays or off days email approval is accepted and the email approval must be attached together with the NCR format during submission. Accepting QA to ensure this is in place prior acceptance of the NCR. In any case, where in depth investigations or evaluations are needed to be carried out to determine root cause and corrective action by the auditee. (e.g. process fall out, parameter optimization or D.O.E'S need to be done or brainstorming), the NCR response due date may be extended with agreement between the auditor and auditee (agreement should be documented on the back page of the NCR). However, containment plans must be reviewed as to be in place until problem is fully addressed with the NCR (e.g.: action plans for containment).

5.1.1.6 Any discrepancy or conformance, which affect product quality, shall be subjected to the Material Review Board, for disposition. (to be brought to the attention to the QA engineering)

5.1.2 Lot rejection at QA Gate

5.1.2.1 1st submission for QA Acceptance of prime lots.

5.1.2.1.1 If lots accepted during 1st submission QA shall stamp ACCEPT and release for next process operation.

5.1.2.1.2 If lots failing the 1st submission (1st time at QA gate), NCR shall be issued. The NCR, traveler and rejected lot shall then be returned to production for appropriate action.

5.1.2.1.3 QA to scrap the unit upon received disposition from the Engineering Team (As per QCG-073 Product Scrappage Procedure)

5.1.2.2 2nd submission of 1st rejected lots for QA Buy-Off.

5.1.2.2.1 The production supervisor / process technician shall provide containment action at the appropriate column (fill up elements 1 to 4) on the NCR before resubmission of the rejected lot for QA buy-off

5.1.2.2.2 QA shall only proceed to buy-off this lot after clause 5.1.2.2.1 is met.

5.1.2.2.3 If the lot is accepted, QA shall release the lot for next operation.

5.1.2.2.4 If same lot failing the 2nd submission (2nd time at QA Gate), the lot shall be rejected and return to production together with the same NCR. The production supervisor shall notify the process engineer / production superintendent of the rejection. & fill up the balance element # 5, 6, 7 & 8 in the same NCR.

5.1.2.2.5 QA to scrap the unit upon received disposition from the Engineering Team (As per QCG-073 Product Scrappage Procedure)

5.1.2.3 Procedure for 3rd submission of rejected lots for QA Buy-Off.

5.1.2.3.1 The process engineer / production superintendent shall provide root cause analysis, containment, corrective and preventive action of the recurrence at the appropriate column on the NCR before 3rd submission of the rejected lot for QA buy-off.

5.1.2.3.2 The corrective action taken have to be approved by the area quality engineer or assigned deputies.

5.1.2.3.3 QA shall proceed to buy-off, after clause 5.1.2.3.1 & 5.1.2.3.2 is met.

5.1.2.3.4 If the lot is accepted, QA shall release the lot for next operation.

5.1.2.3.5 If the lot is failing the 3rd submission (3rd time at QA Gate), the lot shall be held back by QA personnel at identified area for MRB disposition. QA personnel shall notify the QA Engineer / Process Engineer & Superintendent on the rejection by issuing a MRB for to the production personnel.

5.1.2.3.6 The quality personnel or the production / engineering personnel may Call for MRB (See MRB Spec)

5.1.2.3.7 The MRB shall also decide on the further appropriate corrective action to be taken and documented before release of the affected material to production.

5.1.2.3.8 QA shall proceed to buy-off the said material only after receipt of the MRB disposition form. Prior to buy-off QA shall verify the actions taken as to be in place. QA buy-off data shall be documented if in the MRB form.

5.1.2.3.9 QA to scrap the unit upon received disposition from the Engineering Team (As per QCG-073 Product Scrappage Procedure)

QA Gate buy-off results summary as below –

QA Buy-Off (Same lot)	Results	NCR Issuance
1st submission	Pass	No
	Reject	Yes (To fill element 1-4)
2nd submission	Pass	No
	Reject	Yes (To fill element 5-8)
3rd submission	Pass	No
	Reject	MRB

5.1.3 Quality Feedback from Internal Customer(NCR Test)

5.1.3.1 The NCR form shall be generated when:

5.1.3.1.1 A lot is experience overall low test yield / high open-short reject due to a particular assy non-conformance.

5.1.3.1.2 Any abnormal defect / non-conformance detected which is deemed necessary to request corrective action under discretion of requestor's supervisor.

5.1.3.1.3 Assembly defect detected at Final Test.
Note: For electrical test low yield, test will feedback to assembly when the test rejects are due to assembly defect and this particular assembly defect is exceeding 0.5% (threshold).

5.1.3.1.4 NCR requestor from test PE/finishing team shall fill up the NCR form(including accept and reject photo, measurement readings, reference spec# and etc) and submit to test engineering.

5.1.3.1.5 The approved NCR by test engineering shall be returned to the CAR coordinator within 24 hours of the date issue, who will then issue NCR# and update the NCR status.

5.1.3.1.6 For assembly, completed NCR (with containment actions, root cause analysis and CAPA) with area managers approval shall be returned to the coordinator within 10 working days time from the date issued.

5.1.3.1.7 A copy of NCR will be given to the requestor. The NCR will be reactivated again if the requestor not satisfy with the corrective action.

5.1.3.1.8 The following EOL assembly defect caught at QA Outgoing gate in Final Test shall feedback through the flow (Attachment 2).

5.1.3.1.8.1 Marking Defect
5.1.3.1.8.2 Package Defect
5.1.3.1.8.3 Lead Contamination
5.1.3.1.8.4 Flashes on lead.

5.1.3.1.9 Above defects will be verified by Process Engineer (Final Test), QA Engineer Final Test & Assembly) before feedback to assembly. This verification is to determine defect are genuine assembly defect.

5.1.3.1.10 Standard NCR subject to be used when for the alert. Eg:
Test alert –
NCR#_SGG_TESTBU4_181114-01_6SOT23H_PEN0017_ 843130_ INCOMING BLUR MARKING; (NCR# - Reference number issued by test dept)
NCR coordinator alert –

NCR# : NCR_MMM_TESKKYYMMDD-XX_LOT#_FAILURE MODE-REASON CODE

NCR	MMM	TES	KK	YYMMDD	XX	LOT#	FAILURE MODE	REASON CODE
Nature of Report	Customer abbr.	Issued by Test	Department Abbr	YY - Year MM - Month DD - Date	Number of reported failures on the same day	Affected Lot #	Defect type (reference to QRA000001)	Reason Code (reference to QRA000001)

5.2 Invalidation of NCR
The NCR will be invalidated with agreement of Test PE/QA and assembly PE/QA. Upon disagreement of invalidation, Test QA Manager's disposition is required.

5.3 Escalation of NCR
The NCR. Will be escalated when the followings are encountered.

5.3.1 When the NCR is not responded within the time frame allowed in and there is no written/verbal notification by the auditee on the reason for delay.

5.3.2 If the NCR response is found to be inadequate, or not preventive to prevent recurrence of the problem, or if all sections of the NCR are not appropriately filled up.

5.3.3 Upon recurrence of the same mode of audit failure /discrepancy during "closure". If the recurrence is due to the actions committed not effectively implemented (no evidence of implementation) and the auditee recommits a new action due date during the NCR closure verification process;" no escalation is required". Escalation shall be done on the 'open' NCR. If the auditee fails to comply to the recommitted date (still no evidence of implementation and failure seen).

5.3.4 All escalation (1st . escalation) will be escalated to the auditee with copies to the area Manager and Quality Manager for appropriate action.

5.3.5 Escalated NCR must be responded within 2 working days from date of escalation. Failing this, will result in the NCR to be escalated (2nd. Escalation) to the area Manager with copies to the Quality, Operation and General Manager for appropriate action.

5.4 Recurrence of problem after NCR issued.

5.4.1 For internal quality incidences that reoccur, the Quality Manager or his assigned deputy responsible for C.A.R. shall be authorized to request for an 8D to be raised instead of a NCR

5.5 Automotive devices required recapturing and why why analysis on the root cause occurrence in NCR format for QA gate failures. The QA inspector is responsible to follow the requirement of this requirement (S-Site only).

SECTION II : Attachment 1

NCR No. : _____

CARSEM QUALITY DEPARTMENT

NON-CONFORMANCE REPORT (NCR)

1. Issue To : _____ Issue By : _____ Department : _____

Operation : _____ Shift : _____ Date/Time : _____

Lot No. : _____ Device : _____ Sub lot ID : _____

Pkg Type: _____ MC/OPT No. : _____ Sample Size / No of rejects : _____

Date Code: _____

2. Failure Category : (Tick where applicable)

a) Non – Conformaties

1) Audit Failures ☐

2) Administration Issues ☐

3) Non-Compliance to procedure ☐

4) Others _____

b) QA Gate lot Rejections :

Critical Defect ☐

Non Critical Defect ☐

Defect Verified By: _____

c) Quality Feedback from Internal Customer

1)Final test to Assembly ☐

2)Others _____

Required On-Hold Material? ☐ Yes ☐ No

MRB Required? ☐ Yes MRB #: _____

Recapture Require? ☐ Yes ☐ No ☐ No

*If yes, pls attached Recapture Form upon NCR issuing.

Note: Recapture applied when there is direct impact to the product quality (Ref: CRQA100H07).
MRB shall only be raised if the defect criteria is in the MRB Issuance Guideline.

3. Problem Description (Attach reject unit or provide photos of failures whenever possible) :

4 Containment/Immediate Actions:

	Who	When (Indicate date of implementation)
		☐☐☐☐ - ☐☐☐ - ☐☐

Verified By : _____ When : ☐☐☐☐ - ☐☐☐ - ☐☐ (Indicate date of closure) Status : _____

5 Root Cause Analysis using "WHY-WHY ANALYSIS"

Note: During Analysis upon identify true cause you may get the corrective/preventive actions from other areas responsible for the failure.

1ST WHY :

2ND WHY :

3RD WHY :

4TH WHY :

5TH WHY :

6. Resubmission Results (Only applicable for lot rejection at QA Gate)

Date	Qty In	Qty Out	Total Defect breakdown	Badge No.	Sup.Sign
2nd submission					

Date	Time	QA Rej / Sample Size	Description of Defects	QA Stamp

Date	Qty In	Qty Out	Total Defect breakdown	Badge No.	Sup.Sign
3rd submission					

Date	Time	QA Rej / Sample Size	Description of Defects	QA Stamp

4th submission shall Be Accompany by MRB.
MRB No……. Date :

Date	Qty In	Qty Out	Total Defect Breakdown	Badge No	Sup.Sign
4th. Submission after MRB approve					

Date	Time	QA Rej / Sample Size	Description of Defects	QA Stamp

Item 7 & 8 applicable for failure category Non – Conformities, QA Gate submission (3rd Submission only) and Quality Feedback from Internal Customer (Final Test).

7. Corrective / Preventive Action

	Who	Implementation Date

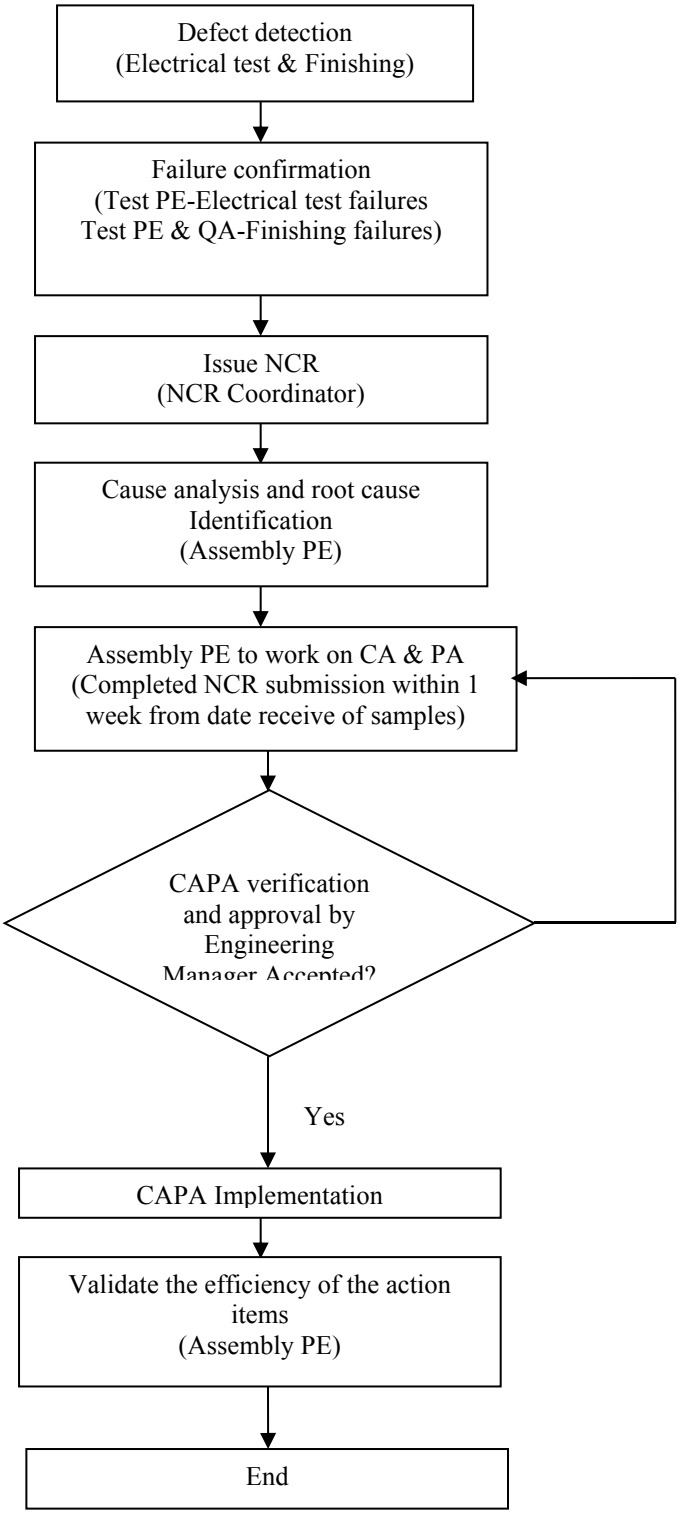
	Non – Conformities, QA Gate submission (3rd Submission only) Quality Feedback from Internal Customer (Final Test)		
	Name	Signature	Date
Responded By:(Production Supervisor)			
Verified By:(Superintendent)			
Approved by: (QA Engineering)			

8.Verification of Corrective / Preventive Action effectiveness & closure

Verified By : _____ (Indicate date of closure) When Status : _____
(QA supervisor and above) (Name and Signature)

Note: Ncr Respond Day – From Date Issued Or Received Of Sample
1. Lot Rejection – Upon Resubmission Of Reject Lot, Immediate.
2. Quality Feedback from Internal Customer – 2 Weeks
3. Audit Failure – Within 2 Working Days

SECTION II: Attachment 2



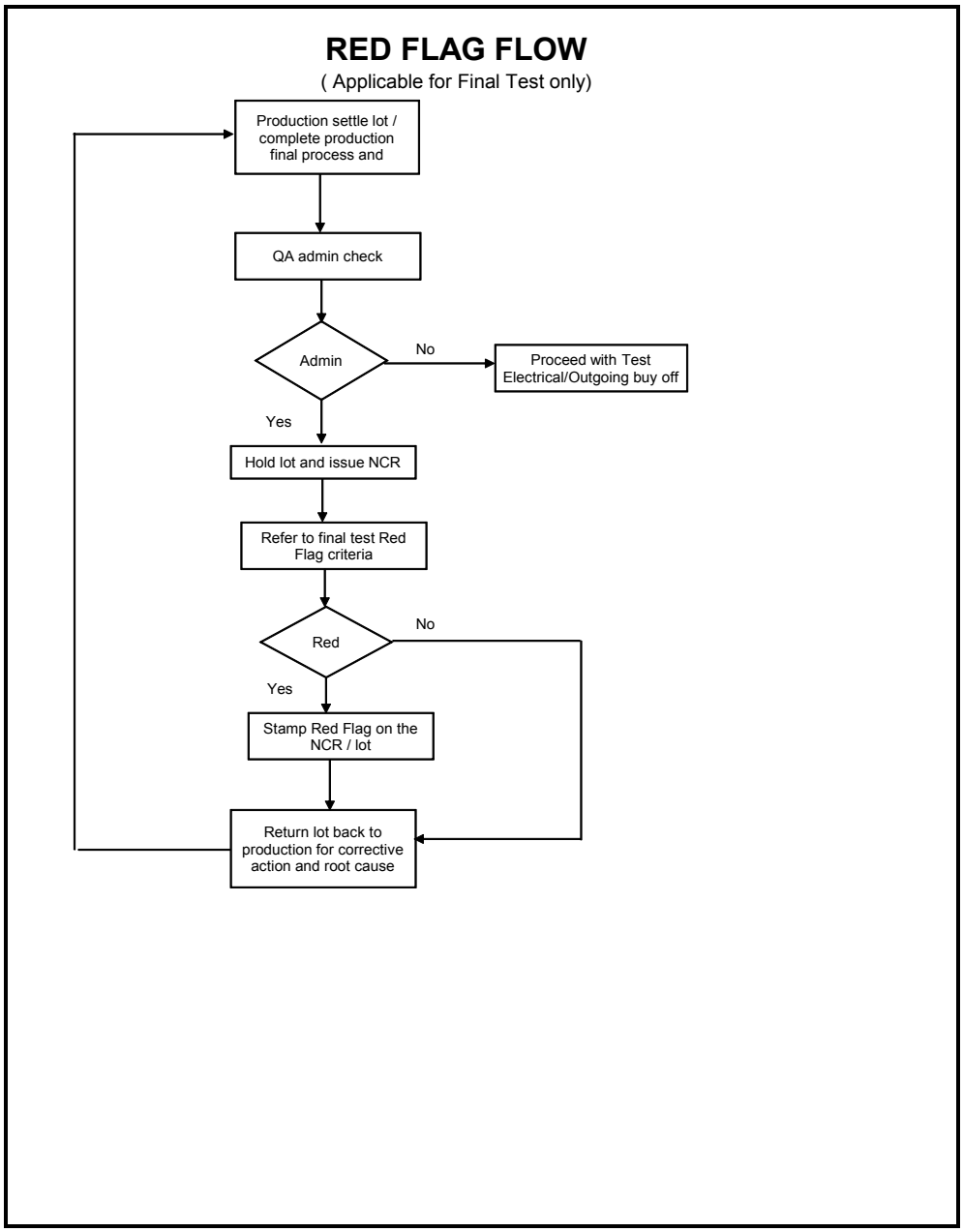
SECTION II : Attachment 3

RED FLAG AT FINAL TEST

1.0 Non conforming admin failures which lead to customer complain CAR / quality incidences will be define as RED FLAG at Final Test.

2.0 QA reporting personal will report the Red Flag to the Management level weekly basis.

3.0 Refer to the below flow for Red Flag NCR issuance.



SECTION II : Attachment 4

RED FLAG DEFINITION

(Applicable for Final Test QA gate only)

Admin failures that lead to customer complaints (CAR) / Quality incidence.

Defect Category	Defect Mode	Description
Test Program issue	Wrong option used	Wrong option on program used
	Wrong program used	Binsum and traveller program not same
	Wrong revision used	Revision stated on binsum program differ from traveller
	Wrong limit used	Wrong limit (QA/FT) used
Documentation issue	Less binsum	Physical qty more than binsum
	Qty variance	Qty stated on traveller and physical not same
	Damage material	Damage tube/ tray/lockreel
Swap defect	Swap lot	Physical box from different lot (not same as traveller)
	Swap label	Physical label from different lot (not same as traveller)
	Swap traveller	Physical traveller from different lot (not same as physical)
	Swap reel	Physical reel from different lot. (not same as traveller)
Baking defect	Over expose bake window	Material expose to air exceeding the time limit
	Wrong baking temperature	material baked with wrong temperature
	Over/ Less baking hours	material baked with less/more baking hours
Mix issue	Mix device	Mix unit from different device
	Mix datecode	Mix unit from different datecode
	Mix tracecode	Mix unit from different tracecode
	Mix lot number	Mix unit from different lot
Marking defect	Wrong marking	marking does not tally with marking instruction
	No marking on unit	dummy unit / unmark units mix
VM / TNR defect	Reverse reeling	Reel in oppsite direction
	Wrong orientation	Unit with the orientation not per traveller
	Units dead bug	Units with the orientation upside down.
	Peel strength error	Fail peel strength / no peel strength / wrong limit
	No protective strip (TNR)	no protective strip on reel
	Quantity variance (tube/TNR / tray)	Total quantity in reel/lot is less or more
Packing defect	Wrong packing method	Wrong packing method not as per BOM
	Wrong packing material	Wrong packing material used not as per BOM
	Open seal bag / seal bag leakage	MBB / Antistatic bag not seal properly / leakage
	No conductive bag (canister)	Units not feed into conductive bag for canister packing
	HIC turns to pink	The HIC turn to pink colour exceeding percentage limit
Label defects	Label error	Wrong packing label information
	No label	Label not available as per the BOM
	Wrong label placement	Label placed at wrong side of box/ reel / bag
	No sticker	Found no sticker (eg. ESD/MSL)
	Wrong sticker	Found wrong sticker used not as per BOM

SECTION II : Attachment 5
NCR TRACKING FOR QA INSPECTION

No	NCR#	Date Issue (YY/MM/DD)	Lot #	Device	MES Lot id	Pkg Type	Issue To (Name)	Failure Mode	Submission Results *(Only applicable for lot rejection at QA Gate)*			MRB #	Date NCR Responded (YY/MM/DD)	NCR Cycle Time	Corrective Action Status (Closed / Open)	Remark
									2nd Submission (result / Sample size)	3rd Submission (result/ samplesize	4th Submission after MRB) (result/ samplesize					

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SECTION II : Attachment 6
NCR TRACKING FOR TEST IQA

ITEM	MONTH	CUST	NCR #	DATE RECEIVE	SAMPLE RECEIVE DATE	ISSUED BY	ISSUED TO	FAILURE MODE	LOT NUMBER	DEVICE	PKG TYPE	D/CODE	IQA DUE DATE TO RESPOND	DATE IQA SENT TO S	CYCLE TIME	ACTION CLOSURE STATUS (OPEN /CLOSED)

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SECTION II : Attachment 7

NCR TRACKING FOR AUDIT

No	NCR#	Dept	Date Initiated (YY/MM/DD)	Issued By	Issued To	NRC Response Date	NCR Cycle Time	Failure Description	Root Cause	Corrective / Preventive Action

SECTION II: Attachment 8
Customer Special Requirement

Avago (All Devices)

A) Source Of Requirement :- As per attached email communication with customer Avago Jee Kwang Mok and Carsem Team dated 2012-Sep-10 & 11

RE MUST READ
Conference Call With

2024) Requirement

2025) Non conformance Report (NCR)

A) Lot rejection at QA Gate – to reject lots that are non-conformance or not meeting to the QA Acceptance Specification, Customer Specifications / Drawings. (Fill up ALL elements in NCR format except element 6. Resubmission result (all items will fill up if there is 2nd or 3rd reject).

SECTION II: Attachment 9
Customer Special Requirement

ADI

TEST NCR FLOW

a) Source of Requirement :

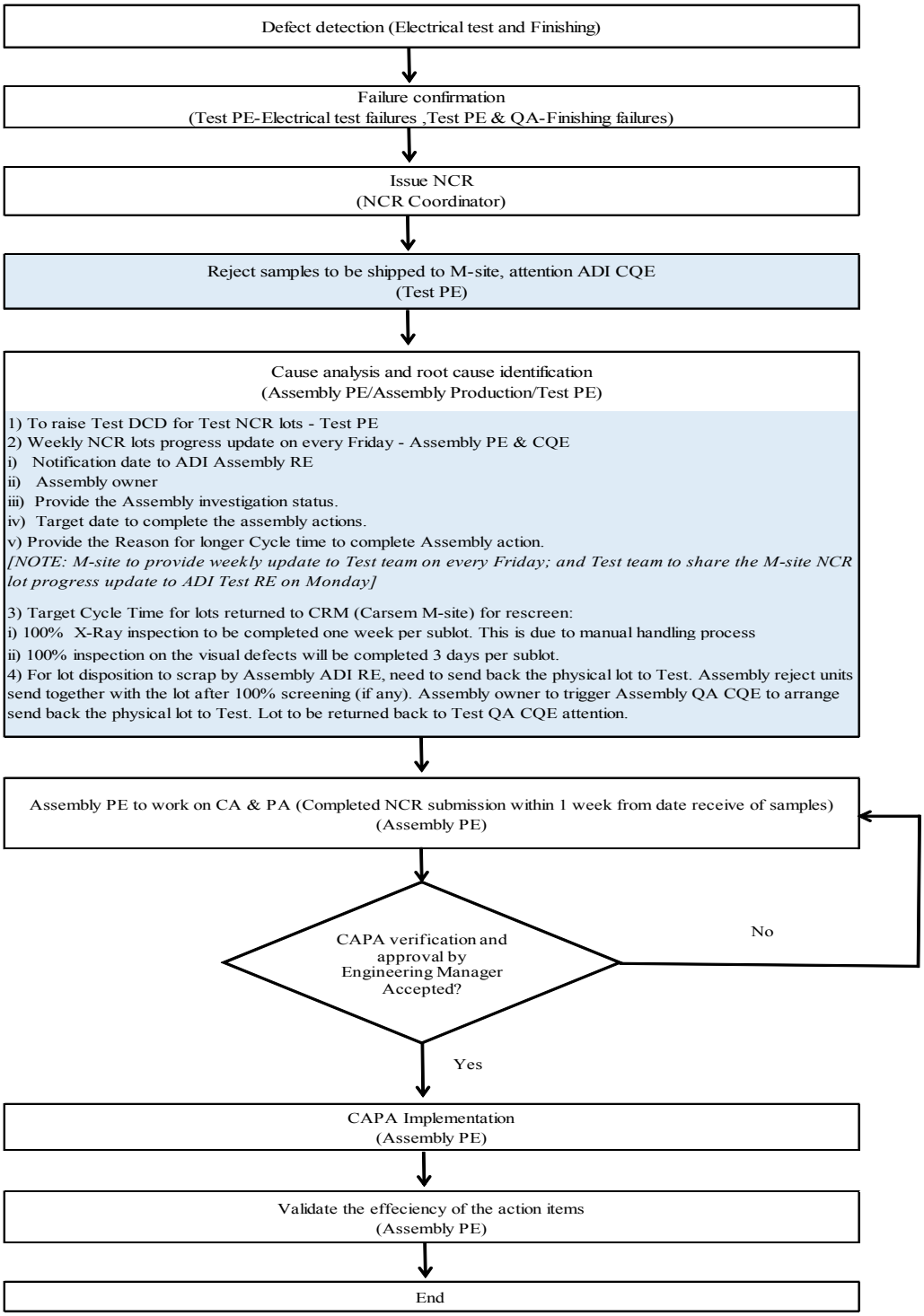
2026) Action item for Q1'19 QOR for the Scorecard point deducted due to "slow interim response for Test DCD".Attached is a meeting minutes between Carsem Team and ADI Test RE (KW Chaw)

E-mail Communication.msg

b) Requirement :

Enhance Test NCR Flow as below

ADI TEST NCR FLOW



SECTION III

1.0 Title :
Supplier corrective action request

2.0 Purpose :
The purpose of this documented procedure is to detail how the supplier corrective action request is applied.

3.0 Scope
Supplier corrective action request is applicable to all suppliers when a non conformance is found on their goods or service supplied to Carsem.

4.0 Applicable form
Material non-conformance, corrective action requests, 8D CAR and 8D evaluation checklist (Attachment 1, 2 & 3).

5.0 Definition
"Suppliers" are defined as "vendors" who deliver their standard catalogue part / products or provide a service to Carsem.

6.0 Procedure :

6.1 Supplier corrective action request (Attachment 1)

6.1.1 When non-conformance of goods (physical material and documentation) is found during IQA inspection buy off or in the process / production line (as a result of raw material) a corrective action request (see Attachment 1) shall be issued to the supplier, stating the discrepancy observed.

6.1.1.1 SCAR Complaint Origin
L : Line Detected Incident (LDI)
C : Customer Complaint (VOC)
I : Incoming Detection (IQA)

6.1.2 The supplier shall respond to the corrective action request by indicating the root cause identification, immediate/interim action, permanent action and verify the effectiveness of the action taken and measures to prevent recurrence. The supplier shall use Carsem 8D format as a guide or reference in responding to any non conformance /correction action request. (As per Section III (Attachment II)

6.1.3 Suppliers ACKNOWLEDGMENT to corrective action request must be done within 24 hours upon received of corrective action request (C.A.R) from Carsem. The acknowledgement may be send in hard or soft copy.

6.1.4 Supplier to respond 3D containment action within 24 hours upon receipt of C.A.R.

6.1.5 The **8D report** & 8 Discipline Evaluation Checklist(As per section III, Attachment III) must be responded by the supplier within **5** working days upon receipt of the corrective action request/samples (if any), **require supplier provide return material authorization number (RMA #) in 5 working days after 8D submission.**
Note : Customer 8D respond cycle time will supersede Carsem requirement (Refer to section 1 – Attachment 8)

6.1.6 When there is any excursion incident (such as fire, chemical spillage, flood, other disaster, etc.) happened which impact the product quality and delivery, the supplier is required to provide an analysis report using the 8D methodology approach with corrective and preventive actions to prevent recurrence or to minimize impact to quality and delivery.

6.2 Penalty for late response to supplier corrective action request.

6.2.1 In the event where a supplier requires more time to investigate the root cause due to its complexity of the defect mode, that requires for additional test or failure analysis, the supplier shall inform Carsem and get approval on the re-commit response date prior to the actual 7 days cycle time.

6.2.2 Failure to comply to 6.2.1 requirement will be escalated to the relevant suppliers manager and further escalation to Carsem & Supplier management involvement when there is potential materials shortage. On other hand, failure to do so will also impact the suppliers service rating for that quarter.

6.2.3 When a completed corrective action report& 8 Discipline Evaluation Checklist are received from the supplier, it shall be reviewed by SQE for adequacy to prevent recurrence of the reported problem. If the corrective action is accepted, the C.A.R will be reported as 'closed' only after all the action items committed are implemented and verified to be effective through minimum three shipments.

Supplier shall provide minimum 1 month monitoring data upon implementation of the corrective and preventive actions for effectiveness verification.

If further actions are considered necessary, the C.A.R will remain as 'open/on going'.

6.2.3.1 In the event where the part become dormant (due to business nature or decision) with no shipment come in for 8.D effectiveness verification, the C.A.R will be automatically closed after one year (from date of complaint).

6.2.3.2 In the event where the part is disqualified due to quality issue and there is no shipment for 8.D effectiveness verification, the C.A.R will be remain 'open' with remarks as 'not active'. However, when the supply is resumed all previous outstanding C.A.R shall be verified for closure per requirement in 6.2.3.

6.2.4 All C.A.R response from supplier must be retained and tracked by the department issuing the corrective action request for tracking purposes.

6.2.5 If complaint by customer, the supplier corrective action will be fan out to other similar commodity supplier.

Note :

1. Disclaimer footnote on the 8D format (Form # CRQA100M01-6) need to be formatted using font type Verdana and the font size must be using size 4.

SECTION III : Attachment 1

CARSEM SEMICONDUCTOR SDN BHD (185524-H)
(A MEMBER OF THE HONG LEONG GROUP MALAYSIA)
Lot 52986 , Taman Meru Industrial Estate , Jelapang , P.O. Box 380 ,

DEST		DATE(YYYY-MMM-DD)		PAGE OF
TO		FROM		
CC		CC		

SUBJECT *MATERIAL NON-CONFORMANCE , CORRECTIVE ACTION REQUEST*

1. CAR NUMBER :

2. PROBLEM / PARTS DEFECTS DESCRITION :

LOT DETAILS :
CARSEM LOT # : VENDOR LOT # : QTY RECEIVED : QTY RECEIVED :
PART # : INVOICE # : QTY REJECTED : QTY REJECTED :
PART DESCRIPTION :

4. RMA # REQUIRED NO [ ] YES [ ] for qty :
SAMPLE REQUIRI NO [ ] YES [ ]
SAMPLE ATTENTION TO:

5.VENDOR ACTIONS REQUIRED

A. Vendor 's acknowledgement to CAR and containment action (3-D) is to reach Carsem within 24 hours.

CAR RECEIVED BY :-
NAME / INITIAL / DATE (YYYY-MMM-DD)

SAMPLE RECEIVED BY :-
NAME / INITIAL / DATE (YYYY-MMM-DD)

B. Corrective and Preventive actions (8-D) to reach Carsem within **5 working days.**

C. Carsem require supplier to provide RMA# in **5 working days** after 8D submission.

D. Vendor is responsible to alert Carsem if any earlier shipments were affected with the same problem.

E. Supplier containment has to be carried out for Carsem M-site , S-site and Suzhou.

CARSEM USE ONLY :	DATE OF CLOSURE(YYYY-MMM-DD) :	BY :
Corrective action reviewed and at Yes	No	
Comment :		

CARSEM's SUPPLIER

8D Corrective Action Report

Carsem Confidential

Type of complaint	: Complaint / Feedback	**Carsem Reference #**	:
Carsem Feedback Date	:	**Customer / Supplier Reference #**	:
3D Report Submission Date	:	**Department**	:
Sample Receipt Date	:	**Package type / Part description**	:
8D Report Submission Date	:	**Part Manufacturing Date /**	:
Point of Detection	:	**Device / Part #**	:
End Customer	:	**Reject Qty**	:
Re-Submission (if applicable)	:	**Failure Rate**	:

Problem Statement : Supplier – Pkg – Defect

D1) Team Formation	**Name**	**Job Title**	**Area of Responsibility**
A) Team Leader:	1)		
B) Team Member:	2)		
	3)		
	4)		
	5)		

D2) Problem / parts defect description.

Background :

Carsem FA analysis :

FA / Sample Photo :

Photo 1 : (Description) Photo 2 : (Description) Photo 3 : (Description)

Supplier FA analysis :

Photo 4 : (Description)

Photo 5 : (Description)

Photo 6 : (Description)

Problem Description Summary:

D3) Interim containment / short term corrective action.

A) Pipeline Analysis

1) Check the existing (potential affected) material in the pipeline for similar defect. Notify customer when there is a need to return affected potential materials

Date	Location	Sorted qty	Rej qty	% of rej	Disposition
	Assembly:				
	Final Test:				
	FG Store:				
	In Transit				
	Shipped to Customer				
	Note : The checking must include partial or residue of affected lot				

B) **Other short term corrective action.**

Actions	Responsible	**Date**
1) Alert QA, production and engineering on the feedback from customer. Status:		
2) Display OPL (one point lesson) and reject photo in production floor to increase people awareness for similar defect. Status:		
3) Check the existing process / machine to ensure no similar defect is generated. Status:		
4) Increase SPC monitoring frequency from _____to _____ per TECN time line (TECN # : _____) Status :		
5) Increase QA gate inspection sampling size from _____ to _____ per TECN time line (TECN # : _____) Status :		
6) Retrieve the assembly history of the affected lot. Status:		
7) Inform Carsem on the potential affected material. Status :		

D4) **Define and verify root causes :**

A) **Fault Tree Analysis :** **Brain Storm Date :**

Problem Defect: XXXXXXXXX ← state the defect

	Primary Contributing Factors	Secondary Contributing Factors	Analysis
1)			**True / Possible / Not possible** (remark : delete extra row which is not utilize)
			True / Possible / Not possible
			True / Possible / Not possible
			True / Possible / Not possible
2)			**Po True / Possible / Not possible**
			True / Possible / Not possible
			True / Possible / Not possible
3)			**True / Possible / Not possible**
			True / Possible / Not possible
			True / Possible / Not possible

B) **1) Why-Why Analysis on the possible technical root cause :**

Why 1:
Because

Why 2 :
Because

Why 3:
Because

Why 4:
Because

Why 5:
Because

2) Why-Why Analysis on the possible systemic root cause :

Why 1:
Because

Why 2 :
Because

Why 3:
Because

Why 4:
Because

Why 5:
Because

C) **Escape Point:**

(Remark : Use Why Why Tools to analyse the escape point, if applicable)

D5) Chosen Permanent Corrective Actions :

Root Cause	Action Taken	Remark
	Stated the action Resp : Date : Status :	
	Stated the action Resp : Date : Status :	

D6) Verification of implemented permanent corrective action's effectiveness :

To monitor _____ ppm for coming _____ weeks upon implementation of permanent corrective action.
Resp:
Status:

D7) Systematic Preventive Actions:

A) FMEA : Mandatory to be filled

Spec # :
Spec title :
Item to be update :
Resp:
Date:

B) Others Relevant Specification :

Spec # :
Spec title :
Item to be update :
Resp:
Date:

C) To proliferate the selected corrective actions from D5 with other Carsem product line

Item to be proliferated :
Resp.:
Date :

D8) **Reviewed and approved by :**

Quality incident		B) Environmental, safety & health incident	
Dept. Manager: Or	_____	Dept. Manager:	_____
Eng. Manager: And	_____	Safety officer: And	_____
Quality Manager:	_____	EMR / safety chairman:	_____

Note:

1) Mandatory signatories:

Quality incident
Signature of the Quality Assurance Manager or his representative during his absence is mandatory.

Environmental, safety & health incident
Signature of EMR (Environmental Management Representative) / Safety Chairman or his representative during his absence is mandatory.

2) Documents to support the closure for an 8D must be attached and verified during verification of closure.

3)All dates shall be written in the format (yyyy-mmm-dd)

SECTION III (Attachment 3)

CARSEM (M) SDN BHD
(A Member of the Hong Leong Group Malaysia)

8-Discipline Evaluation Checklist

The following guidelines is to be used when reviewing/generating an "8D Corrective Action Report".

Assessment Rating for Acceptance of 1st Pass 8D Report:
SCAR#:_____

Category	Definition	Score
Reject	Failed to meet minimum requirement of the 8D checklist	≤ 89.0%
Accept	Meet all the requirements of the 8D checklist with or without revisions	≥ 90.0%

Score guides: YES = 1, NO = 0, N/A = No count

8-Discipline Guidelines		Assessment		
D1	Internal/External Members	Yes	No	N/A
1.1	Considered cross-functional members (QA must always be represented) with the process and product knowledge.	☐	☐	☐
1.2	Members have the skills and required technical disciplines to solve the problem and implement corrective actions. Members can evolve over time.	☐	☐	☐
1.3	Designated Team Leader should be the owner of the issue.	☐	☐	☐
1.4	Wrote the complete name, initials (if applicable) and job position of team members.	☐	☐	☐
D2	Describe the Problem	Yes	No	N/A
2.1	Stated the problem by identifying in quantitive terms by answering the 5Ws,1H.(Attach defect photo) What is wrong? Ans: Product Where did it happened? Ans: Example: Assembly Area Who is complaining? Ans: Internal or External Customer When did it happened? Ans: Date of Complaint What happened? Ans: Defect (Is and Should Be) How big is the problem? Ans: Qty or defect rate	☐	☐	☐
2.2	Stated the failure analysis conclusions with on the Failure Analysis report (if available).	☐	☐	☐
2.3	Performed and stated lot history details (process and shipping history)	☐	☐	☐
D3	Implement & Verify Containment	Yes	No	N/A
3.1	Identified the scope and exposure of material impacted			
	Effective recapturing system for effective containment:			
	- List of Lots (Quantify, Summarize)	☐	☐	☐
	- MRB Hold, Recall, Rework affected lots	☐	☐	☐

	- Consider processed lots before and after affected lot	☐	☐	☐
	- At WIP, Finished Goods	☐	☐	☐
	- Product risk assessment	☐	☐	☐
3.2	Had alert engineering, production, QA, other relevant parties.	☐	☐	☐
3.3	Displayed OPL for reject mode.	☐	☐	☐
3.4	Performed process mapping on the affected lot. Checked what was changed. History of hardware, software, product, process, people changes.	☐	☐	☐
3.5	Established an inspection /screening method and tighten existing inspection method.	☐	☐	☐
3.6	Verified the PPM of the problem internally (at least 3 months data). Revisited internal CARs and MRBs	☐	☐	☐
3.7	Checked if we previously done an 8D on the same/similar failure. If yes, checked why the 8D fails.	☐	☐	☐
3.8	Shutdown the affected machine or fix, adjust, reset-up, PM etc. (if appropriate to stop the problem)	☐	☐	☐
3.9	Decertify then recertify or retrain/re-orient operators/technician/inspectors (if necessary/ appropriate to stop the problem).	☐	☐	☐
3.10	Correct procedure immediately (specify specs number) Review of FMEA and Process Control Plan and OCAP to check the current process control is adequate to prevent recurrence. Why Escape root cause?	☐	☐	☐
D4	Define and Verify Root Causes	Yes	No	N/A
4.1	Performed process mapping on analysis of all potential root causes	☐	☐	☐
4.2	Root Cause: Below are recommended tools:			
	- Fault Tree Analysis	☐	☐	☐
	- Why-Why analysis	☐	☐	☐
	- Cause & Effect (fishbone) Diagrams - SPC charts - DOE/Evaluation/Simulation for root cause validation - Process FMEAs - Process flow diagrams	☐	☐	☐
4.3	Are root causes identified and verified by testing or simulating each potential cause against the problem? Illustration or photos are included for clarity.	☐	☐	☐
4.4	Escape Cause: Why did the problem get to the customer? - Checked or audited conformance/compliance to manufacturing or process procedures, process control requirements, monitors and set-up checklist - Identified escapee point - Use Why Why tools to analyse the escape point.	☐	☐	☐
	- Considered 4 Ms (Man, Method, Machine, Material) in the total analysis	☐	☐	☐
4.5	Is a conclusion summarized on the root cause of the problem and escapee root cause?	☐	☐	☐

D6	Verification of Implemented permanent Corrective Action's Effectiveness	Yes	No	N/A
6.1	Validated the corrective actions over time - How are we going to monitor effectiveness of corrective action for the root cause and escapee cause? Who will do it? When is the completion date?	☐	☐	☐
	- Show the same indicators/trend chart that showed the initial problem and demonstrate effect of corrective action?	☐	☐	☐
	- Consider to include the action as audit item.	☐	☐	☐
D7	Systemic Preventive Actions	Yes	No	N/A
7.1	Reviewed the problem history. Identified which systems, practices, procedures allowed or may allow similar problem to occur and escape. - Are FMEAs and control plan updated	☐	Mandatory to fill up FMEA	
	- Are appropriate system specifications and procedural specs updated for all permanent corrective actions?	☐	☐	☐
	- Are all actions horizontally expanded and proliferated to other applicable processes?	☐	☐	☐
D8	Congratulate Team	Yes	No	N/A
8.1	The report is reviewed and approved by the Departmental and Quality Managers	☐	☐	☐

SCORE (%) = # of provided items required per discipline X 100%
# of total required items per discipline

8D Score	Result (Accept / Reject)
Supplier	
SQE	

Reviewed/Evaluated by:

Supplier: _____ Date: _____

SQE: _____ Date: _____

SQE Manager: _____ Date: _____

Remark :

Appendix 1A

Customer Special requirement

(ST)

A) Source of requirement:

1) Spec No. : 7044694 - Specific Quality & Technical Requirement for GOBM Subcontractors.

2027) Requirement :

BKM Fan Out Flow for Malaysia & Suzhou Site

Refer to Flow in Appendix 2B.

Fan Out Action Review Tracker in Attachment 14.

C) Fan Out will be initiated to gain from;

2028) Product recalls related to field returns from customers triggered by customer complain VOC-SCAR.

2) Product audits failures trigger line down issues during internal product audit.
3) Whole lot scrap due to RMA triggered by customer complaint VOC- SCAR.
4) Customer audit major finding from other sites.

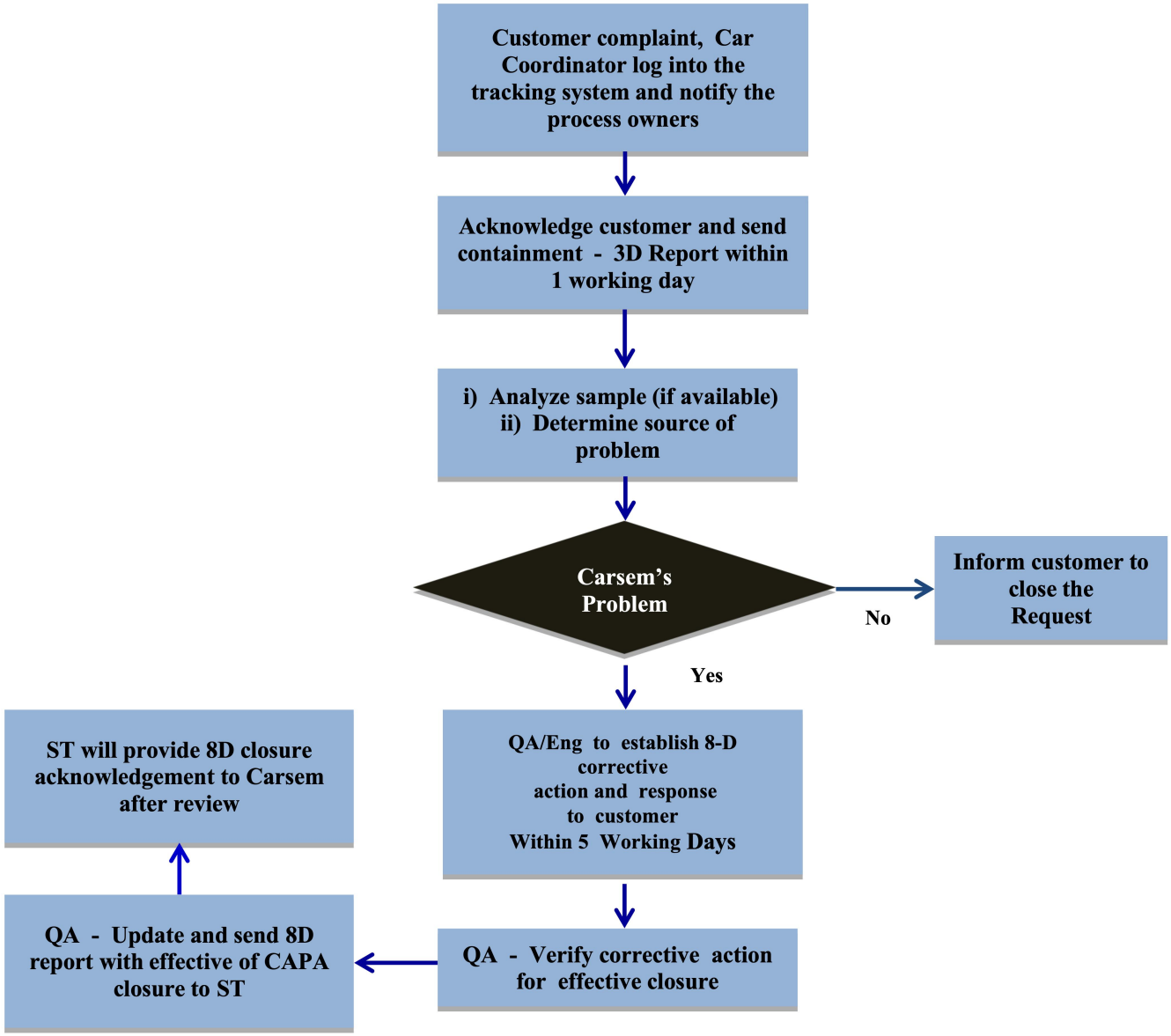
Appendix 1B

Customer Special Requirement

(ST)

a.) Source of requirement : AUD_17_019302 (OFI 19 VDA 7.5) & SSGGC2020101631

b.) Requirement : - Customer Feedback For ST



Note:
Committed action must deployed and executed even though the affected system being improved
In the situation cannot compliance/execute the committed action, action owner/QA need notify customer and get approval.

Appendix 1C

Customer special Requirement

(ST)

A) Source of requirement :
Audit report# : AUD_20_026867 (OFI 03 – 7.4)
AUD_23_034007 (NC06-7.4)

B) Requirement : -
8D Action Update to ST and 8D action tracking system

1. 8D committed action summary tracking shall be update through ST Transformation program.

2. Completed action closure shall be update into 8D report ad submitted to ST.

3. The 8D actions tracking system procedure for ST customer requirement refer to below.

   (Applicable to Carsem Malaysia only)

   Procedure:

   a) QA Clerk/Engineer update D5, D6 and D7 action per customer format requirement into 8D action tracking system upon 8D accepted by Customer.

   Remarks: Official 8D report shall be referred to the report that shared to customer/Car Coor Team and documented in the 8D database. Record in e8D system is to serve as the purpose for action closure notification only due to system limitation.

Appendix 1D

CUSTOMER SPECIAL REQUIREMENT

ST AUTOMOTIVE

A) SOURCE OF REQUIREMENT :-

1.) PER ST AUTOMOTIVE READINESS AUDIT FINDING, FINDING CODE: OFI04 OR REFER TO BELOW ATTACHMENT FOR RELEVANT FINDING.

Auto Cert Audit Report- Carsem M .pc

2.) PER ST QUALIFICATION AUDIT FINDING, FINDING CODE: OFI 20 OR REFER TO BELOW ATTACHMENT FOR RELEVANT FINDING.

Qualification Audit Report _ Carsem Mal

B) REQUIREMENT:-

1. 3X5WHY METHODOLOGY IS REQUIRED FOR ANY 8D FOR AUTOMOTIVE DEVICES.

2. ALL NEW INCOMING CAPA TO BE REVIEWED FOR POLIFERATION TO OTHER PROCESSES TO BENCHMARK BEST PRACTICES.

Appendix 1E

CUSTOMER SPECIAL REQUIREMENT

ST

A) SOURCE OF REQUIREMENT :-

i. PER ST SPEC 7085545_39.0 –SPECIFIC QUALITY AND TECHNICAL REQUIREMENTS FOR GOBM SUBCONTRACTORS

2) Per Alignment Meeting with ST GOBM. Refer to email below.

ST–BEMT checklist.pdf

B) REQUIREMENT:-

1. Carsem required to fill up the page number accordance to 8D report at column H (in BEMT checklist) in order to justify scoring.

Appendix 1E (CON'T)

	ITEM	Score 0	Score 1	Score 2	Assessment score	NOTES
PROBLEM DESCRIPTION	Completeness and relevance	not present *Note: copy paste of Customer failure note is not the description of the failure*	Description of the failure not supported by data	- Description of the failure at Customer supported by data (failure rate, source/conditions of failure, fail test value vs ref value where applicable, otherwise explanation of the functional failure) - traceability of all the claimed samples - timeline to describe the problem when applicable	1	
	Confirmation of Customer symptom	not present	Data reported but no evidence/explanation of matching between electrical failure and Customer reported symptom	Electrical analysis present and matching with Customer symptom according to an explained model	1	*If the failure analysis is done by the Customer with clear description of the electrical failure, clear evidence of the defect and clear match between the electyrical failure and the physical defect, put full score*
	Defect evidence (through optical/SEM picture, FIB, EDX..)	not present	Indirect evidences	Defect evidence with full characterization (EDX...)	2	
	Matching between electrical failure and defect found	not explained	hypothesis not fully supported by data	Model of matching supported by evidences	1	
ACTIONS	Containment actions	not present *Note: lot history, commonality analysis, testing results, risk assessment are not considered containment actions*	Effectivess < 90% or not supported by data	Effectivess >= 90% supported by data	1	*Although quantified risk assessment of material is a type of containment by 8D definition, it will not be considered in scoring here.*
					Max 24	

Appendix 1E (CON'T)

	ITEM		Score 0	Score 1	Score 2	Assessment score	*NOTES*
ROOT CAUSE	Ishikawa Analysis leading to technical occurrence/escape RCs and evidences		not present	Ishikawa present with only few root causes highlighted and/or no evidence for exclusion of branches	Exhaustive Ishikawa supported by evidences for each branch *Note: for new failure modes it has to be built with process expert s*	2	
	Coherence and matching of root cause with all available informations (physical defect, failure rate, commonality…)		weak explanation leaving open question	explanation is present but some evidences are not coherent	Model explaining the mechanism with all data	1	
	5-WHY analysis leading to systemic occurrence/escape RCs		not present	present but stopping at technical level and/or missing escape root cause	5-why reaching system/managerial root cause for both occurrence and escape	1	
ACTIONS	Corrective actions aligned with findings and identified RCs		OPL, training, operator debrief are considered not valid actions	Actions addressing only a subset of technical root causes	Actions covering all technical root causes	1	*When exact technical root causes cannot be defined, corrective actions need to cover all potential root causes identified.*
	Effectiveness of corrective actions		not present	actions not ensuring eradication (<90%)	Effectiveness >=90% on all technical root causes actions	1	* To give action verification effectiveness % when action is still open.
	preventive actions	Actions in addition to FMEA and CP/RTP/PM review/update	not present	actions addressing all systemic root causes	actions addressing all system root causes with 100% effectiveness	1	*Here we check the preventive actions in addition to FMEA and related document review/update* * To give an estimated effectiveness when action is still open
		FMEA and CP/RTP/PM review/update	not present	mentioned only	Evidence of FMEA and CP/RTP/PM review/update for the relevant item (snapshot)	1	*If FMEA has been updated, related documents like Control Plan/RTP/PM need to be updated. Consider full score as long as evidence of update or review (when update is not required) is presented in 8D/ QIP tool.*
						8	
						Max 24	

Appendix 1F

CUSTOMER SPECIAL REQUIREMENT

(ST)

A) Source Of requirement:

ii. 8D Report#: SSGTP2019041051

iii. Requirement

8D action Validation Guideline

	Correclative action category		Validation Guideline
1	Procedure/method	1.1	Procedure documented into spec ?
		1.2	Operator briefing on this spec ?
		1.3	Audit for compliance to this procedure
		1.4	Is this a correction type action? If yes, effectiveness trend is required.
2	Machine modification	2.1	Are all the machines installed ?
		2.2	Is it set-up checklist revise to include checking on this item ? If not , pls state justification.
		2.3	Is it PM checklist revise to include checking on this item ? If not , pls state justification.
		2.4	Is this a correction type action? If yes, effectiveness trend is required.
3	New machine	3.1	PCN raise ?
		3.2	Qualification report approved ?
4	Jig/tool	4.1	Procedure establish for this Jig/tool and documented into the spec
		4.2	Compliance audit by interview technician on this procedure
		4.3	Jig/tool registered into Tooling room
		4.4	If measurement jig, is it register for calibration ?
		4.5	Is the part have a safe guard storage ?
5	Material change	5.1	PCN raise ?
		5.2	Qualification report approval

Appendix 1G

CUSTOMER SPECIAL REQUIREMENT

(ST)

C) SOURCE OF REQUIREMENT :-

ST- CARSEM TECHNICAL REVIEW REMOTE AUDIT OFI 5 and OFI 6

D) REQUIREMENT:-Why Why analysis self assessment

1. Do self assessment for WHY WHY analysis as per below attached before submit to ST.

Why Why Analysis Assessment Checklist

Instruction: This checklist applicable to VALID 8D cases.
The checklist is required to be evaluate and attached along at D7 in the 8D report.

		Drop Down List Selection		
Section	Criteria	Y/N	4M+1E element	SRC Number
TORC	Is Why Why start with Possible cause.			
TORC	Is it explain "why did the problem happen"?. Please stated related"3M element" that your root cause linked to.			
SORC	Is it explain" why did the system allow this problem to happen?"			
SORC	Is the SORC are linked to any QMS Process?Please stated SRC Number from QMS Process Reference list.			
TERC1	Is it explain " why did TORC escaped?" Please stated related"3M element" that your root cause linked to.			
TERC2	Is it explain " why did failure mode escaped?" Please stated related"3M element" that your root cause linked to.			
SERC	Is it explain " why did failure mode was undetected?"			
SERC	Is the SERC address the organization cause of the probem. If Yes, pls stated SRC Number from reference list.			

4M+1E element- Machine,Measurement, Method, Material & Environment

QMS Process Reference List

QMS Process	SRC #	Systemic Root Cause
Package Development process	SRC01	Design of Product, bill of material is not robust, or risk management is incomplete, to meet requirements
	SRC02	Design of Process, Equipment, Tooling is not capable, or risk management is incomplete, to meet requirements
	SRC03	Requirement management (from Customer to Carsem) is incomplete
	SRC04	Requirement management (from Carsem to Supplier) is incomplete
Control of Production process	SRC05	Process / Equipment/ Tool & control system is not production capable, or risk management is incomplete, to meet requirement
	SRC06	Safe Release / clearance after machine stoppages/maintenance is incomplete
Total Productive Maintenance process	SRC07	Maintenance system (definition, risk management, prevention & detection) is incomplete
Change Management Process	SRC08	Change Management Process (risk management, templates, work instruction, checklist, decision) has gaps
Quality Documentation	SRC09	Work Instruction or training material definition is incomplete (for shop floor)
	SRC10	Documentation (procedure, specification, process description & document structure, etc) or evidence of implemented quality management (checklist, record, forms, etc) is incomplete
Quality Leadership	SRC11	Management review systematic has gaps
	SRC12	Quality Culture/ Policy implementation is incomplete
	SRC13	Allocation of resources (budget, headcount, etc) is insufficient (due to prioritization)
Competency / People development	SRC14	Personnel training and certification system (competency , awareness) is incomplete
	SRC15	Human resources policy (hiring, reward, disciplinary) has gaps
Supplier Quality Management process	SRC16	Supplier / vendor management process (selection, development, qualification, improvement, assessment) has gaps
Audit Management Process	SRC17	Audit management process has gaps
Deviation Management Process	SRC18	Deviation Management process has gaps
Continuous Improvement Process	SRC19	Continuous Improvement Management process
Product Identification and traceability	SRC20	Product traceability system is incomplete
Knowledge Exchange Process	SRC21	Knowledge exchange is incomplete
Laboratory Requirements	SRC22	Laboratory management has gaps
Calibration	SRC23	Calibration system is incomplete

Appendix 2

Customer Special Requirement

(IFX)

A) Source of requirements:

1) As per email between Gene.de-Roca (cust.IFX) and Laila(Carsem) dated 2017-Mar-07
2) As per audit response from POM audit 12 Jan 2017 from IFX Gene de Roca on IFX 8D investigation guideline procedures.

Update POM Audit in Carsem S.msg

iv. Requirements:

IFX 8D investigation guideline procedure

1. Pre-8D investigation, 8D team leader & QA to refer to the 8D guide line, obtainable from IFX 8D format folder through below link.
   M:\8D REPORTS(PIC_QASYSTEM)\8D Format (Master List)\Customer IFX
2. During 8D investigation, manufacturing persons follow the 8D guideline
3. Post-8D investigation, QA verification for the 8D guideline compliance

Aditional 8D Investigation Guideline 1) Review existing FMEA database, look for similar defect mode and the root causes and : a) Validate/Simulate the defect through each of these root causes and check any similarity of defect pattern generated as per complaint.
v. If simulate the same defect pattern, check any weakness in current control.
2) Process mapping and commonality study a) If complaint more than 2 lots, perform process mapping and commonality study to identify the same factors.
vi. Utilize these common factors for root cause investigation
vii. Review lesson learnt and existing control from past similar defects.
Check any similar 8D for these defects in the past 12 months. Review the root causes and the control to see any similarity which can help in current root cause investigation.

Appendix 3A

Customer Special Requirement

(ROBERT BOSCH)

viii. Source of requirement :

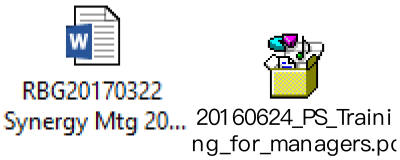
ix. RBG Synergy Meeting on 2018-Mar-28 between RBG (Arndt Norman) and Carsem (Sharon Ko, Inderjeet, KK Wong, Nazri, KH Ong, Wayne Lai, Alvin Seow, Karenjitt, KH Ho, Regis, David Wong, Daniel Hong, CL Hang, Jeffler, Manimaran, Muhammad Yusof, Steven Chu, YK Thien, MW Soo, LS Tan, Angel Gan, ZX Lee, BS Ding, Ivy Chin, LY Ang.)

B) Requirement:

1.0) Managerial Root Cause

1.1) It is mandatory to include the Managerial Root Cause for every 8D submission to RBG.

1.2) Managerial Root Cause guideline per attachment.



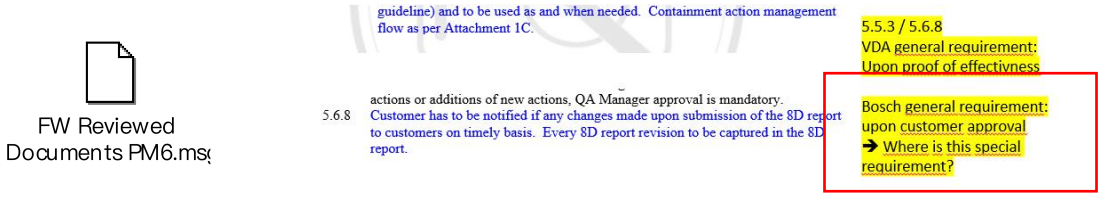
Appendix 3B

Customer Special Requirement

(ROBERT BOSCH)

x. Source of requirement:

1.0) RBG PM6 managerial systemic solution discussion meeting. Email communication received on 2021 Dec 14 @ 3.03pm from Heyer Jens-Martin.



B) Requirement:

xi. RBG's requirements –

1.1) To notify and obtain RBG's approval if any changes made upon submission of the 8D report to customers on timely basis. Every 8D report revision to be captured in the 8D report.

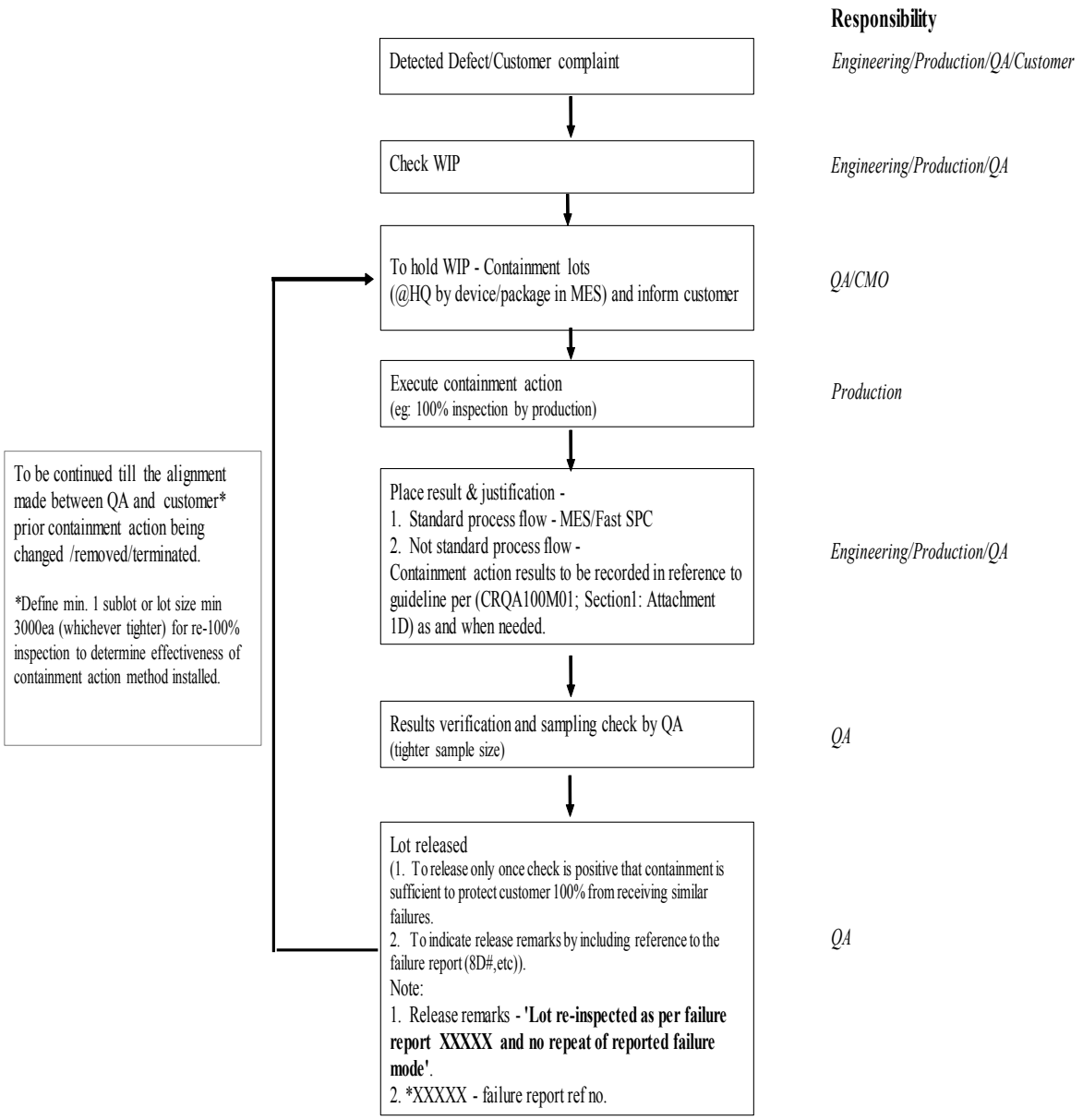
1.2) To ensure effectiveness of implementation of 8D corrective actions, assurance need to be performed on a reasonable basis, needs to be aligned with RBG.

xii. To have Containment Lot Management Flow to manage containment lots –

to hold all lots and release only once check is positive that containment is sufficient to protect customer 100% from receiving additional failures and with traceability. (Refer Appendix 3B-1 for RBG Containment Lot Management Flow)

Appendix 3B-1

Containment Lot Management Flow (RBG)



Appendix 3C

Customer Special Requirement

(ROBERT BOSCH)

A Source of requirement:

1.0) As per RBG CSR, QAA_EN as at 01/2011, Quality Assurance Agreement, it is always in the automotive sector and preferably in the non-automotive sector, the SUPPLIER shall process all 8D reports using the supplier portal from SupplyOn.

2.0) Email communication received on 2021 July 07 @ 4.58pm as waiver of compliance from Svetank with confirmation from Eser Batuhan (RBG).

RBG EDI Document.msg

QAA_EN-5.4 Complaints, Solving

B) Requirement:

1.0) Carsem had confirmed the EDI document (SupplyOn) is not applicable to how Carsem is currently working with Bosch.

2.0) Carsem will be running with current Carsem complaint management system using Carsem-RBG 8D format with latest revision and not implement new changes, therefore supplier portal from SupplyOn will not be used and therefore EDI document will not be completed.

Appendix 4

CUSTOMER SPECIAL REQUIREMENT

(ALLEGRO)

(Applicable to CARSEM S-Site)

A) SOURCE OF REQUIREMENT/S :

1. 8D NUMBER: COMP4848/ SALLC2022012112

B) REQUIREMENT :

1. As per communication between KY Leong (ALL) and Fatin Nurnisha (QA) on 2022-MAR-07 for COMP#4848 (Stray Wire Issue), to implement Lot History Assessment Checklist (below) for any VOC feedback.

RE EXTERNAL D–Ten
A8519-1 line failure

Checklist is compulsory to be compiled by QA and to be send separately to ALL CQE when and upon needed.

Item#	Checking Description	WHO	Applicable? Yes/No	Status & finding
1	Containment Action (Future Hold Lots, for requirement needed)	QA	Y	
2	Machine down time	Maintenance	Y	
3	Machine log view	Maintenance	Y	
4	Machine related critical alarm (define per case basis)	Process Engineer	Y/N	
5	Machine alarm frequency	Process Engineer	Y	
6	Machine commonality/variation	Process Engineer	Y	
7	SPC checking Ball Shear & Wire Pull, etc (define per case basis)	QA	Y/N	
8	Recapturing 3 lots from affected lot (before & after), (further expand on case basis)	Process Engineer	Y	
9	Package/Device commonality (define per case basis)	Process Engineer	Y/N	
10	3rd opt inspection yield & defect mapping	Production/ Proc engineer	Y	
11	Material batch to batch variation (e.g wire, epoxy, wafer, etc)	IQA	Y	
12	Stagging process time commonality	Process Engineer	Y/N	
13	Man commonality	Production	Y	
14	Shift commonality	Production	Y	
15	Gap analysis from the method (e.g. change capillary, wire threading, etc)	Process Engineer/ Production	Y	
16	Summary Time frame of data analysis, define per case basis (e.g month to month)	All Team above	Y	
17	Summary Low, Medium & High risk lot level	All Team above	Y	

Appendix 5

CUSTOMER SPECIAL REQUIREMENT

(TI)

(Applicable to Carsem SZ-Site)

A) Source of requirement :
As per VDA 6.3 audit findings on 2022-Jul.

CARZ VDA6_3
audit result_2022

B) Requirement :

1) For automotive customer complaint , if NTF (No Trouble Found) need to use 4D report to include the process lot history review and current production control summary.

Appendix 6

CUSTOMER SPECIAL REQUIREMENT

(TI)

(Applicable to Carsem M & S-Site)

A) Source of requirement:

As a managerial systemic solution during meeting discussion. Email conversation received on 2023 June 10 @ 11:02PM from Lee, Vai Yuen.

RE_ Carsem Outlier
DAC8814ICDBT 3218

B) Requirement:

For customer complain (non-Automotive),

1. Allocate sufficient resources to complete defect analysis task with in specific time.
   - 2 days (sample size less than 10u)
   - 3 days (sample size more than 10u)
   - Pre alert TI teams if required extra days

xiii. Refer below escalation table as reference

		8D CT 10 calendar days									
Feedback Date	Sample received	Day1	Day2	Day3	Day4	Day5	Day6	Day7	Day8	Day9	Day10
* Check Prosess Mapping * Machine error log * Interview tehnician if required	* Analyse affected units. * Send sample to FA	Write D4			Write D5			Write D6 & D7	REPORT COMPILATION REVIEW & PUBLISH		
	* Pre alert Customer if required extra days.	a) 2 days(sample size less than 10u) b) Escalate if not done		a) Simulation b) Escalate if not done			a) Start draft D5 b) Escalate if not done		a) D5 review with management b) Escalate if not done	a) 8D report review with management b) Escalate if not done	
		a) 3 days(sample size more than 10u). b) Escalate if not done			a) Simulation b) Escalate if not done			a) Start draft D5 b) Escalate if not done		a) 8D report review with management b) Escalate if not done	

History of changes :

Ecn#/Rev	Description	Effective Date
880087-A	Change spec no QCG-014 to CRQA100M01 Rev: A Originator : LT Wong	01FEB92
920429-B	Complete Revision Originator : YC Choo	19MAY92
920889-C	Section 5.0 add note Originator : Lt Wong	15SEP92
921166-D	Complete rewrite Originator : Seevagan	17MAR93
930337-E	Revise 8D format Originator : Seevagan	14MAY93
930603-F	Revise para 9.6.5 add in para 9.7.1,9.7.2,9.7.3, 9.7.4,9.7.5. Originator : Seevagan	07JUN93
930991-G	Revise Attachment 3 (D.P.R format) remove "critical – yes, no from Section I of form. Originator : Seevagan.	29OCT93
950784-H	Amend item 9.4.1, 9.4.3, 9.5.3. Add in item 9.4.3 and renumber.Change Attachment 2 format for suppler C.A.R. Originator : YM Chin.	20OCT95
96070085S-I	Add sec.4.1 / 4.4 / 6.2.2 /6.3.1/8.6 / add new para 10.0 insert section their,ii,iii customer fault feed back. Section their section ii – Discrepant process report and section iii – supplier corrective action request. Rewrite para 7.0 of section I, include RDR format escalation format, in sec i. Make permanent the AEC 8D format,insert 8D valuation form and include special req on 8D response . Update para 5.1 of section ii. Originator : Ambrose Revise section iii and Attachment 1 – supplier corrective action request Originator : Nancy Ng.	30JAN97

History Of Changes :

Ecn#/Rev	Description	Effective Date
97030157S-J	xiv. Revise 8D format – add corrective action verification status column for item 3,5 & 7 in the 8D Attachment in section 1. Originator : Ambrose. xv. Add in item 6.1.6. Originator : Albert	05APR97
97080570S-K	xvi. To add in po number into supplier corrective action format (Attachment 1). 2) Section 3 : a) Revise item 4.0, 5.1.1, 5.1.3, 5.1.4, 5.1.5, 5.2.3. b) Revise supplier corrective action request format completely (Attachment 1) Originator : YL Pun xvii. Revise :- Section 1: a) Item 5.2.5, Att.1, 1a, 2 and 5. Section 2: b) Att.1 & 2. Section 3: c) Item 1.0, 2.0, 3.0, 5.1, & 'note'. Originator : Seevagan	22SEP97
98070448S – L	1)Change specification on title from "Corrective and preventive action" to "corrective and preventive action procedure". 2)Change all "QC" terminology to "QA". Originator : BL NG 3)Edit item number 3.1 – remove sentence "(with specific exception to captive lines)", 6.2.1, 6.3, 6.3.1, 6.3.2, 1.1,11.2 4) Add item number 6.4, 6.4.1, 6.4.2, 7.2, 8.7 5)Edit section their – customer fault feedback - Delete item number 4.2 and renumber the following item numbers. - Revise item number 5.1 as containment action response requirement, 5.2.5, Attachment 1(a), Attachment 1(b) – Section 2, Section 5, Attachment 2, Attachment 3, Attachment 4, Attachment 5 - Add item number 5.2.6.1, Attachment 6(a) and 6(b). - Delete Attachment 1their - Carsem quality department 8D corrective action report (form # CRQA100M01 – 2). 6)Edit Section II, Attachment 3. 7)Edit Section III, Attachment 1. Originator : Seevagan Processed by : Ruby	1998-NOV-23

History of changes :

Ecn#/Rev	Description	Effective Date
99040313S-M	xviii. Revise item 1.2,5.2.5,5.2.8.3,5.3.1, Attachment 1b (section 5) and section 1 Attachment 6b) xix. Remove Hong Leong Logo from formats. Originator : Shanthi xx. revise item 5.1.1 at section i Originator : Siva k Processed by : Saras	1999-APR-21
99070535S-N	1) Delete item 5.2.8.1 – 5.2.8.4 2) Revise Section 1, Attachment 2, Att 5, Section II – Att 3, Att 6A & 6B, xxi. Revise item 6.1.5,6.1.7, item 4.0 & 5.1.2 at Section I. 4) Add in item 5.2.4,5.2.6,5.2.8 – 5.2.11, 6.1.7 and note at item 5.3.1 at Section 1 5)Add in Attachment 6, 7, 8 , 9 and Attachment 10 at section 1. xxii. Revise Attachment 3 at Section II Originator : Shanthi Processed by : Saras	1999-JUL-29
99090723S – P	1)Revise Attachment 1 in Section II – Discrepant Process Report (Page 33 of 42). Originator : BL NG Processed by : Ruby	1999-OCT-19
99100841S-Q	xxiii. Revise Attachment 2 in section 1 – (customer fault feedback management flow under item 2.1.(Page 18) xxiv. Revise Attachment 5 in section 1 – (Add in item 3 and 4) and add in new customer ELMOS & SIPEX requirements. (Page 22) Originator : Shanthi Processed by : Saras	1999-DEC-16
00030152S – R	1) Update Section III : a) Revise item 6.2, 6.2.1, 6.2.2, 6.2.3, 6.2.4. - *(Page 36 – 37 Of 42)* b) Add in item 6.2.3.1 & 6.2.3.2. – *(Page 37 Of 42)* xxv. Revise Attachment 1. – *(Page 38 Of 42)* 2) Change “Vendor” terminology to “Supplier”. Originator : YL Pun Processed by : Amy	2000-Mar-08

History Of Changes :

Ecn#/Rev	Description	Effective Date
00040473M – S	1) Add in item 9.4, 9.4.1, 9.4.2 and 9.4.3 *(Page 3 & 4 of 42)* Originator : Sashikala Processed by : Eme	2000-APR-26
00070791S – T	xxvi. Revise Attachment 5 of Section 1 – *(Page 22 of 43)* 2) Add in item 7.2 of Section II – *(Page 31 of 43)* Originator : Shanthi Processed by : Amy	2000-Jul-31
00080897S – U	1) Revise Attachment 1 of Section III *(Page 38 of 43)* - Change 3-D and 8-D response time : 3D from 2 working days to 24hrs, and 8D from 10 working days to 10 calender days. 2) Revise item 6.1.2,6.1.4,6.1.5 *(Page 36 of 43)* Originator : YL Pun xxvii. Revise Section 1 : Attachment 5- at Texas Instrument and add item 5. *(Page 22 of 43)* xxviii. Revise Global 8D Corrective Report *(Page 8 – 11 of 43)* Originator : Seeva Processed by : Saras	2000-Oct-19
00111148S-V	1.) Revise Section : Attachment 5, 2.) Revise note 1-add in IRC/ MAXIM and note 3 . 3.) Revise 107roliferate action response req from 3 to 2 days and customer Lucent Tech from 24 HR to 3 days. 4.) Remove SEMIGEM and ELECTRONICS 5.)Add in customer PERICOM., AIRCHILD,ALPHA *(Page 22 of 43)* Originator : Seeva Processed by : Lin	2000-Dec-22
01010069S – W	Section 1 – Attachment 5 – Special Customer's Requirements – Change of Final 8D Response Req. for Customer Analog (All Sites) from 8 Calendar Days to 7 Calendar Days and for Customer Intersil from 10 Working Days to 8 Calendar Days. *(Page 23 of 44)* Originator By :- Seevagan Processed By :- Githa	2001-Feb-12

History Of Changes :

Ecn#/Rev	Description	Effective Date
01090688S – X	1)Update Section 1 - Para 4.0 - Include 'Definition : 8D – Global 8D'. - Add item numbers 5.1 – 5.5 and renumber all other item numbers accordingly. *(Page 6 of 45)* 2)Update Section 1 – Attachment 1 8D Format *(Pages 9 – 12 of 45)* 3)Update Section 1 – Attachment 1A Example of a Complete 8-D *(Pages 13 – 16 of 45)* 4)Update Section 1 – Attachment 1B G8D Evaluation Form. - Add item (b) and update item I. *(Page 17 of 45)* 5)Update Section 1 – Attachment 5 Special Customer's Requirements. - Update Customer Corrective Action Response Requirements for customer ON SEMI, LTC, ANALOG (All sites), TEXAS INSTRUMENT, AGERE SYSTEMS, ALLEGRO and SEMTECH. - Update item numbr 2, 3, 4 and add item number 6.0 *(Page 23 of 45)* Originator : Seevagan Processed by : Ruby	2001-Oct-12
02020150S – Y	1. Add in 'Global' at item 5.1 – *(Page 2 of 47)* 2. Add in reference spec # EMSWK0004 at item 8.8 – *(Page 4 of 47)* 3. Update Section 1 : - Change '8D' to 'G8D' at item 4.1 & 4.9 - *(Page 6 of 47)* - Remove 'Att. 1" at item 4.1 – *(Page 6 of 47)* - Remove "Defination : G8D – Global 8D" at item 4.0 – *(Page 6 of 47)* - Revise item 5.0 (1) to change 'Att. 1' to 'Att. 1a" – *(Page 6 of 47)* - Remove item 5.3 and renumber the following items accordingly – *(Page 5 – 9 of 47)* - Revise item 5.3.1 to change 'special' to 'specific' – *(Page 6 of 47)* - Revise item 5.4.1 to change '8D' to 'G8D', add in reference spec # : EMSWK0004 and add in 'Note' – *(Page7 of 47)*	2002-Mar-07

History Of Changes :

Ecn#/Rev	Description	Effective Date
	- Revise item 5.4.1 to change ‘8D’ to ‘G8D’ and add in ‘Note’ – *(Page 7 of 47)* - Revise item 5.5.2 to change ‘every 7 working days’ to ‘every 7 calendar days’ – *(Page 7 of 47)* - Revise item 5.6.5 to change ‘customer notification is required’ to ‘customer is to be notified’. – *(Page 7 of 47)* - Revise item 5.6.8 to remove ‘However’ . – *(Page 8 of 47)* - Revise item 5.6.9 to add in ‘process’ and ‘product’. – *(Page 8 of 47)* - Revise item 5.6.10 to remove ‘for management review’ – *(Page 8 of 47)* - Revise item 5.6.10.1 & 5.6.10.2 to remove ‘(see paragraph 5.3)’ – *(Page 8 of 47)* - Add in item 5.6.12 – “CAR Co-ordinator will provide the weekly report on customer dissatisfaction trend’ – *(Page 8 of 47)* - Revise item 5.7.1 to remove ‘Note’ – *(Page 9 of 47)* - Remove Att. 1 - Revise Att. 1A – Sample of a completed 8D format . – *(Page 10 – 13 of 47)* - Devide Att. 5 to Att. 5A, 5B, 5C, 5D & 5E. – *(Page 20 – 24 of 47)* Originator : Seevagan Processed by : Amy	

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
Z	03040328S	2003-Apr-16		1) Change all D.P.R to N.C.R xxix. Page 35 & 36 of) Revise Section II : Attachment 1(Non Conformance Report) format Originator : BL Ng Processed by : Saras
AA	04010048M	2004-May-19	Page 1 of 50 Page 16 of 50 Page 41 of 50 Page 20 –25 Page 26 Page 27	1. Changed Carsem (M) and Carsem (S) to Carsem (M) Sdn. Bhd. M-Site and S-Site. 2. Attachment 2 Section II – Item 2.2 change company name from Carsem M and Carsem S to Carsem (M) Sdn. Bhd. M-Site, S-Site or Carsem Suzhou. Originator By :- Seevagan Processed By :- Githa 3. Section III – Revise item 6.1.3. Originator By :- Shanthi Processed By :- Githa 4. Revised attachment 5A to 5F 5. Added in attachment 5G Originator : Eme Yusnita / Megala 6. Revised Attachment 6 8D Corrective Action Implementation Review Form (IRF) - Form # CRQA100M01-7 Rev. B Originator : Sasikala Processed by : Sugeeta

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
AB	04100668S	2004-Dec-30	Page 22 of 53 Page 24 of 53 Page 26 of 53 Page 27 & 28 of 53 Page 40 & 41 of 53	1. Add in item 6 under Attachment 5C. 2. Remove note 3 & renumber the rest of item from Attachment 5E 3. Revise Attachment 5G - remove IR from the title - remove IR/Unisem column - remove note 3 & 4 4. Add in Attachment 5H (IR customer special requirement) & Attachment 5 I (Flextronic Semiconductor customer special requirement) Originator : BL Ng 5. Change all " N.C.R." to "D.P.R" 6. Complete rewrite Attachment 1 (Discrepant Process Report) Under section II Originator : Sasikala Processed by : Saras
AC	06110837M	2006-Dec-08	Page 3 Page 4 Page 4 Page 6 Page 7 & Page 8 Page 15 – 16 Page 17 Page 18	1. Revise 6.1.7 2. Revise item 8.5 – Remove spec # QCG-075 and replaced with spec # QCG000110. 3. Revise item 8.7 – Change Quality System Requirement QS-9000 – AIAG Manual to Quality System Requirement ISO TS16949. 4. Section I – Revise item 4.1 , 4.5 to 4.10 5. Revise item 5.6.4, 5.6.6, 5.6.7, 5.6.8, 5.6.9, 5.6.9.1 – 5.9.6.4, 5.6.10 and 5.6.11 6. Revise attachment 2 on Customer fault feedback management flow. 7. Revise attachment 3 – 8D administration flow. 8. Revise attachment 4 – C.A.R Verification closure flow.

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
				9. Revise the following attachment:-
			Page 19	i Attachment 5A – Add in QA Engineer / Electronic Mail.
			Page 20	ii. Attachment 5B - Removed customer Microtune/Transilica requirements. - Revised note # 3 & 4.
			Page 21	iii. Attachment 5C - Add in customer Wolfson and Flextronic requirements and revised note # 3 & 4 to add in 'QA Engineer ', 'by electronic mail' & 'and preventive'.
			Page 22	iv. Attachment 5D – Revised note # 1 & 3 to add in 'of complaint', QA Engineer ' and 'by electronic mail
			Page 23	v. Attachment 5E - Removed customer Zarlink (UK), Beckman, Zarlink (Canada) and Adaptec requirements. - Transfer customer Fairchild and MIP requirement from attachment 5G into attachment 5E.
			Page 24	vi. Attachment 5F - Revised note # 2 to add in 'of complaint', QA Engineer ' and 'by electronic mail. - Add in SMSC customer requirement. 10 Removed attachment 5G and renumber the following attachments.
			Page 25	11. Attachment 5G – Add in note # 3.
			Page 27 – 29	12. Add in attachment 6 – How to assign C.A.R number.
			Page 30	13. Add in on list of customer with their own 8D format – Attachment 7
			Page 31	14. Add in 8D Closure tracking – Attachment 8. 15. Renumber attachment 6 – attachment 9
			Page 32	16. Revise attachment 9 – 8D Corrective Action Implementation Review Form (IRF) Form # CRQA100MO1-7 Rev. C
			Page 33 – 34	17. Revise attachment 9A & 9B – Implementation Review flow.

Revision History

11.5	ECN #	Effective Date	Affected Pages	Description Of Change
			Page 14	17. Obsolete the following attachment:- i. Attachment 8 – Quality Assurance Department (8D Response) – Form # CRQA100M01-8 Rev. A ii. Attachment 9 & Attachment 10 – 10 Calendar / Working Days Cycle Time and Express 8D! – Form # CRQA100M01-9 Rev. A 18. Remove Section II : Attachment 3 – Sample of an Escalation Notice 19. Section 1 : Attachment 1b 8D Evaluation Form – Section 8 Reviewed and Approved – Add in "(This inclusive all the revise 8D.)" Originator : Malathi Processed by : Sugeeta
AD	07100636S	2007-Oct-24	Page 42 of 53	1. Section III : Item 6.2.2 – add "be escalated to the relevant suppliers manager . On other hand, failure to do so will also". Originator : Nameeta Processed by : Prema
AE	07120828S	2008-Jan-08	Page 2 of 56 Page 6 – 7 of 56 Page 9 – 14 of 56 Page 15 – 16 of 56 Page 17 – 18 of 56 Page 19 of 56	1. Item 5.0 – Change from "Global 8" to 8" 2. SECTION I : – Item 4.1 – Change from "G 8" to 8D" – Item 5.2 – Change from "G 8" to 8D" – Revise Item 5.4 and Change from "G 8" to 8D" Change from "G 8" to 8D" – Item 5.5.1 – Change from "G 8" to 8D" 3. Complete re-write of SECTION 1 : ATTACHMENT 1A – SAMPLE OF A COMPLETED 8D FORMAT – QUALITY INCIDENT – FORM # CRQA100M01-1 REV. H 4. Revise SECTION 1 : ATTACHMENT 1B – 8D EVALUATION FORM . Revise SECTION 1 : ATTACHMENT 2 CUSTOMER FAULT FEEDBACK MANAGEMENT FLOW Revise SECTION 1 : ATTACHMENT 3 8D ADMINISTRATION FLOW

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
			Page 31 of 56 Page 32 of 56 Page 43 of 56	SAMPLE OF DEPARTMENT ABBREVIATION LIST : - - Add in "SOICN/M" and "PLATING" - Revise AP abbreviation from"CS to "AP" Attachment 7 - delete " Maxim" SECTION III : Item 6.1.2 – Change from "G 8" to 8D" Originator : YH Cheong Processed by : Prema
AF	08050290S	2008-Jun-06	Page 2 Page 4 Page 7 Page 9 – 10 Page 17 – 19 Page 21 Page 23 Page 24 Page 26 Page 27	1. Add in item 5.3 & 5.4 2. Item 8.5 – add in "VOP". 3. Section I - Item 1.0 _ add in ".. – Voice of Customer" - item 4.1 – arrange renumber attachment - item 5.3 – change "public folder" to "QA public folder" - Add in item 5.4.1 - Item 5.6.9 – revise attachment 8 to attachment 9. 4. Add in Section 2 5. Section 1 : attachment 1b - revise item c)customer reference # Section 1 – Attachment 2 – item 1 – add in ".. – Voice of Customer". 6. Add in section 1 : Attachment 3 Internal fault feed back management flow and renumber following attachments. 7. Section 1 : attachment 4 (8D administration flow) - Add in "internal feedback" at title. 8. Section 1 : attachment 6a (customer's special requirements on C.A.R. cycletime) - Add in item 3. 9. Section 1 : attachment 6b (customer's special requirements on C.A.R. cycletime) - remove "item 2 & 3.

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
			Page 28	10. Section 1 : attachment 6c (customer's special requirements On C.A.R. cycletime) - add in "4d – 48 hrs upon sample received 8D". Revise note : - - Item 2: Remove "Maxim". - Remove item 6. - Change "Public Folder" to "QA Public Folder"
			Page 29	11. Section 1 : attachment 6d (customer's special requirements On C.A.R. cycletime) Revise note : - Item 2 : remove "KL" - Change "public folder" to "QA public folder"
			Page 30	12. Section 1 : attachment 6e (customer's special requirements On C.A.R. cycletime) - Add in "Exar" Revise note at item 2 :- - Change "Zarlink (UK)/ Canada" to "Sipex". - Change "public folder" to "QA public folder".
			Page 31	13. Section 1 : attachment 6f (customer's special requirements on C.A.R. cycletime) - add in "Vishay"
			Page 32	14. Section 1 : attachment 6G (customer's special requirements IR) - Item 1 – change "Public Folder" to "QA public folder". - Remove item 3
			Page 37	15. Revise attachment 8 : list of customer with their own 8D format
			Page 40	16. Attachment 10A 8-D Corrective Action – Implementation Review Flow (8D's requiring IRF for other site) - Revise note to add in "ST and Intersil" Originator : Malathi Processed by : Sue

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
AG	08060346S	2008-Jun-17	Page 4 Page 5 Page 47 Page 48 Page 49 Page 50 – 55	1. Revise item 9.4.1 to add in "eliminate causes of potential nonconformities in order to prevent their occurrence" and "Preventive actions shall be appropriate to the effects of the potential problem". 2. Add in item 9.4.3 and renumber the following item. 3. Add in item 9.5 Originator : Julia Zhu 4. Revise item 6.1.2 to add in "(As per Section III (Attachment II)" 5. Item 6.1.5 – revise 8d cycle time from 10 to 8 calendar days 6. Item 6.2.1 – revise 8d cycle time from 10 to 8 calendar days 7. Section III : Attachment 1 - FORM # CRQA100M01-5 REV. H - Revise SCAR format to add in " Supplier containment has to be carried out for Carsem M –site , S-site and Suzhou. 8. Add in Section III : Attachment 2Supplier 8d Corrective Action Report – Form # CRQA100M01-6 Rev. A Originator : LY Ang / Nameeta Processed by : Prema
AH	08080483S	2008-Sep-15	Page 28 Page 30	1. Add in TI Automotive Customer Requirement 2. Add in Cypress 8D Cycle time requirement Originator : Malathi Processed by : Prema
AI	08090522S	2008-Oct-08	Page 39	1. Revise Attachment 10 : 8-D CORRECTIVE ACTION IMPLEMENTATION REVIEW FORM (IRF) – FORM # CRQA100M01-7 REV. D - Add in 8D Title Originator : Malathi Processed by : Prema

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
AJ	09020040S	2009-APR-03	Page 2	1. Revise item 2.0 2. Revise item 3.1 3. Add in item 3.2 & 3.3 4. Remove item 4.3 and renumber other item 5. Remove item 4.5 & 4.10 and renumber other item 6. Add in item 4.4 – 4.14
			Page 3 & 4	7. Add in item 5.5 & 5.6 8. Revise item 6.1.4 8. Add in item 6.18 – 6.1.12
			4	10 Add in word 'Production' under item 6.2 11. Complete rewrite item 6.2.1 & 6.2.2 12. Add in item 6.2.3
			Page 5	13 Revise item 7.1 14. Add in item A,B & C under item 7.2
			Page 6	15. Remove item 8.0 -8.8 and renumber other item 16. Revise item 8.1 17. Remove item (IV) and renumber other item
			Page 7	18. Add in word 'SECTION 1A'
			Page 10	19. Add in word 'SECTION 1B'
			Page 19 – 21	20. Add in word ' D1,D2,D3(a),D3 (b),D4,D5,D6,D7 & D8
			Page 34,35,38,39, 40,41 & 42	21. Add in word ' SECTION 1'
			Page 43	22. Revise item 2.0 23 Add in item A,B & C under item 3.0 24. Revise item 4.1 25. Complete Rewrite item 4.2 & 4.3 26. Add in item 4.4 27. Revise and add item A,B & C under item 5.1
			Page 44	28. Revise item 5.1.1.1,5.1.1.2,5.1.1.3 & 5.1.1.4
			Page 44 & 45	29. Add in item 5.1.2 -Lot rejection at QA gate
			Page 45 & 46	30. Add in item 5.1.3 Quality Feedback from Internal Customer
			Page 46 & 47	31. Revise item 5.2,5.2.1,5.2.2,5.2.3,5.2.4 & 5.2.5 , 5.3.1 & 5.3.2
			Page 48 & 49	33. Complete rewrite Attachment 1

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
			Page 51 Page 52	34. Add in Attachment 2 35. Add in Attachment 3 36. Obsolete QCG000018 (LRR PROCEDURE) & QRA000023 (IQA PROCEDURE) already consolidate into CRQA100MO1 Originator : KW Foo Processed by : Karen
AK	09040163S	2009-JUNE-24	Page 55 Page 58	2. Revise Section III : Attachment 1 3. Revise Carsem's Supplier 8D corrective action report Originator : Wanie Processed by : Karen
AL	09070328S	2009-NOV-09	Page 9 Page 10 Page 11 Page 18 Page 22 Page 25 P age 44 Page 46 Page 47 Page 48 Page 51 Page 52	1. Add in item 1 & 2 under item 5.7.1 2. Add in item 5.5.1 3. Add in item 1 & 2 under item 5.6.1 4. Remove Rodziahm name and replace with Fatimah under item 9 5. Revise item 3.2 6. Revise item 5.1 7. Revise item 6.4 8. Revise item 3.2 9 Revise item 5.1 Originator : Malathi Training to conducted by originator / respective department head (s) 10. Remove word " Metal Finishing to…….Back grind etc under item (C) for item 3.0 11. Remove item 4.3 & 4.4 under item 4.0 12. Remove word " Assembly to Metal Finishing etc" under item THEIR for item 5.1 13. Add in word " NCR Test " under item 5.1.3 14. Add in item 5.1.3.2 & 5.1.3.5 15. Revise item 5.1.3.6,5.1.3.7 & 5.1.3.11 16. Remove item 5.3.1 and renumber other item 17. Revise Section II : Attachment 2 18. Remove Section II : Attachment 3 and update with Section III in page 52

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
			Page 53	19. Remove Section II : Attachment 2 – Carsem Recurring Deficiency Report and update with item 6.2 – 6.2.5 Originator : KW Foo 20. Add in item 6.2.5 Originator : LY Ang Processed by : Karen Training Not Required
AM	1004052S	2010-JUL-19	Page 3 Page 8 Page 9 Page 11 Page 12-17 Page 20 Page 22 Page 23 Page 24 Page 26 Page 27 Page 29 Page 30 Page 35 Page 37 Page 38 Page 39 Page 40 Page 41 Page 42	1. Revise item 6.1.7 2. Complete rewrite item 5.5.2 , 5.6.2 & 5.6.5 3. Revise item 5.6.91 4. Revise item 5.5.2 Originator : Malathi Maniam 5. Remove the entire old 8D format and replace with New format 6. Revise item D3 (a) Training to be conducted by Originator / respective Department head (S) Originator : Sun Lian 7. Revise item 5.3 8. Revise item 6.3 & 7.0 9. Revise item 6.1 10. Revise section 1-attachment 5 – C.A.R verification closure flow 11. Add in customer AOS under section 1 – Attachment 6A 12. Revise section 1-Attachment 6C 13. Add in item 4 & 5 – 5.5.4 under section 1 – Attachment 6D 15. Add in Product Line Column under Section 1 – Attachment 7 16. Add in word HD1 & HD2 under Abbreviation List 17. Remove Customer Sipex / Exar, Infenion & PHK 17. Remove the old 8D Closure tracking and replace New tracking 18. Revise 8-D Corrective Action Report under Section 1 – Attachment 10 19. Revise 8-D Corrective Action Flow under Section 1 –Attachment 10A 20. Revise Attachment 10B

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
			Page 43 Page 44 Page 46 Page 50 Page 52	21. Add in Section 1-Attachment 11-8-D Corrective Action –Carsem (M) 8D approval format Training Not Required Originator : Malathi Maniam 22. Complete Rewrite word under item 5.1 (B) 23. Revise item 5.1.2.1, 5.1.2.2, 5.1.2.2.1 ,5.1.2.2.4 , 5.1.2.3 , 5.1.2.3.1 & 5.1.2.3.5 24. Revise Non –Conformance Report (NCR) Training to be conducted by Originator / respective Department head (S) Originator : KW Foo 25. Revise item 6.1.1 Training Not Required Originator : LL Hoo Processed by : Karen
AN	1010006S	2010-NOV-18	Page 44 Page 52 Page 53 Page 54 Page 55 Page 56	1. Add in word " Non-Conforming……..(Refer Attachment 3 & 4) under item 3.0 2. Add Attachment 3 – Red Flag at final test 3. Add in Attachment 4 – Red Flag definition Originator : Parminder 4. Revise item 6.1.5 under Section III 5. Revise item 6.2.1 6. Revise Section III Attachment 1 (Form # CRQA100M01 -5 Rev I) Originator : LY Ang Processed by : Karen

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
AP	1106266S	2011-Jun-30	Page 44 Page 44 Page 44 Page 45 Page 51 Page 54 Page 55 Page 56	1. Complete rewrite item under 4.2 & add in item 4.3 & 4.4 2. Add in item 5.1.1 3. Remove word “admin issuance & admin under item (A) 3.0 4. Add in item 5.1.1.4 5. Revise Section II –Attachment 2 6. Add in Section II – Attachment 5 –NCR tracking for QA inspection (CRQA100M01 – 10 Rev A) Add in Section II – Attachment 6 – NCR 7. tracking for test IQA (CRQA100M01 – 11 REV A) 8. Add in Section II – Attachment 7 – NCR tracking for audit (CRQA100M01 -12 Rev A) Originator : Johthi Dass Processed by : Karen
AQ	1106271S	2011-Jul-14	Page 2 Page 3& 4 Page 6 Page 6 Page 7 Page 8 Page 9 Page 10 Page 12 Page 19 Page 20 & 21 Page 22 &23 Page 24	1. Revise item 5.1 & 5.2 2. Complete rewrite item 6.1-6.6.1 3. Complete rewrite item 8.1 4. Remove item (i) – renumber other item ,remove item VI & remove word” and all communication till” under item (iii) 5. Remove word ” quality system” under Note Revise item 5.5.2-5.6.7 6. Revise item 5.6.9.2 – Note item 1 & 2 7. Revise item 5.5.1 8. Section 1 : Attachment 1A:- 9. Revise tittle from : Sample of a completed 8 Format Quality Incident To : 8D Format. There is no change in the content of the format (Form # CRQA100M01 -1 Rev I) Add in item 17 & revise item D1 10. Revise item D4 & D8 11. Revise item 12. 2.3,2.4,3.2,4.0,6.0,6.3,6.4,8.0,8.2,9.0,9.2 & Note item 1 under Section 1 : Attachment 2 13. Revise item 2.0,2.1,3.2,4.0,4.1,5.1,6.1,8.0,8.2,9.0 & 9.2 under Section 1 : Attachment 3

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
			Page 25 Page 26 Page 30 Page 37 Page 38	14. Revise 8D Administration Flow under Section 1 Attachment 4 15. Revise C.A.R verification closure flow under Section 1 Attachment 5 16. Revise Section 1 : Attachment 6D & remove note, item 3 & renumber other item 17. Remove Abbreviation for ICA,SOICN/M & combined to 1 abbreviation 18. Add in item 5 & 6 and complete rewrite item under note Originator : Malathi Maniam Processed by : Karen
AR	1111077S	2011-Nov-28	Page 46 Page 47	1. Add in item 5.1.2.1.3 & 5.1.2.2.5 2. Add in item 5.1.2.3.9 Originator : Jason Tan Processed by : Karen
AS	1112074S	2012-Mar-20	Page 7 Page 7 Page 7 Page 8 Page 9 Page 10 Page 11 Page 12 Page 20 & 23 Page 24 Page 26 & 27 Page 28 Page 30	1. Complete rewrite clause 4.9 & 4.10 & add in clause 4.11 & 4.12 2. Remove word “sample of “ under clause 5.0 3. Revise clause 5.2 4. Revise clause 5.5.2 & 5.6.5 5. Revise clause 5.6.7,5.6.8,5.6.9.2,5.6.9.3 & remove word “the “ under clause 5.6.11 6. Complete rewrite “sample of ,1B under clause 5.1 & remove number “7” under clause 5.2 & complete rewrite clause 4.9 & 4.10 and add in clause 4.11 & 4.12 7. Add in clause 5.5.3 – 5.5.8 & Revise clause 5.5.2 8. Revise clause 5.5.9 & add in clause 5.5.10 - 5.5.10.4 , 5.5.11 & 5.5.12 9. Revise item 8 & D8 under section Attachment IB 10. Revise clause 3.1 , 3 under Section 1 :Attachment 2 11. Revise clause 6.1 & add in Note under Section 1 Attachment 3 12. Revise Section 1: Attachment 4 – 8D Administration flow 13. Revise Section 1:Attachment 6A
			Page 31	14. Remove Section 1:Attachment 6B – Customer special requirement – AGERE system, ZILOG, PERICOM

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
			Page 32	15. Revise & add in item under Section 1 : Attachment 6C
			Page 33	16. Add in Section :Attachment 6C (cont'd)
			Page 34	17. Revise Section 1 :Attachment 6D & complete rewrite item 3
			Page 35	18. Revise ,add in & remove Note under Section 1: Attachment 6E
			Page 36	19. Remove customer LTC ,Vishay,add in customer Freescale, add in source of requirement for SMSC & remove Note
			Page 37	20. Add in Source of requirement & item 3-5 under Note Section 1 Attachment 6G
			Page 38	21. Remove Section 1 :Attachment 6H – Customer special requirement Flextronic Semiconductor
			Page 39	22. Add in Section 1 :Attachment 6I
			Page 42	22. Add in department HDPOWER under sample of department abbreviation list
			Page 44	23. Add in form # under 8D closure tracking Section 1 : Attachment 9 as previous rev don't have the form#
			Page 45	24. Remove 8-D corrective action implementation revise form (IRF) (Form#CRQA100M01–7 Rev E)
			Page 46	25. Remove Section 1 : Attachment 10A – 8-D corrective action – implementation revise flow (8D's requiring IRF for other site)
			Page 47	26. Remove Section 1 :Attachment 10B – 8-D corrective action implementation review flow (8D's requiring IRF for other site
			Page 48	27. Add in Section 1 :Attachment 11-Poliferation forum flow
			Page 49	28. Add in Section1 :Attachment 11A-Poliferation forum tracking format (Form # CRQA100M01-13 Rev A)
			Page 50	29. Add in Section 1 :Attachment 11B-Forum meeting minutes format (Format # CRQA100M01 – 14 Rev A)
			Page 51	30. Complete rewrite Section 1:Attachment 12 – Carsem(M) -8D approval format (Form # CRQA100M01-9 Rev B)
			Page 52	31. Add in Section 1 :Attachment 13A – ADI 8D corrective action closure validation flow
			Page 53	32. Add in Section 1 : Attachment 13B – ADI proliferation system flow Originator : Malathi Maniam Processed by : Karen

Revision History

Rev.	ECN #	Effective Date	Affected Pages	Description Of Change
AT	1204187S	2012-May-18	Page 22 Page 28 Page 30 Page 31 Page 32 Page 33&34 Page 36 Page 37 Page 38 Page 42	1. Add in item IV & Note under clause 5.2 2. Add in word "Calendar" under Section 1 :Attachment 6A 3. Add in word "Huge Important" & item 4, 5 & 6 under Section 1 : Attachment 6C 4. Add in word "Calendar" under Section 1 Attachment 6 C(Cont'd) & remove item 7 5. Add in Section 1:Atatchment 6C (Cont'd) –Customer Special Requirement Elmos 6. Add in item 3, 4 & 6 and word" working" under Section 1 Attachment 6D 7. Add in "Calendar" under Section 1 :Attachment 6F 8. Add in word "30 working / calendar days" under Section 1 : Attachment 6G 9. Remove Section 1 :Attachment 6I – Customer Special Requirement Elmos & update under in page 34 10. Remove department Power & Micro and update 2 under HD department. Originator : Malathi Maniam Processed by : Karen
AU	1206044S	2012-Jul-02	Page 64	1. Revise Section II – Attachment 5 –NCR Tracking for QA inspection (Form # CRQA100M01 - 10 REVB) Originator : Noora Processed by : Karen
AV	1211018S	2012-Nov-26	Page 5 Page 30 Page 41 Page 35 Page 33 Page 30-37 Page 41 Page 62	1. Revise typo error from Poke to Poka under item (b) clause 7.4.3 2. Revise CT for customer ADI from 7 days to 6 days & Revise working days from 6 to 7 under Section 1 :Attachment 6B 3. Add in item 7 under Section 1 : Attachment 8 4. Incorporate Customer Allegro spec # 900014 into this spec under Section 1 :Attachment 6E 5. Incorporate customer Intersil spec # Sub-Corrective-Action under Section 1 : Attachment 6D item 3 Revise alphabet at attachment as per sequence 7. Incorporate customer Maxim spec # 18-0100 & customer IDT spec # FRC-0004 under 8. Section1 Attachment 8-List of customer with their own 8D format 9. Originator : Malathi Maniam Add in item (D) under Section II 5.1 10. Add in Section II : Attachment 8 Customer Special Requirement Avago (All devices) Originator : SC Chiang Processed by : Karen

REVISION HISTORY

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
AW	1312022S	2014-Jan-10	Page 22 of 93	1. Add in item – The 8D must be signed off by the Dept Manager followed by the QA Manager approval. Approval of the Quality Assurance manager or his representative during his absence is mandatory. (If the 8d is revised with changes on root cause or to committed actions or additions of new actions, Dept Manager & QA Manager approval is Mandatory.) under D8 Reviewed and approved ***Reference to TECN # 1312010S**

			Page 48 of 93	2. Complete revise Section 1 : Attachment 13B- ADI Proliferation System Flow
				- Remove the below scope:
				  - CAPA proliferation action in the D7 of the 8D report.
				  - BKM (shared by ADI)
				- Add in new ADI Proliferation System Flow
				
				***Reference to TECN# 1312014S**
				
				Originator : Ruby

			Page 63 of 93	3. Revise Section III - Add in below note: Note: For IR customer, corrective action request and return material authorization number (RMA#) must be responded by the supplier within 5 calendar days upon receipt of the corrective action request/samples. ***Reference to TECN# 1311047S** Originator : Nameeta Processed by : Syima

REVISION HISTORY

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
AX	1408068S	2014-Sep-24	Page 72 of 96	1. Add in Appendix 1-MELEXIS Lesson Learn Fan Out Flow *Reference to TECN # 1405045S Originator : Malathi
			Page 2 of 96	2. Add in item 4.13- QRA0044 – e8D System User Manual
			Page 23 of 96	3. Revise Section 1 : Attachment 2. - Add in word "upon sample received" under item 2.4
			Page 29 of 96	4. Revise Section 1 : Attachment 6A Customer ST Microelectronic (All Sites) - Change final 8D response req from 4 working days to 5 working days.
			Page 31 of 96	5. Revise Section 1 : Attachment 6C Customer Maxim - Add in spec# 18-1347 under source of requirement. - Add in below note: Item no 3-All maxim 8D reports must be submitted with subconassembly 8-discipline checklist only for scar. Item no 4 – When responding to maxim customer complaint, 3x5 analysis will be used in analysis process of every maxim scar. INCORPORATE CUSTOMER MAXIM SPEC # 18-1347 INTO THIS SPEC
			Page 33 of 96	6. Revise Section 1 : Attachment 6D. Customer Melexis - Add in spec# 2014-06 under source of requirement. - Add in word "Quality Assurance Agreement Rev.008" under type of report issued. - Add in word "8D response time – time between receiving of failed devices from Melexis and submission of final report." Under final 8D response req.

REVISION HISTORY

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
			Page 35 of 96	7. Revise Section 1 : Attachment 6E Customer Sipex/EXAR - Add in "(respond < than or within 5 days – 2 pts, respond within 5- 10 days – 1 pt, more than 10 days – 0 pt)" under final 8D response req. - Add in note : Carsem is required to update mct on failure verification report (4D) within 48hrs upon sample receipt.
			Page 38 of 96	8. Add in Attachment 7 under how to assign C.A.R Number.
			Page 39 of 96	9. Complete revise sample of customer abbreviation list.
			Page 40 of 96	10. Revise sample of department abbreviation list. - Remove IC/ICA/SOICNM & ICB. - Add in CSMO & Backgrind. - Change from MLP to MLP_L & AP to MLP_N.
			Page 41 of 96	11. Revise Section 1- Attachment 8- List Of Customer With Their Own 8d Format - Remove customer SILABS – CYG Originator : Ruby Processed by : Syima
AY	1412008S	2015-Jan-12	Page 73 of 102 Page 74 of 102 Page 75 of 102 Page 76 of 102 Page 77 of 102	1. Add in Appendix 2-Customer Special Requirement (ST) 2. Add in Appendix 3- BKM Focus team for 3 Sites 3. Add in Appendix 4-BKM Escalation Guideline Add in Appendix 5- 8D Proliferation action guideline matrix 4. Add in Attachment 14- Fan Out Action Review Tracker (Form# CRQA100M01-16 Rev : A) Reference to TECN# 1412001S

REVISION HISTORY

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
			Page 6 of 102 Page 9 of 102 Page 48 of 102	5. Section 1A Revise item 4.12 to add in 13C 6. Section 1B Revise item 4.12 to add in 13C 7. Add in Attachment 13C-ST – CAPA Validation 0 ~ 6-Month Tracking (2014) (Form# CRQA100M01-17 Rev : A) Originator : Sasikala Processed by : Syima
AZ	1502005M	2015-Mar-19	Page 64 of 102	1. Section III- Revise item 6.2.3- add in below statement Supplier shall provide minimum 1 month monitoring data upon implementation of the corrective and preventive actions for effectiveness verification. Reference to TECN# 1502020S Originator : Wong NC Processed by : Sham

REVISION HISTORY

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BA	1505089S	2015-May-28	Page 74 of 104	1. Revise Appendix 2- Customer Special Requirement for ST Revise item 1 under activities column as per below From:- 8D CAPA Action – upon 8D submitted to customer, BKM Coordinator will work with QA Engineer / QA Manager to identified the FAN OUT action CIP (Quality Improvement) – done on quarterly basis, Site BKM Champion will work with OM and QA Manager to identify CIP actions To: 8D CAPA Action – BKM activity on the 129roliferate129n action will only be 129 roliferate to other department upon complete 8D closure.BKM Coordinator will work with QA Engineer / QA Manager to identified the FAN OUT action.CIP (Quality Improvement) – done on quarterly basis, Site BKM Champion will work with OM and QA Manager to identify CIP actions REFERENCE TO TECN # 1504031S TRAINING NOT REQUIRED Originator : Reetha
			Page 70 of 104	2. Revise Section III(Attachment II) -Add in WHY-WHY analysis on the possible systemic root cause Form # CRQA100M01-6 REV: C TRAINING NOT REQUIRED Originator : LY Ang Processed by : Sham

REVISION HISTORY

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BB	1506072S	**2015-Jul-23**	Page 7 of 106 Page 27 of 106 Page 41 of 106	1. Add in "test" for item 5.5.1 2. Revise Section 1: Att:5 – Add in wording "customer need to be informed on all recommit action" 3. Add in Section 1 : Att: 8 ST Automotive special requirement for 3X5why methodology. REFERENCE TO TECN # 1506034S Originator : Reetha Processed by : Lyn
BC	1509003C	2015-SEP-07	Page 11 of 106	1. Add in "8D actions closure in QA CAT system" for item 5.5.11.1 (.(Applicable to Carsem Suzhou only) Originator : Yao Jinhua Processed by : Li Qin

REVISION HISTORY

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BD	1510035M	2015-NOV-20	Page 13 of 108 Page 65 of 108	1. Revise 8D Corrective Action Report Form # CRQA100M01-1 REV : J - Add in Note: check partial of affected lot under pipeline analysis in 8d report 8D REF # : STCT150619-1 Training Not Required Originator : BD Lau 2. Revise Section III- Supplier corrective action request -Add in item 6.1.6- When there is any excursion incident (such as fire, chemical spillage, flood, other disaster, etc.) happened which impact the product quality and delivery, the supplier is required to provide an analysis report using the 8D methodology approach with corrective and preventive actions to prevent recurrence or to minimize impact to quality and delivery. Training to be conducted by originator/ respective department head (s) REFERENCE TO TECN # 1510021S Originator : Wong NC Processed by : Sham
BE	1511017C	2015-DEC-04	Page 30 of 108	1. Update Section 1 : Attachment 6C- CUSTOMER'S SPECIAL REQUIREMENTS FOR UPI Originator : Yao Jinhua Processed by : Li Qin

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BF	1603001M	2016-Mar-29		1. Revise Attachment 1A- SECTION 1 : 8D FORMAT
			Page 13 of 111	a) Item D3-(1) – Add a statement "Notify customer when there is a need to return potential affected materials"
			Page 17 of 111	b) Item D7-Add in item C- To proliferate the selected corrective actions from D5 with other Carsem product line. Item to be proliferated : Resp: Date: FORM # CRQA100M01-1 REV K Originator : BD Lau
			Page 80-81 of 111	2. Add in Appendix 6 – Customer Fan In Flow Originator : Reetha Processed by : Sham
BG	1604003C	2016-Apr-07	Page 8 of 112	1. Add Note 3: Process engineer must update TECN (temporary engineering change) within 2 working days after corrective action submitted if corrective action involved engineering change.
			Page 11 of 112	2. Add Note 3: Process engineer must update TECN (temporary engineering change) within 2 working days after corrective action submitted if corrective action involved engineering change. Originator : Daisy Lu
			Page 3 of 112	3. Revise item 6.2-Add in item 6.2.3- To use the Data Retrieval Checklist to ensure the recaptured data in the lot listing is complete and accurate with a peer validation performed.
			Page 83 of 112	4. Add in Attachment 15- Data Retrieval / Lots & Material Recapturing Checklist FORM # CRQA100M01-18 REV:A REFERENCE TO TECN # 1603048S Originator : Mugilan Processed by : Li Qin/Sham

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BH	1605010M	2016-May-31	Page 69 of 113	1) Revise Section III(Attachment II) 8D Corrective Action Report a) Revise item D3 under item 1 Add in word :- "(potential affected)" Notify customer when there is a need to return affected potential materials Revise column location-Add in In Transit Shipped to Customer Note : The checking must include partial or residue of affected lot Add in column Disposition b) Remove item D2 as below:- 2) Check potential affected materials shipped Date Location Vendor Lot # Shipped date Quantity RMA # Carsem S: Carsem M: Carsem Sz: Affected lot:
			Page 73 of 113	Revise item D7- Add in item C To proliferate the selected corrective actions from D5 with other Carsem product line Item to be proliferated : Resp.: Date : Form # CRQA100M01-6 REV: D Originator : LY Ang Processed by: Sham

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BI	1702002M	2017-Feb-23	Page 50 of 114	1. Revise SECTION II item 5.1- From:- The Non Conformance Report shall be issued to the appropriate personnel of the area by any employee when a discrepancy, non-conformance or any issue that may affect the company's image or Product Quality. The format is divided into 8 elements (see attachment 1) To:- The Non Conformance Report shall be issued to the appropriate personnel of the area by any employee when a discrepancy, non-conformance or any issue that may affect the company's image or Product Quality. . The format is divided into 8 elements (see attachment 1). Based on the NCR issuance, the appropriate contents must be completely filled. The use of hard copy or system NCR or equivalent is permitted. 2. Revise item (B):- From:- Lot rejection at QA Gate – To reject lots that is non-conformance or not meeting to the QA Acceptance Specification, Customer Specifications / Drawings. (Fill up element 1, 2, 3, 4) for lots rejec[t] 1st submission. If same lot rejected again, fill up element # 5, 6, 7 & 8 durin[g] 2nd submission). To:- Lot rejection at QA Gate – To reject lots that is non-conformance or not meeting to the QA Acceptance Specification, Customer Specifications / Drawings. Originator : Laila Processed by: Sham

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BJ	1703015M	2017-Mar-28	Page 82 of 117 Page 88 of 117	1. Add in Appendix 7-Customer Special Requirement for IFX Originator : Kannan 2 Add in Attachment 16 – ST special requirement on customer feedback for ST flow Originator : Reetha Processed by: Sham
BK	1704038M	2017-May-09	Page 65 of 119 Page 75-77 of 119	1. Revise SECTION III- Supplier corrective action request a) Revise item 6.1.5-Add in word "& 8 Discipline Evaluation Checklist(As per section III, Attachment III)" b) Revise item 6.2.3-Add in word "& 8 Discipline Evaluation Checklist are" and change IQA to SQE 2. Add in Attachment III- 8-Discipline Evaluation Checklist CRQA100M01-19 Rev:A Originator : LY Ang Processed by: Sham
BL	1707044M	2017-Aug-16	Page 32 of 120	1. Revise Section 1- ATTACHMENT 6D CUSTOMER'S SPECIAL REQUIREMENTS ON C.A.R. CYCLETIME Change wording from SUBCONTRACTOR RATING SYSTEM to SUB-RATING Originator : Reetha Processed by: Sham

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BM	1802045S	2018-Mar-02	Page 2 of 121 Page 5 of 121 Page 6 of 121 Page 27& 29 of 121 Page 40 of 121 Page 43 and 48 of 121 Page 75-77 of 121	1. Revise item 4.0-Revise item 4.11 and add in item 4.14-4.15 2. Revise item 7.4.3(c)-add in wording "Lesson learn(LL)" 3. Revise Section 1A- a) Revise item 4.9 and 4.10 5. Revise Section1 – Attachment 5 and Attachment 6(B) 6. Revise Section 1-Attachment 8 Add in wording "8D CYCLE TIME MATRIX" and CUSTOMER 8D CYCLE TIME & 8D REPORT FORMAT MATRIX 7. Revise Section 1-Attachment 11 and Attachment 13B Originator : Reetha 8. Revise Attachment III-8-Discipline Evaluation Checklist *Form # CRQA100M01-19 Rev: B* Originator : LY Ang Processed by: Marteena
BN	1803036M	2018-Mar-28	Page 11 of 121 Page 12-17 of 121 Page 46 of 121 Page 66 of 121	1. Add in note no 4- for Disclaimer footnote 2. Revise 8D format : CRQA100M01-1 Rev: L (Section 1:Attachment 1A) -Add in note for disclaimer footnote 3. Revise CARSEM (M) 8D Approval Format CRQA100M01-9 Rev: C (Section 1: Attachment 12) -Add in note for disclaimer footnote 4. Add in note no.1 for disclaimer footnote Originator : Rukumani/Reetha Processed by : Sham

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BP	1805005M	2018-May-14	Page 17 of 123 Page 21 of 123 Page 73 of 123 Page 77 of 123 Page 89 of 123	1. Revise Section 1 : Attachment 1A 8D format under item D7- FORM # CRQA100M01-1 REV: M a) Add in word "Mandatory to be filled" 2. Revise Section 1 : Attachment 1B 8D EVALUATION FORM under item D7- a) Add in word "FMEA is mandatory to be filled" 3. Revise Section III-(Attachment II) – CARSEM's SUPPLIER 8D Corrective Action Report under item D7 (a) Form # CRQA100M01-6 REV: E a) Add in word " Mandatory to be filled" 4. (Attachment III) 8-Discipline Evaluation Checklist under item D7-7.1 Form # : CRQA100M01-19 Rev:C a) Add in word " Mandatory to fill up FMEA" 8D REF # : ALLCMN170804 Originator : Patricia Kong 5. Add in Attachment 17- Special Requirement for Robert Bosch REFERENCE TO TECN # 1804027S Originator : MW Soo Processed by : Sham
BQ	1807001M	2018-July-13	Page 40 of 123	1. Revise Section 1-Attachment 8- LIST OF CUSTOMER WITH THEIR OWN 8D FORMAT & 8D CYCLE TIME MATRIX Originator : Sharmila/Ruby Processed by : Sham

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BR	1808022M	2018-Sep-06	Page 3 of 125 Page 79 of 125	1. Revise item 6.4- Risk Analysis a) Add in item 6.4.3(d) 2. Revise Appendix 2-Customer Special Requirement for ST a) Add in item C Originator : Chris Liu Processed by : Sham
BS	1901026M	2019-Feb-13	Page 2-3 of 129 Page 5 of 129 Page 7-8 of 129 Page 9 of 129 Page 11 of 129 Page 17 of 129 Page 20-21 of 129 Page 32-33 of 129 Page 36 of 129 Page 37 of 129 Page 41 of 129	Complete re-write 1. Revise item 6.1.12 2. Add in item 6.3.4 3. Remove 6.4-6.4.4 and re-naming the following item 4. Revise item 7.5- a) Remove item 7.5.3(a-d) b) Renaming the following item and revise item 7.5.3 5. Revise Section 1A-Customer fault feedback – Voice of Customer a) Revise item 5.5.2 b) Revise item 5.7-Add in note no 3 and 4 and renaming the following item 6. Revise Section 1B- Internal Fault Feedback (VOP – Voice of process) a) Item 5.5.2 b) Item 5.6-Add in note no 3 and 4 and renaming the following item 7. Revise Section 1- Attachment 1A-8D FORMAT Form # CRQA100M01-1 Rev:N a) Revise item C 8. Revise Section 1-Attachment 1B 8D EVALUATION FORM a) Revise item D3(b) and D7 9. Revise Section 1- a) Attachment 6D-Customer Special Requirement on CAR cycletime for Renesas b) Attachment 6G- Customer Special Requirement on CAR cycletime for Infineon c) Add in Attachment 6H- Customer Special Requirement on CAR cycletime for Melexis _HH 10. Revise Attachment 8- List of customer with their own 8D format & 8D cycle time matrix

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
			Page 42 of 129 Page 56 of 129 Page 56 of 129 Page 60 of 129 Page 43-44 of 129 Page 68 of 129	11. Revise Attachment 8A- Customer Special Requirement for ST Automotive 12. Revise Section II- Non Conformance Report (in-house corrective action request). a) Revise item 5.1.3.1.4 - 5.1.3.1.6,5.1.3.1.10 b) Revise item 5.2 13. Revise Section II-Attachment 2 Originator : Thilagavathy 1. Add in Section 1: Attachment 8B- Customer Special Requirement for ST Originator : Daisy Lu 1. Revise SECTION III- Supplier corrective action request a) Revise item 6.2.2 Originator : Ivy Chin Processed by : Sham
BT	1903031M	**2019-Mar-30**	Page 5 of 134 Page 41 of 134 Page 46 of 134 Page 72 of 134	1. Add in item 7.5.3 and re-naming the following item 2. Revise SECTION 1 : ATTACHMENT 6G-Customer Special Requirement for Infineon 2. Revise Section 1-Attachment 8- List of customer with their own 8D format & 8D cycle time matrix Revise column 8D format under customer Melexis Originator : Thila Processed by : Sham 3. Revise Section III – Supplier corrective action request a) Add in item 6.1.1.1 Originator : Ivy Chin

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
BU	1905107M	2019-Jun-14	Page 27-28 of 136 Page 50 of 136 Page 47-49 of 136 Page 88-89 of 136	1. Revise Section 1 : Attachment 2 Customer Fault Feedback Management Flow a) Add in item 9.3- For ST 8D action validation, refer to guideline stated in attachment 8C. The validation to be performed by personnel from other group serves as 3rd pair of eyes. 2. Revise Section 1 : Attachment 3 Internal Fault Feedback Management Flow a) Add in item 9.3- For ST 8D action validation, refer to guideline stated in attachment 8C. The validation to be performed by personnel from other group serves as 3rd pair of eyes. 3. Add in Section 1- Attachment 8C- Customer Special Requirement for ST a) Add in 8D action Validation Guideline 8D Report#: SSGTP2019041051 Originator : Nurhidayah 4. Revise Attachment 8B- 8D Self Assessment 5. Renaming Appendix 2 to Appendix 2A 6. Revise Appendix 2A- Customer Special Requirement for ST Refer to Flow in Appendix 2B. Fan Out Action Review Tracker in Attachment 14. 7. Add in Appendix 2B-Flow REFERENCE TO TECN # 1904036S Originator : Thilagavathy Processed by : Marteena

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE

BV	1906035M	2019-Jul-31	Page 7 of 139 Page 59 of 139 Page 73-74 of 139	1. Add-in attachment 9 (ADI Test NCR Flow) in Section II 2. Add-in Attachment 9 in item 5.1 (D) 3. Add-in Section II: Attachment 9 Customer Special Requirement for ADI-TEST NCR FLOW REFERENCE TO TECN # 1906010M Originator : Azizah Bt. Ramli Processed by : Sham
BW	2008039S	2020-Oct-06	Page 38 of 139	1 Revise Containment Action Response REQ. from “48 HRS” to “24 HRS “ for Customer ALLEGRO at Section 1 : Attachment 6E Originator : Sabiha Processed by : Marteena
BX	20100015M	2020-Oct-15	Page 76 of 139	1. Revise item 6.1.5 From: Note: For IR customer, corrective action request and return material authorization number (RMA#) must be responded by the supplier within 5 calendar days upon receipt of the corrective action request/samples. To : Note : Customer 8d respond cycle time will supersede Carsem requirement(Refer to section 1 – Attachment 8) Originator : LY Ang Processed by : Rodzi
BY	2101033M	2021-Feb-08	Page 2-3 of 138 Page 6 of 138 Page 11-14 of 138 Page 26-27 of 138 Page 28 of 138	1. - Add in item 5.7 - Revsie item 6.1.3 - Remove item 6.1.5 2. Revise summary list of Attachment - Remove item 24 - Renumbering the following items 3. Revise items 5.5.1, 5.5.2, 5.6.5, 5.4.1 & 5.5.2 4. Revise Attachment 2 – CUSTOMER FAULT FEEDBACK MANAGEMENT FLOW - Revise items2.4 & 6.3 5. Revise Attachment 3 – INTERNAL FAULT FEEDBACK MANAGEMENT FLOW - Revise item 6.1

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
			Page 35 of 138 Page 45 of 138 Page 49 of 138 Page 55 of 138 Page 69 of 138 Page 88 of 138 Page 85 of 138	6. Revise Attachment 6C – Customer Special Requirement (Elmos) - Revise from '7' to '10' working days 7 Revise Attachment 8 – LIST OF CUSTOMER WITH THEIR OWN 8D FORMAT & 8D CYCLE TIME MATRIX - Revise from 'calendar' to working days 8. Remove Attachment 11B – Forum meeting minutes format Form#: CRQA100M01-14 Rev. A 9. Revise Attachment 13B – ADI 8D PROLIFERATION/LL SYSTEM FLOW - Revise item 5.1.3.1.6 10. Revise Attachment 9 – Customer Special Requirement (ADI) - Revise item 6.1.5 11. Revise Appendix 2f – Customer Special Requirement (ST) - Revise item 1 Originator : Thilagavathy Processed by : Rodzi 12. Revise Appendix 2C – Customer special Requirement (ST). • Added new requirement. Originator : Nurhidayah Processed by : Nur Dalila
BZ	2102053S	2021-Feb-19	Page 2 of 134 Page 6 of 134 Page 6 of 134 Page 9 of 134	1 Add in item 4.16- QCG0000133 – Best Known Method (BKM)/Lesson Learnt (LL) Proliferation Procedure 2 Removed Attachment 11,11A & Attachment 13B at list of table 3 Renumber from 24-31 to 22-28 at list of table 4 Removed Appendix 1 &Appendix 2B at list of table

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
			Page 9 of 134	5 Renumber from 1-9 to 1-7 at list of table
			Page 10 of 134	6 Delete item 4.9- Proliferation / Lesson Learned forum flow (Attachment 11) at section 1
			Page 10 of 134	7 Delete item 4.10- Proliferation / Lesson Learned forum tracking (Attachment 11A) & Forum Meeting minutes format (Attachment 11B) at section 1
			Page 11 of 134	8 Renumber from 4.11-4.12 to 4.9-4.10 & 5.6.9 to 5.6.8 at section 1
			Page 11 of 134	9 Removed sentence "Upon approval the hardcopy 8D must be submitted to C.A.R CO-Ordinator for retention purpose" from item 5.6.2
			Page 11 of 134	10 Removed wording "C.A.R CO-Ordinator" from item 5.6.4
			Page 11 of 134	11 Removed item 5.6.8- If corrective action is taken on one product/process, the proliferation / LL forum must be done on other similar processes / products. Refer to attachment 11 for more details.
			Page 11 of 134	12 Re moved sentence "or the C.A.R. CO-Ordinator" 5.6.9
			Page 13 of 134	13 Remove sentence "Proliferation / Lesson Learn forum flow (Attachment 11)" from item 4.9
			Page 13 of 134	14 Removed sentence "Proliferation / Lesson Learn forum tracking (Attachment 11A) & Forum Meeting minutes format (Attachment 11B)" from item 4.10

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE	
			Page 13 of 134	15	Renumber from 4.11-4.12 to 4.9-4.10 & 5.6.9 to 5.6.8 at section 1B
			Page 13 of 134	16	Removed sentence "Upon approval the hardcopy 8D must be submitted to C.A.R CO-Ordinator for retention purpose" from item 5.5.1
			Page 14 of 134	17	Removed sentence "Upon approval the hardcopy 8D must be submitted to C.A.R CO-Ordinator for retention purpose" from item 5.5.4
			Page 14 of 134	18	Removed sentence "If corrective action is taken on one product/process, the proliferation/LL forum must be done on other similar processes / products. Refer to attachment 10 for more details" from item 5.5.9
			Page 31 of 134	19	Revise Section 1 :Attachment 5 C.A.R. Verification Closure Flow
			Page 33 of 134	20	Removed item #6. ADI 8D Proliferation/Lesson Learn Flow as Per Attachment 13B" from SECTION 1 : Attachment 6B
			Page 34 of 134	21	Removed "Attachment 11" from SECTION 1 : Attachment 6C
			Page 35 of 134	22	Removed "Attachment 11" from Section 1 : Attachment 6D
			Page 41 of 134	22	Removed "Attachment 11" from Section 1 : Attachment 6G
			Page 48 of 134	23	Complete removed "Section 1: Attachment 11-8D Proliferation/ LL Forum Flow
			Page 49 of 13	24	Complete removed SECTION 1: Attachment 11A-Poliferation Forum Tracking Format

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
			Page 52 of 134 Page 83 of 134 Page 85 of 134 N/A	25. Complete Removed Section 1 : Attachment 13b-Adi 8d Proliferation / LL System Flow 26. Complete removed Appendix 1 MELEXIS Lesson Learn Fan Out Flow 27. Complete removed Appendix 2B- Remove all item related to BKM and refer to QCG0000133. 28. Remove all item related to BKM and refer to QCG0000133. Originator : Thilagavathy Processed by : Marteens
CA	2108017M	2021-Nov-11	Page 2 of 134 Page 54 of 134	1. Add in spec.#EMS000024 – Hazardous substances management system manual in all quality system spec Originator : LY Ang Processed by : Rodzi 2. Revise Section II- Attachment 1 – **NON-CONFORMANCE REPORT (NCR)** Form# CRQA100M01-8 Rev. C a) Add in critical and non critical column for item 2(b) Originator : Annie Processed by : Sham
CB	2112002S	**2022-Jan-17**	Page 2 of 135 Page 3 of 135 Page 10 – 15 of 135 Page 16 – 21 of 135 Page 22 – 25 of 135 Page 26 of 135 Page 27 of 135	1. 6.1.2 – revise from Attachment 9 to Attachment 8. 2. Delete item# 6.1.4 3. Add in item# 6.5.5 & renumber the following. 4. Revise Section 1A & 1B. 5. Revise Section 1 : Attachment 1A. (CRQA100M01-1 Rev.P) 6. Revise Section 1 : Attachment 1B. 7. Add in Attachment 1C. 8. Add in Attachment 1D.

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
			Page 28 – 29 of 135 Page 30 -31 of 135 Page 32 of 135 Page 33 of 135 Page 44 of 135 Page 45 of 135 Page 48 of 135 Page 54 – 55 of 135 Page 80 of 135 Page 92 of 135	9. Revise Section 1: Attachment 2 Item 8.2 -change 10k to 15k 10. Revise Section 1: Attachment 3. Item 8.2 -change 10k to 15k 11. Revise Section 1: Attachment 4. 11. Revise Section 1: Attachment 5. 12. Add in Section 1: Attachment 6I. 13. Add in Section 1: Attachment 7. 14. Add in Section 1: Attachment 10. 15. Revise Section II: Attachment 1. (CRQA100M01-8 Rev.D) 16. Revise Appendix 1B. 17. Add in Appendix 3B. Originator : Thilagavathy Processed by : Dylla
CC	2202016M	2022-May-27	Page 22 of 139 Page 27 of 139 Page 47 of 139 Page 85-87 of 139 Page 94-95 of 139 Page 9 of 139 Page 96 of 139	1. Revise Section 1 : Attachment 1A-8D Format a) Add in requirement to review Control Plan under D7 FORM # CRQA100M01-1 REV: Q 2. Revise Attachment C- Containment Lot Management Flow Revise Section 1-Attachment 7 Customer 8D Cycle Time & 8D Report Format Matrix a) Revise EPC 8D C/T Revise Attachment 1E 3. Revise Attachment 3B- Customer Special Requirement for RBG 4) Add in Attachment 3B-1 Containment Lot Management Flow (RBG) Originator: Thilagavathy Processed by : Sham 5 Add # 12- Appendix 4 at table list of 6 Add Appendix 4-Customer Special Requirement (ALLEGRO) *Reference to 8D # COMP4848 Originator: Fatin Nurnisha Processed by : Marteena

REV.	ECN #	EFFECTIVE DATE	AFFECTED PAGES	DESCRIPTION OF CHANGE
CD	2206003M	2022-Jul-27	Page 25-26 of 141 Page 47 of 141 Page 51 of 141	1. Revise SECTION 1 : Attachment 1B 8D Guideline Form a) Revise item D2 and item D2(b) b) Revise item D4 2. Revise SECTION 1: Attachment 7 CUSTOMER 8D CYCLE TIME & 8D REPORT FORMAT MATRIX a) Revise Customer Skyworks-under 8D cycletime-from 7 calendar day to 10 working day 3. Add in Attachment 11 Guideline for process mapping and commonality analysis Originator: SS Ng Processed by : Sham
CE	2208003C	2022-Aug-10	Page 9 & 100 of 144	1. Add Appendix 5 for TI customer special requirement. (SZ Site only) Originator: Ashley Lu Processed by: Chris Liu
CF	2208028S	2022-Nov-17	Page 58 of 146 Page 2 of 146 Page 9 of 146 Page 10 of 146 Page 10 of 146 Page 10 of 146 Page 13 of 146 Page 29 of 146 Page 31 of 146 Page 33 of 146 Page 34 of 146	1. Add in item 5.5 **Originator: Kalai Selvi** 2. Delete item 4.9 & renumber. 3. Update table list of Appendix 4. Revise item 4.0 5. Add in word at 5.2 6. Add in word at note 5.4.1 7. Revise item 4.0 8. Section 1 : Attachment 2. • Add in word at 5.1 9. Section 1 : Attachment 3. • Add in word at 5.1 10. Revise Section 1 : Attachment 4. 11. Revise Section 1: Attachment 5.

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			Page 37 of 145 Page 41 of 145 Page 45 of 145 Page 46 of 145 Page 51-52 of 145 Page 70 of 145 Page 98 of 145	12. Section 1 : Attachment 6C – revise requirement for SEMTECH, Texas & Maxim 13. Removed "Note : Carsem is required to update MCT on failure verification report (4d) from within 48hrs upon sample receipt. from Section 1: Attachment 6e and change doc number for Allegro 900014 to PRC-0001658 14. Add in Section 1: Attachment 6J. 15. Section 1: Attachment 7 – revise requirement for TIM & Maxim. 16. Add in Attachment 12 - Guidelines on VOC and VOP Categorizing for Internal Reporting. 17. Section III • Revise item 4.0 18. Add in Appendix 3C-Customer Special Requirement (Robert Bosch). Originator: Thila Processed by : Dylla
CG	2304004S	2023-APR-06	Page 3 of 147 Page 3 of 147 Page 6 of 147 Page 6 of 147 Page 6 of 147 Page 6 of 147 Page 10 of 147 Page 11 of 147 Page 11 of 147 Page 11 of 147	1 Revise item 6.1.2 from "To track and follow up on the 8D C.A.R closure & escalate when necessary. (Attachment 8)" to "To track the 8D C.A.R closure (Attachment 8)." 2 Add sentence "open and close status" at item 6.1.3 Table list of Attachment- 3 -Removed "Attachment 6B" & renumber 4 -Removed Attachment 10- Data Retrieval / Lots & Material Recapturing Checklist 5 -Removed Attachment 11- Guideline for process mapping and commonality analysis 6 -Add Attachment 6K Revise Section I 7 a.) Remove item 4.4, 4.5 under Section 1 8 b.) Remove item 5.5.2 and renumber 9 c.) Add item 5.6.2, 5.6.7 and renumber 10 d.) Revise item 5.6.8- Add in QA Eng. responsible to update customer 8D action any recommit and completion date

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			Page 12 of 147	11 e.) Add item 5.6.11-5.6.16
			Page 13 of 147	12 f.) Revise 5.7.1, add in 5.7.2 Revise Section 1B:-
			Page 14 of 147	13 a.) Remove 4.4, 4.5
			Page 28 of 147	14. Add in form number CRQA100M01-20 Rev A for Section 1: Attachment 1D Revise Section1 : Attachment 2
			Page 31-32 of 147	15. a.) Add item 10- 8D closure note under 10.1 16. Remove Section 1: Attachment 6B- 6 working /calendar days
			Page 37-38 of 147	17 Add note : NOTE :For Customer ST
			Page 37-38 of 147	18 Revise Section 1: Attachment 6C -Add in 5 and 10 working/calendar days -Add in Analog -Revise TIM from 7 to 10 for non-auto -Add in note no 9 & 10
			Page 40 of 147	19 Revise Section 1: Attachment 6C cont - Elmos 7 working days priority and 10 w/days as standard
			Page 45 of 147	20 Revise calendar days from “8 calendar days to 10 calender days
			Page 47 of 147	21 Add in Section1: Attachment K – Car cycle time for Little Fuse
			Page 48 of 147	22 Revise Section 1: Attachment 7-Customer 8D Cycle Time & 8d Report Format Summary MATRIX - Revise ADI from 7 to 10 cal/days - ELM 7 wo/days (prio) 10 wo/days (standar) - Correct typo error Melexis & Semtech spec number - TIM 7 to 5 cal day for auto , 10 cal day for non suto - 7 cal day to 7 wo/days for NXP - Add in little fuse - Revise 8D cycle time SEMTECH from : 7 working Days” to “5 Calendar Days (Auto) 10 calendar days (Non-Auto) 23. Remove Attachment 9 Form # CRQA100M01-9 Rev C 24. Remove Attachment 10 Form # CRQA100M01-18 Rev A

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				25 Add in QA gate buy-off results summary item 5.1.2.3.9 Originator :Thila
			Page 11 of 147	26. Add item 5.5.4 under section 1- Once received VOC, respected QA engineer will request to increase QA sample size based on defect mode.
			Page 12 of 147	27 Revise clause 5.611 from ”Upon verification for closure as of target date; if all actions are found to be in place and effectively carried out, the 8D C.A.R shall then be defined as ‘closed’. Any C.A.R found not effective, or actions not carried out shall be defined as ‘Open’ or ‘Pending’ to “The QA Engineer shall lift the increase sampling size setting in system (if require), before close the 8D. Upon verification for closure as of target date; if all actions are found to be in place and effectively carried out the 8D C.A.R shall then be defined as ‘closed’. Any C.A.R found not effective or actions not carried out shall be defined as ‘Open’ or ‘Pending’.
			Page 13 of 147	28 -Add item 5.4.4- Containment action on increasing QA sample size through “Sampling Instruction” shall be updated in MES system (if required) as per below guideline:
			Page 14 of 147	29 Add sentence “The QA Engineer shall lift the increase sampling size setting in system (if require), before close the 8D” at item 5.5.11.1 Originator : Nor Suhaila Processed by : Marteena
CH	2304001C	2023-May-23	Page 5 of 145	1. Add in item 7.5.1 (b,c)- Risk Analysis. Originator: Mary Wu Processed by: Chris Liu

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CI	2309027S	2024-Feb-05	Page 15 of 147 Page 47 of 147 Page 84 of 147 Page 99 of 147 Page 36 of 147 Page 42 of 147 Page 45 of 147 Page 47 of 147	1. Add in item 5.5.12.1 2. Section 1: Attachment–7 - Customer 8D Cycle Time & 8d Report Format Summary Matrix • Revise customer ST 3. Complete rewrite Appendix –C - Customer special Requirement (ST) 4. Appendix –6 - Customer Special Requirement (TI) (Applicable to Carsem M & S-Site) Originator : SS Ng / Norhayati Fuzi 5. Section 1 : Attachment 6C • Revise customer Semtech 6. Revise Section 1 : Attachment 6G 7. Revise Section 1 : Attachment 6J 8. Section 1: Attachment–7 - Customer 8D Cycle Time & 8d Report Format Summary Matrix • Revise customer ICS / IDT & Semtech Originator : Thila Processed by : Dylla
CJ	2402006S	2024-Feb-16	Page 2 – 5 of 147 Page 10 – 12 of 147 Page 22 & 25 of 147 Page 29 of 147 Page 45 of 147 Page 58 of 147	1. Add in item 4.17 2. Revise item 6.2 3. 6.3.2 – add in wording 4. Revise item 6.4 & 6.4.1 5. Delete item 6.4.11 & 6.4.12 6. Add in item 6.5 7. Delete item 7.4.5 8. Delete item 5.4 & renumber the following 9. Revise SECTION 1 : Attachment 1B 8D GUIDELINE 10. Revise SECTION 1 : Attachment 2 Customer Fault Feedback Management Flow 11. Revise SECTION 1 : Attachment 6J 12. Revise SECTION II : Attachment 1 (Form# CRQA100M01-8 Rev.E) Originator : Thila Processed by : Dylla

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CK	2501075S	2025-Feb-17	Page 6 of 150 Page 26 of 150 Page 47 of 150 Page 48 of 150 Page 49 of 150 Page 54 of 150 Page 59 to 150 Page 60 to 150 Page 59 to 150	1 Update table list of Attachment 2 Add "validation results to be recorded in 8D report" at D6 3 Add Section 1: Attachment 6L-Customer's Special Requirements On C.A.R. Cycletime MACOM 4 Add Section 1: Attachment 6M-Customer's Special Requirements On C.A.R. Cycletime LEM 5 Add for Customer MACOM at SECTION 1: Attachment 7 6 SECTION II 7 -Add sentence "During holidays or off days email approval is accepted and the email approval must be attached together with the NCR format during submission. Accepting QA to ensure this is in place prior acceptance of the NCR. " under item 5.1.1.5 Revise SECTION II: Attachment 1 8 -To add in corrective / preventive action respond, verify and approval 9 -To add in Verification of Corrective / Preventive Action effectiveness & closure verify by QA supervisor and above. 9 -Revise to *Notes: During weekend and public holiday approval via email is allowable and the approval must be attached together with this NCR format.* (Form# CRQA100M01-8 Rev.F) Originator : Thila / Fatin Processed by : marteena
CL	2506010C	2025-AUG-11	Page 3 of 151 Page 71of 151 Page 73 of 151 Page 60-61 of 151	1. Add Item 4.18 VDA6.3 Standard, Item 5.8 NTF, Item 5.9 0 km complaints Definition Originator: Wu Yanqing 2. Update Item 6.1.5 3. Update Section III; Attachment 1 (CRQA100M01-5 REV.J) Originator: Lily Meng 4. Revise Non-conformance report(NCR) *(FORM#CRQA100M01-8 REV.G)* -To revise corrective / preventive action respond, verify and approval -To add in Clause "applicable for failure category Non – Conformities, QA Gate submission (3rd Submission only) and Quality Feedback from Internal Customer (Final Test)" at item 5,7 & 8

				Originator: Fatin Processed by: Chris Liu
CM	2508037C	2025-SEP-17	Page 6 of 152 Page 49 of 152 Page 50 of 152	1. Update Section 1 Attachment list. 2. Add Section 1 Attachment 6N -Customer Special Requirement on C.A.R. Cycletime-JOU 3. Update Section 1 Attachment 7 CUSTOMER 8D CYCLE TIME & 8D REPORT FORMAT SUMMARY MATRIX -Add JOU Customer requirement. Originator: James Yan Processed by: Chris Liu

ECN Checklist